Reaching for Yield in the Dark

Bank Competition for Household Savings in China

Kevin Zheyuan Cui[1] Tao Shen[2] Tianqi Zhi[2]

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Abstract

Household savers in China's RMB 33 trillion market for bank wealth-management products respond strongly to the returns that products report and to how much those returns fluctuate, but not to the risk behind them. Using the products' holdings, we show that smaller banks exploited this bias to compete for savings, because they lack the reputation that draws savers to larger banks. Their products reached for yield by holding riskier debt than larger banks' products, and the largest source of the extra yield was the debt of local government financing vehicles (LGFVs) in the banks' home provinces. Yet smaller banks rated these products as safe, locked savers in with restricted redemptions, and reported returns that barely moved, so the risk stayed in the dark. The strategy needed both the risky yield and the concealment of its risk, so it was fragile. China's 2023 debt-resolution program absorbed much of the LGFV risk and lowered the yield with it. Smaller banks relied on that yield the most, so they lost returns and fundraising. When banks have little reputation to lose, they compete through a fragile strategy that exploits the behavioral bias of households.

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[1] Department of Economics, Princeton University

[2] School of Economics and Management, Tsinghua University

1 Introduction

How small banks compete with large banks, and whether the way they compete is good for the economy, is a long-standing question in banking. In the United States, one view holds that small banks survive by knowing local borrowers whom large banks cannot evaluate, so they bring credit to small firms that would otherwise go without (Stein 2002, Berger et al. 2005). Another view holds that small banks lack the scale, reputation, and expected government support of large banks, and that they make up for these disadvantages by taking more risk (Gandhi and Lustig 2015, Buch et al. 2014, Wang 2017). China's regulators intend small banks to play the first role. They direct rural commercial banks to lend mainly to small firms and farmers in their home counties, and they tolerate higher loss rates on small-business loans (China Banking and Insurance Regulatory Commission 2019a,b). Small banks must also compete with large banks for savings. This side of the competition is harder to study, because deposits fund a bank's whole balance sheet, so savers' choices cannot be linked to the assets behind them. China's bank wealth-management products (WMPs) make this link possible. WMPs are a central part of Chinese banks' business. Banks sell them to households as an alternative to deposits, manage them directly or through wealth-management subsidiaries, and have used them to compete for savings (Hachem and Song 2021, Acharya et al. 2025). Each product discloses its holdings and its net asset value, from which we measure its size, its returns, and its savers' net flows, so we can see both what savers choose and what their savings buy.

The market is large. At the end of 2025 WMPs held RMB 33 trillion, nearly a quarter of China's GDP. They had about 143 million investors, and almost all of them were individuals.[1] Since 2018 these products cannot guarantee principal, so their losses fall on savers. Most of these savers are not financially sophisticated. They see a product through what its bank reports and not through the assets it holds. The bank reports the product's returns, how

[1]Industry statistics are from the Banking Wealth Management Registration and Depository Center's annual market reports (2025, 2026), referred to below as the annual WMP market reports. China's 2025 GDP of RMB 140.2 trillion is the National Bureau of Statistics' preliminary estimate.

much those returns fluctuate, and a risk rating that it sets itself and that decides who may buy the product. The banks that sell these products differ widely in reputation, from large national banks with nationwide franchises to small city and rural banks with little reputation to lose. Yet smaller banks' products looked better than larger banks' products on everything that savers watch. They paid higher returns, their returns moved less, and more of them were rated safe, at one of the two lowest of five risk levels. We ask how banks with little reputation could offer such products, and what risk this competition places in household savings.

Our answer is that smaller banks compete by taking risk that their savers cannot see, and that this strategy is both exploitative and fragile. Smaller banks face larger banks whose reputations attract cautious savers. Households move their savings quickly toward higher reported returns, yet they are risk averse and avoid products that look risky. To win these savers, smaller banks need a higher return that looks safe. We show how they achieve it by linking the holdings of China's WMPs to their reported returns, ratings, and investor flows. Savers respond strongly to the returns that products report and to how much those returns fluctuate, but not to the risk behind them. Smaller banks exploit this bias. Their products reach for yield by holding riskier debt than larger banks' products. The largest source of their extra yield is the debt of local government financing vehicles (LGFVs) in their home provinces (Chen et al. 2020, Acharya et al. 2025). LGFVs are state-owned companies that borrow outside formal government debt to fund local infrastructure. Yet smaller banks rate these products as safe, lock savers in with restricted redemptions, and report returns that barely move, so the risk stays in the dark. In the 2022 bond sell-off, market yields and spreads implied a loss of about 1.6% on their holdings in two months, while their reported returns only dipped to about zero. Smaller banks need this strategy to survive. Without the risky yield they cannot draw savers away from more reputable banks, and if the risk showed, savers would run. The strategy needs both the risky yield and the concealment of its risk, so it is fragile. China's 2023 debt-resolution program absorbed much of the LGFV risk and lowered the yield with it. Smaller banks relied on that yield the most, so they lost returns

and fundraising. A model of competition between banks that differ in reputation explains these patterns.

Our data are the quarterly portfolio disclosures of 60,396 WMPs issued by 171 banks. The disclosures run from 2022Q1 to 2025Q3 and list each product's largest holdings. We link them to each product's reported returns, risk rating, and investor flows, to the issuers of the bonds it holds, and to its bank's balance sheet, branches, and WMP balances. Figure 1 shows that the main patterns are already visible in the raw data. It compares national banks, larger local banks, and smaller local banks. We call a bank smaller when its interest-bearing assets at the end of 2021 are at or below the median of the banks in our main sample, which is RMB 0.44 trillion. All smaller banks are local banks. Before the 2023 program, smaller banks' WMPs held the riskiest bonds. Their bonds had been issued at higher spreads than those of larger local banks and national banks, and more of their holdings were home LGFV bonds. Valued at market prices, their holdings implied the most return volatility, yet their products reported the least. National banks' products reported nearly all of the volatility their holdings implied, whereas smaller banks' products reported about half. Nearly all of smaller banks' products also restricted redemptions, and nine in ten carried a safe rating.

Savers reward the returns and stability they see, not the risk behind them (Section 3). Net inflows into products in the top fifth of past returns exceed those into the bottom fifth by 12% of assets. Savers punish even small dips below the usual quarterly return of 0.5 to 1%. Daily-open products can be redeemed on any business day, and those that reported 0.25 to 0.5% had net flows 43 percentage points of beginning assets below those of products of the same bank that reported the usual return. This response comes from savers' decisions to withdraw. It holds within the same product, and it is not statistically significant where savers cannot redeem. Savers also penalize the volatility they see, but not the risk that products hold. One standard deviation more of reported volatility lowers flows by 2.3% of beginning assets, whereas the relation of flows to a product's holdings of home LGFV credit is small and not statistically significant. The 2022 bond sell-off gives the sharpest test of this

contrast, because the same market losses appeared to very different degrees in the returns that products reported. Among products of the same bank, each 10 basis points a month of reported loss in November and December 2022 came with withdrawals of 1.2% of assets in the next quarter, whereas the loss implied by the holdings made no statistically significant difference. Savers thus ran from the losses that products reported, not from the losses their holdings took. A bank can therefore attract savers with a higher reported return, as long as the risk behind that return does not show.

Smaller banks earn their higher returns by holding riskier debt (Section 4.1). Their products report about 0.5 percentage points a year more than larger banks' products with the same redemption terms and rating. To find where this extra return comes from, we price every holding by its credit spread over Treasury yields. The holdings of smaller banks' products carry a spread of 205 basis points, against 149 at national banks. Home LGFV bonds are the largest source of this gap. They make up 20% of the holdings of smaller banks' products, against 1% at national banks, and smaller banks' home LGFV bonds carry a spread of 317 basis points. Smaller banks therefore earn 61 basis points more of spread than national banks from home LGFV bonds. They also earn 14 basis points more from the capital bonds of other local banks and 14 more from corporate bonds of their home provinces. National banks earn more of their yield elsewhere, mainly 26 basis points more from private credit, such as trust loans, and from asset-management plans and funds. The extra spread on home LGFV bonds also showed up in the returns that products reported, as these bonds earned the largest return premium of any asset class we examine.

Smaller banks also hide the risk of this debt from their savers (Section 4.2). Their products hold more risk than larger banks' products with the same redemption terms and rating, and a bank half as large holds bonds with a risk premium 16 basis points higher. Smaller banks put this debt in products that restrict redemptions. The average smaller bank restricts withdrawals on 98% of its products, against 89% at larger banks, and such products need not sell their holdings when markets fall. The returns these products report smooth out

the swings in the value of their holdings. A product of a bank half as large reports 10% less volatility, although its holdings imply 5% more. Smaller banks also rate these products as safe, and this rating is what savers see when they buy. At the average smaller bank, 90% of fixed-income products carry a safe rating, against 78% at the average national bank. The 2022 sell-off shows how much stayed hidden. Measured against their usual monthly returns, smaller banks' products showed a third of the fall in the value of their holdings, while larger banks' products showed three-fifths. Smaller banks' closed-end products still reported positive returns in the sell-off months.

This was reaching for yield, not better local information (Section 4.3). Most of smaller banks' extra yield comes from borrowers in their home provinces, so it could reflect the better knowledge of local borrowers that the first view credits to small banks. However, a larger bank headquartered in the same city has at least the same access to them, because it has more branches there. We therefore compare banks from the same city. Larger and smaller banks from the same city hold similar shares of their portfolios in the city's LGFV bonds, but a bank half as large holds more bonds of the city's riskier county-level issuers. In other cities of its home province, it holds bonds with coupons 24 basis points higher. The coupon gap is larger still, at 53 basis points, in other provinces where neither bank is at home, and those issuers later had more credit events. Smaller banks thus take more risk for more yield even where they have no local advantage.

Debt resolution removed the main source of the yield on which smaller banks competed (Section 5.1). After the 2023 program, average LGFV issuance spreads over Treasury yields fell from about 200 basis points to 97. The extra return that products earned from home LGFV bonds fell by an even larger share. A product with 20 percentage points more of its holdings in home LGFV bonds earned about 21 basis points more a year before the program and only 5 more after it.

Only banks with enough savers could replace the lost yield (Section 5.2). Private credit to LGFVs kept more of its yield, but it is a negotiated loan, and only a bank with many

savers can cover the fixed cost of arranging and monitoring it. Larger local banks therefore replaced part of the lost yield with private credit, whereas the smallest third of local banks did not. The banks that had relied most on home LGFV bonds lost the most. A bank whose initial home LGFV bond share was ten percentage points higher lost about 12 basis points more of annualized return and raised 15.3% less through newly launched WMPs. Bank size predicts these losses on its own, but its coefficient is no longer statistically significant once we hold fixed a bank's reliance on home LGFV bonds and its use of private credit. Smaller banks thus lost because they had relied the most on that yield and could not replace it.

The program also absorbed the risk that smaller banks' products had hidden (Section 5.3), and the market's price of that risk fell sharply. Nearly all smaller banks run their WMPs without a wealth-management subsidiary, so we measure this price on the home LGFV holdings of local banks without one. On these holdings, provincial LGFV spreads fell from 297 basis points in June 2023 to 87 a year later. As the most reliant banks withdrew, national banks' products took over their borrowers, and these borrowers now pay lower coupons.

We explain these patterns with a model in which a larger bank with more reputation and a smaller bank with less compete for savings (Section 6). Savers choose a bank by its reputation and by the returns and volatility its products report, and they take these reports at face value. Cautious savers care more about reputation, so they favor the larger bank, and the savers whom the smaller bank can attract are less cautious. Each bank chooses how much risky debt to hold and whether to hide part of its risk from what its products report, and a bank that hides puts its reputation at stake if the hiding is exposed. Hiding pays only for a bank that holds enough risk to hide, and exposure costs the most at a bank whose savers are cautious. The smaller bank therefore reaches for yield and hides its risk, while the larger bank takes less risk and reports it in full. Both banks face the same investments, so the smaller bank needs no better information to do so. Debt resolution lowers the yield and with it the gain from hiding. When the lower yield ends the smaller bank's hiding, it cuts a larger share of its risky holdings than the larger bank and loses more of its return and its savers.

The broader lesson is that when banks have little reputation to lose, they compete through a fragile strategy that exploits the behavioral bias of households, who care about returns and observed volatility but not underlying risk. Such a strategy lasts only as long as the risk both exists and stays hidden. Reporting at market prices would expose the risk, and removing the risk would remove the yield, so either one ends it.

Our paper contributes to four strands of the literature. First, we contribute to work on competition between large and small banks. Small banks have an advantage in lending to opaque local firms (Petersen and Rajan 1994, Berger and Udell 2002, Stein 2002, Berger et al. 2005, Nguyen 2019), and this lending adds value to community banks (Holod et al. 2024). Large banks gain from economies of scale, deposit market power, and expected government support (Hughes and Mester 2013, Drechsler et al. 2017, O'Hara and Shaw 1990, Gandhi and Lustig 2015). Competition for deposits shapes banks' risk taking and fragility (Boyd and De Nicolò 2005, Martinez-Miera and Repullo 2010, Egan et al. 2017), and in 2023 uninsured depositors moved from regional banks to the largest banks (Cipriani et al. 2024, Caglio et al. 2024). Whether smaller banks reach for yield more is unsettled. Small US banks took more loan risk after monetary easing and lengthened their asset maturities more when rates were low (Buch et al. 2014, Wang 2017), and the response of bank risk taking to low rates depends on bank capital (Dell'Ariccia et al. 2017). In China, the entry of national banks into local markets intensified competition for borrowers and led incumbent banks to lower their screening standards (Gao et al. 2019), and small and medium-sized banks used wealth-management products to compete for deposits (Hachem and Song 2021, Acharya et al. 2025). We show that smaller banks compete for uninsured household savings through the returns they report. They reach for yield and hide its risk because they lack reputation, not because they know their borrowers better.

Second, we contribute to work on what retail investors respond to and how intermediaries cater to them. Fund flows chase past performance (Sirri and Tufano 1998) and simple, visible signals such as ratings and raw returns rather than risk-adjusted performance (Del Guercio

and Tkac 2008, Barber et al. 2016, Evans and Sun 2021, Ben-David et al. 2022). Investors demand claims that look safe (Gennaioli et al. 2013), rely on intermediaries they trust (Gennaioli et al. 2015), and buy retail products whose costs and risks are hard to see (Gabaix and Laibson 2006, Henderson and Pearson 2011, Célérier and Vallée 2017). The reported returns of illiquid portfolios are smooth (Getmansky et al. 2004, Cassar and Gerakos 2011), and flow sensitivity makes funds fragile when losses appear (Chen et al. 2010, Schmidt et al. 2016, Goldstein et al. 2017, Falato et al. 2021). We show that savers in a market of RMB 33 trillion respond strongly to reported returns and their fluctuations and punish even small dips below the usual return. In the 2022 sell-off they ran from the losses that products reported and not from the losses that the holdings took. Banks with little reputation exploited this bias by rating risky products as safe, locking savers in, and reporting returns that barely moved.

Third, we contribute to work on reaching for yield and on what restrains it. Competition (La Spada 2018, Choi and Kronlund 2018), regulatory constraints (Becker and Ivashina 2015), and low interest rates (Di Maggio and Kacperczyk 2017, Lian et al. 2019) encourage reaching for yield, and money market funds took more risk when investors rewarded higher yields (Kacperczyk and Schnabl 2013). Franchise value restrains banks' risk taking (Keeley 1990, Hellmann et al. 2000). We show that reputation determines which banks reach for yield in products sold to households, and that the strategy is fragile because it needs both the risky yield and the concealment of its risk. Proximity can also give lenders and investors an information advantage (Petersen and Rajan 2002, Degryse and Ongena 2005, Hauswald and Marquez 2006, Agarwal and Hauswald 2010, Coval and Moskowitz 2001, Kacperczyk et al. 2005). In our setting, larger banks from the same city have at least the same access to local borrowers, so smaller banks' riskier holdings reflect reaching for yield and not such an advantage.

Fourth, we connect China's LGFV debt and shadow banking to banks' competition for household savings, and study how government support for borrowers affects the intermediaries

that hold their debt. Existing work traces off-budget fiscal expansion through LGFVs (Bai et al. 2016), the refinancing of stimulus loans through bonds and shadow banking (Chen et al. 2020), and the crowding out of private investment (Huang et al. 2020). It also shows how deposit competition (Acharya et al. 2025), liquidity rules (Hachem and Song 2021), monetary tightening (Chen et al. 2018), competition with insured deposits (Xiao 2020), and implicit guarantees (Allen et al. 2023, Huang et al. 2023) shaped shadow banking in China. Governments influence credit allocation (Sapienza 2004, Khwaja and Mian 2005) and banks' holdings of government debt (Becker and Ivashina 2018, Ongena et al. 2019). In China, creditor political power and expected government support shape defaults and credit pricing (Gao et al. 2021, Geng and Pan 2024). Geng et al. (2025) show that debt resolution narrows local banks' subordinated-bond spreads while reducing profitability at banks in riskier cities. We show that LGFV debt from banks' home provinces was the largest single source of the extra yield with which smaller banks competed for household savings. We also show that debt resolution absorbed the risk hidden in their products, lowered that yield, and moved both savings and borrowers toward larger banks.

2 Institutional Background and Data

2.1 Wealth-management products

A wealth-management product pools savers' money in a separately accounted portfolio run by a bank or its wealth-management subsidiary. WMPs are a major alternative to bank deposits for Chinese households and a central part of the country's shadow banking system (Acharya et al. 2025, Chen et al. 2020). Since the 2018 asset-management reform, products must report net asset values (NAVs) and cannot guarantee principal or returns.[2]

Savers see a product mainly through what its bank reports. Each product advertises a performance benchmark, which is the return it aims to deliver. It also publishes its NAV,

[2]See the asset-management reform guidelines (People's Bank of China et al. 2018).

which shows past returns and how much they fluctuate. The issuing bank rates each product's risk on a five-level scale from 1 for the safest to 5 for the riskiest. We call ratings 1 and 2 safe ratings. The bank sets this rating itself under its own method. The rules name qualitative criteria such as the portfolio, the past performance of similar products, and the risk level, and they require neither an external rater nor a quantitative standard (China Banking and Insurance Regulatory Commission 2018b, Article 27). Savers see the rating in the risk disclosure statement of the sales documents when they buy (China Banking and Insurance Regulatory Commission 2021, Article 49). The rating also decides who may buy. Banks rate each saver's risk tolerance on at least five levels and may sell a product only to savers whose tolerance is at or above its rating (China Banking and Insurance Regulatory Commission 2018b, Articles 28–29), so a safe rating opens a product to more cautious savers. The rules require a second, independent rating only when a product is sold by an institution other than its issuer, and the higher of the two ratings is then disclosed (China Banking and Insurance Regulatory Commission 2021, Article 30). Otherwise the issuing bank's own rating is the only one, and in practice its scale is coarse. In April 2026, 26,709 of the roughly 33,900 rated products in the national registry were rated 2, and products with very different portfolios shared a grade (Li 2026).

What savers do not see is the risk of the holdings. When they buy, the sales documents give a product's investment scope, risk rating, and performance benchmark, but not its holdings. The holdings appear only in quarterly and annual reports published after the sale, and these reports list only the largest positions by name. Ordinary savers seldom read these reports. To learn from them how much of a portfolio depends on one local government's credit, a saver would have to identify each issuer, find its location, and judge its credit risk. Most savers lack the financial knowledge to do so.

Redemption terms vary. Daily-open products allow withdrawals on any business day, periodic-open products only on set dates, and closed-end products only at maturity. Periodic-open and closed-end products thus restrict redemptions. Most of the market consists of

fixed-income products, which regulation requires to keep at least 80% of their assets in debt claims.

National banks are the large state-owned banks and the nationwide joint-stock banks. They have extensive branch networks and nationwide name recognition. Local banks are the provincial and city commercial banks and the rural banks. They compete for savings with national banks and with each other. Regulation also ties local banks to their home markets. Rural commercial banks, for example, must make loans to small firms and farmers their core business and keep their branches and lending within their home county (China Banking and Insurance Regulatory Commission 2019b). Supervisors tolerate higher loss rates on small-business loans, up to 3 percentage points above a bank's overall nonperforming-loan ratio (China Banking and Insurance Regulatory Commission 2019a). Most national banks and the largest local banks run their WMPs through wealth-management subsidiaries. The other local banks are mostly smaller and run their WMPs directly.

WMPs invest mainly in fixed-income and similar assets such as government and policy-bank bonds, deposits, negotiable certificates of deposit, and corporate debt (Figure A1).[3] Most readily accessible investments offer limited credit spreads, and the public high-yield corporate bond market is narrow (Schipke et al. 2019).

LGFV bonds are one source of higher yield. LGFVs are state-owned companies that finance local infrastructure and public services. They borrow in their own names and outside formal government debt. Repayment depends on project assets, local fiscal resources, and expected government support (Hendy et al. 2024, Yang and Jiang 2021). In the issue-level sample of Table A6, average pre-program fixed-rate issuance yields were 4.26% for LGFVs, 3.03% for other nonfinancial issuers, and 2.77% for financial issuers. These differences persist after controlling for issue characteristics and ratings. Because these bonds trade nationally, a bank can hold LGFV bonds from any province. We estimate that WMPs held about RMB

[3]The commercial-bank WMP rules list the permitted investments (China Banking and Insurance Regulatory Commission 2018b, Article 35).

3.1 trillion in LGFV bonds at year-end 2024. This amount is 10.2% of total WMP net assets and 28.1% of outstanding LGFV bonds (Figure A2 and Internet Appendix D.2).

WMPs can also finance the same borrowers through private credit (PC), which includes identified trust loans, leasing receivables, and claims on identified borrowers held through trust or asset-management plans. Unlike a marketable bond, private credit must be arranged, documented, and monitored without a continuously observable trading price, and this work is costly even for a small investment.

Regulatory caps on private credit were far from binding for local banks without a subsidiary. Chinese rules treat private credit as non-standard debt. A bank's private credit across all its WMPs may not exceed 35% of their total net assets, and for WMPs that a bank runs itself it may also not exceed 4% of the bank's total assets (China Banking and Insurance Regulatory Commission 2018b,a). At the median bank without a subsidiary in June 2023, the second cap allowed private credit of up to 61% of the bank's WMP balance. Private credit made up about 3% of the holdings these banks disclosed, far below either limit, so the caps cannot explain why they made little use of it (Section 5.2). What these banks faced instead was supervisory guidance on the size of their WMP business. In September 2021, the financial press reported that banks without a subsidiary would not be allowed to expand this business and would wind down existing balances (Fang 2021). In June 2024, regulators in several provinces asked these banks to shrink it, in some cases to zero by the end of 2026 (Li 2024, Li and Zhang 2024). Because this guidance could also lower their fundraising, we confirm that a bank's initial home LGFV bond share predicts its later fundraising in a sample that ends before the June 2024 guidance (Section 5.2).

2.2 Debt resolution and data

The policy announcement of July 24, 2023, called for comprehensive local-debt resolution. Provinces began issuing special refinancing bonds in October and raised about RMB 1.4 trillion in 2023Q4 (Xinhua News Agency 2023, Wang 2023, Wong 2024), and an expansion

announced in November 2024 raised the debt-swap limit by RMB 6 trillion for 2024–2026 (Ministry of Finance 2024). The proceeds repaid selected obligations within a broader program that also restructured loans and restricted new borrowing. The refinancing bonds are provincial-government obligations, not the LGFV bonds that WMPs hold. The program reached WMPs instead through expected government support, which it strengthened without formally guaranteeing every LGFV claim. Stronger support lowers LGFV spreads. A fall in spreads raises the prices of bonds already held, and it lowers the yields available when banks reinvest. We date implementation to 2023Q4 in quarterly tests and to October 2023 in monthly tests.

Our data start with product-level WMP holdings from a private data vendor. The holdings identify each product's largest disclosed positions at the individual-asset level. Quarterly disclosures over 2022Q1–2025Q3 contain 2,279,252 holding-level records for 60,396 products and 253,484 product-quarters. The 171 banks are headquartered in 30 provinces and include state-owned and joint-stock national banks, provincial and city commercial banks, and rural banks.

We link the holdings to several other sources. Another private data vendor provides product NAVs and bond issuance terms, including issuer ratings and administrative levels. It also provides the financing records in which LGFVs disclose their private credit and its contract rates. Product filings give each product's type and redemption terms. Company registration records locate the borrowers of private credit, and an LGFV roster identifies LGFV issuers. We also observe issuers' defaults, rating warnings, and guarantor events. For banks, we use their financial statements, the locations of their branches by city from a bank-branch registry, and their total WMP balances, which combine institution-level statistics with the banks' own disclosures. Market prices come from Treasury yields by maturity, the China Securities Index credit-bond indices and valuation yields, and provincial LGFV spreads. Provincial special-refinancing bond issuance, provincial GDP and LGFV debt, and city statistics on GDP, growth, LGFV debt, and urban income describe the regions. Industry

totals come from the annual WMP market reports and the Asset Management Association of China. Tables A1–A3 define the variables and report sample coverage and descriptive statistics.

We measure a bank's size by its interest-bearing assets at the end of 2021, and Figure 2 shows that size is highly skewed. The median national bank had RMB 7.8 trillion of these assets, against RMB 0.19 trillion for the median local bank. National banks hold 76% of sample WMP assets. Most banks are thus small local banks, while most savings sit with the few large national banks.

A bank's holdings and product returns are those of its WMPs, whether run directly or through a subsidiary. An investment is home when its borrower is in the bank's headquarters province, so national and local banks' WMPs can both hold home and non-home assets. In the product regressions, i denotes a product, a its issuing institution (the bank or its subsidiary), and b the parent bank. We index quarters by q and months by m, and p_b is the bank's headquarters province. The parent bank is the issuing bank itself when the bank runs its products directly. Log bank size$_b$ always refers to fixed FY2021 bank size, whereas log bank assets in the control rows vary over time. Because bank size is so skewed (Figure 2), we compare banks by size in three ways, by log size across all banks, by a split at the median size, and by size terciles among local banks. We take the median over the banks in the product sample of Section 4.2, where it is RMB 0.44 trillion of FY2021 interest-bearing assets. We call the banks at or below it smaller banks, and all of them are local banks. Local banks without a wealth-management subsidiary include nearly all smaller banks.

Figure 1 previews the patterns that the rest of the paper examines. It splits the banks above the median size into national banks and larger local banks, and it compares each group with the smaller local banks. Before the 2023 program, smaller local banks' WMPs held the bonds with the highest spreads. From 2022Q1 to 2023Q3, the bonds they held had been issued at an average of 224 basis points over Treasury yields, against 183 at larger local banks and 126 at national banks (Panel A). After the program these spreads fell for

all three groups, to 131, 115, and 91 basis points by 2025Q3. When each holding is marked to market prices, the holdings of smaller local banks implied the most return volatility, yet their products reported the least. From April 2022 to June 2023, their holdings implied a standard deviation of monthly returns of 46 basis points, against 44 at larger local banks and 35 at national banks. The products themselves reported 24 basis points, against 32 and 34 (Panel B). National banks' products thus reported nearly all of the volatility their holdings implied, whereas smaller local banks' products reported about half. Smaller local banks also held the most home LGFV bonds. These bonds made up 31% of the bonds and private credit held by their products, against 19% at larger local banks and almost none at national banks, which held 24% in private credit (Panel C). Finally, nearly all of smaller local banks' products restricted redemptions and carried a safe rating. At the average smaller local bank, 98% of fixed-income products restricted redemptions and 90% carried a safe rating, against 86% and 78% at the average national bank (Panel D).

3 What Savers See

Savers chase the returns that products report, and they penalize reported volatility and any dip below the usual return. They do not respond to the risk behind these returns, which is largely hidden from them. When savers buy a product, they see its rating and its past returns. Its holdings appear only in quarterly and annual reports that ordinary savers seldom read (Section 2). Even savers who read these reports often lack the financial knowledge to judge the risk of the underlying investments. This reliance on what products report is the bias that smaller banks went on to exploit. The bond sell-off of 2022 provides the sharpest test. In it, savers ran from the losses that products reported, not from the losses their holdings took.

We rank products' prior-quarter returns within the same risk rating and redemption type and pool products of local and national banks. Following the performance–flow approach of

Sirri and Tufano (1998) and La Spada (2018), we estimate

$$\begin{aligned}\text{Net flow}_{iq} = &\sum_{k=2}^{5} \beta_k \mathbb{1}\{\text{Return quintile}_{i,q-1} = k\} \\ &+ \gamma' \,\text{Controls}_{i,q-1} + \lambda\, \mathbb{1}\{\text{Local bank}\}_b + \alpha_{g(i,q-1),q} + \varepsilon_{iq}.\end{aligned} \tag{1}$$

The outcome $\text{Net flow}_{iq} = 100\,[A_{iq} - A_{i,q-1}(1 + r_{iq})]/A_{i,q-1}$ is asset growth net of investment gains, as a percentage of beginning assets. A product's size A_{iq} is the median over its disclosed positions of the position's value divided by its portfolio weight, and r_{iq} is its quarterly return compounded from monthly changes in log distribution-adjusted NAV. Net flows are capped at the 1st and 99th percentiles, and a product that matures or stops disclosing holdings leaves the sample rather than recording zero assets. $\text{Return quintile}_{i,q-1}$ is the prior return quintile, and the bottom quintile is omitted. The index $g(i, q-1)$ identifies the product's category, which is its risk rating and redemption type in the prior quarter. Category–quarter effects hold common demand conditions fixed.[4]

Flows rise across the performance ranks (Table A5). Net inflows into top-quintile products exceed those into bottom-quintile products by 12.0% of beginning assets, and top-quintile products are 8.0 percentage points more likely to receive positive flows. Among products of banks in the same home province and quarter, the flow difference is a similar 12.6%.

To separate what savers can see from what they cannot, we add two variables to equation (1). The first measures the risk in the product's disclosed portfolio at the end of the prior quarter. It is the home LGFV credit share, which is home-province LGFV bonds plus private credit to home-province LGFVs as a percentage of disclosed holdings, and it is zero for products that hold none. The second is the volatility of the product's reported monthly

[4]The regression includes the local-bank indicator and eleven controls. The controls are three product characteristics (size, risk rating, and redemption terms), four bank characteristics (log bank assets, nonperforming-loan ratio, cost–income ratio, and net interest margin), and four home-city characteristics (GDP, GDP growth, LGFV debt/GDP, and urban income). The sample is 2022Q3–2025Q3, and errors cluster by parent bank.

returns over the preceding year, which savers can see. Each variable enters per standard deviation.

Flows respond to reported volatility but not to the risk held (Table 1). Adding the home LGFV credit share leaves the top-to-bottom quintile flow difference at 12.1% of beginning assets (column 2). One standard deviation of this share is 16.7 percentage points of disclosed holdings. A share that is higher by this amount is associated with flows 0.54% of beginning assets lower, which is not statistically significant, and the 95% confidence interval excludes reductions larger than 1.2%. By contrast, a one-standard-deviation higher volatility of past reported returns lowers flows by 2.3% of beginning assets, while the coefficient on the home LGFV credit share is still not statistically significant (column 3).

Savers punish any dip below the usual return, even when the return stays positive. We group daily-open products by the return they reported for the quarter. We then compare their flows with those of products of the same bank in the same quarter that reported the usual 0.5 to 1%, using the controls and category effects of Table 1 (Table 2, Panel A). Daily-open products that reported 0.25 to 0.5% had net flows 43 percentage points of beginning assets lower in the same quarter, and those that reported smaller gains or losses had net flows 46 to 52 points lower. The pattern also holds for flows in the quarter after the return was reported, so the outflows did not cause the low returns. In that quarter, products that had reported 0.25 to 0.5% had net flows 19 points lower, and those that had reported losses had net flows 28 to 34 points lower. The pattern also holds within the same product. When product fixed effects compare each product with itself in other quarters, daily-open products that reported 0.25 to 0.5% had net flows 30 points lower in the same quarter and 16 points lower in the next.

The response comes from savers who can withdraw. In Panel B of Table 2, the slope of net flows on the reported quarterly return differs by redemption type. For daily-open products, each percentage point of reported return comes with net flows 53 percentage points of beginning assets higher in the same quarter and 58 points higher in the next. With product

fixed effects the two slopes are 21 and 27 points. Savers can redeem periodic-open products only on set dates, and for these products the slopes are at most 8 points. Savers cannot redeem closed-end products before maturity, and for these products the slopes are small and not statistically significant in any column. Net flows subtract the reported return from asset growth, so an error in a reported return would link the two mechanically for every redemption type. The flat slopes of closed-end products show that such errors do not drive our estimates. These results imply that a product whose reported returns showed its holdings' losses would have lost savers.

The bond sell-off of late 2022 tests this implication directly. The losses came from a market-wide fall in bond prices that no single bank caused, and the same market losses appeared to very different degrees in the returns that products reported (Section 4.2). Because WMPs report net asset values (Section 2), a fall in bond prices can show up in the returns that savers see. Bond prices fell sharply, and provincial LGFV spreads rose by about 110 basis points. Between the end of October and mid-December 2022, the one-year Treasury yield rose by 60 basis points and the three-year yield by 36. Yet the losses that products reported were small. In early December 2022, 4,932 of 28,038 WMPs had net asset values below par, and more than 90% of these were below par by less than 2% (Wang 2022). Even these small reported losses were followed by large withdrawals. China International Capital Corporation (CICC) estimates seasonally adjusted net redemptions of about RMB 720 billion in November 2022 alone and reports that the most risk-averse savers led them (Wang et al. 2022). Redemptions forced funds and WMPs to sell a record RMB 1.3 trillion of interbank bonds that month (Bloomberg News 2022). According to industry estimates, WMP balances stood at RMB 31.3 trillion in October 2022 before the redemptions began and at 25.0 trillion at the end of March 2023 (Liu and Xie 2024). The official market report puts them at 27.65 trillion at the end of 2022, between these two dates (Banking Wealth Management Registration and Depository Center 2023). Our data show the same outflow. Products that report in consecutive quarters lost 12.6% of their assets in 2022Q4 and another 14.7% in

2023Q1. Daily-open products took in 6.2% of their assets in 2022Q3 and then lost 10.9% and 15.9% to net withdrawals in the two following quarters.[5]

Savers ran from the losses that products reported, not from the losses their holdings took. In 2022Q4, daily-open products that reported a quarterly loss lost 25 to 37% of their assets, while those that reported the usual 0.5 to 1% took in 9% (Table D4). To measure the losses that holdings took, we mark every disclosed asset each month to the market series that fits it. For an LGFV bond, this series is the Treasury yield of its maturity plus the LGFV spread of the issuer's province. We call the resulting monthly returns implied returns, and Section 4.2 gives the construction. We relate each product's net flow in 2023Q1 to its reported and implied returns in November and December 2022. Products that had reported larger losses lost more of their assets than other products of the same bank, whereas the losses in their implied returns made no statistically significant difference (Table D5). Each 10 basis points a month of reported loss came with net outflows larger by 1.2% of beginning assets, and by 8% among daily-open products.

Savers' sensitivity to what they see gave banks a strong incentive to offer products whose reported returns are high and fluctuate little. A product that reports a high return attracts savings, and one that shows even a small dip loses them, whatever it holds. Banks with little reputation could compete for savers only through what their products reported. They needed a higher return than more reputable banks, and they needed its risk not to show.

4 Reaching for Yield in the Dark

Savers reward the returns and stability that products report and do not see the risk behind them (Section 3). A bank with little reputation can therefore attract savings only with a higher reported return, and it keeps them only if the risk behind that return does not show.

[5]Our sample covers most of the market. It holds RMB 21.7 of 27.65 trillion at the end of 2022 and 19.7 of 25.0 trillion at the end of March 2023. The 12.6% decline in 2022Q4 is close to the fall of about 12% in industry balances from October to December 2022, and total sample assets fell 9.0% in 2023Q1, against about 9.6% in the market figures above (27.65 to 25.0 trillion).

This section shows that smaller banks did both. Section 4.1 shows that they earned their higher returns by holding riskier debt than larger banks, and that the largest source of the extra yield was the debt of LGFVs in their home provinces. Section 4.2 shows that the risk stayed in the dark, because smaller banks put this debt in products that restrict redemptions, report returns that barely move, and carry the banks' own safe ratings. Section 4.3 shows that this was reaching for yield rather than better local information.

4.1 Higher returns from local credit

Smaller banks' products reported about 0.5 percentage points a year more than larger banks' products with the same redemption type and risk rating (Section 4.2). They earned these higher returns by holding riskier debt throughout their portfolios, and the largest source of the extra yield was the debt of LGFVs in their home provinces. Their WMPs held more LGFV bonds and less private credit than larger banks' WMPs, concentrated these bonds at home, and held them in products that restrict withdrawals. Home LGFV bonds carried the largest return premium of any asset class we examine.

Where the yield comes from. Smaller banks hold riskier debt throughout their portfolios, but most of their extra yield comes from credit in their home provinces. To locate the extra yield, we price every disclosed holding by its credit spread over Treasury yields. A bond carries its issue spread over the Treasury yield of the same maturity. Private credit carries its contract rate over the Treasury yield of its remaining maturity. Asset-management plans and funds carry the mean issue spread of AA+ or AAA bonds with one to three years to maturity, and cash, deposits, and repurchase agreements carry zero. A product's all-in spread is the value-weighted mean over its holdings. It is therefore the sum over asset classes of each class's holding share times its spread. The size gradient of the all-in spread thus splits exactly into the contributions of the classes (Table 3, Panel B). Section 4.2 uses the same all-in spread to measure the risk a product holds.

Per log point of bank size, the all-in spread falls by 12.8 basis points (Table 3, Panel B, column 12). Home LGFV bonds drive this gradient. Their contribution to the all-in spread falls by 17.0 basis points per log point of size. Two other kinds of credit work in the same direction. The contribution of other local banks' capital bonds falls by 4.3 basis points, and that of bonds of non-LGFV corporate issuers from the home province falls by 4.6. Larger banks earn more of their yield elsewhere. The contribution of private credit rises by 8.2 basis points per log point of size, and that of asset-management plans and funds by 5.2 basis points, which is not statistically significant. These offsets make the all-in gradient smaller than the home LGFV component alone. The same pattern appears in levels. The holdings of smaller local banks' products carry an all-in spread of 205 basis points, against 149 at national banks. Smaller local banks earn 61 basis points more of spread than national banks from home LGFV bonds. They also earn 14 basis points more from other local banks' capital bonds and 14 more from home corporate bonds. National banks earn 26 basis points more from private credit and from asset-management plans and funds, and 6 more from the remaining classes (Table 3, Panel A). Smaller banks also hold higher-spread claims within most asset classes. Their home LGFV bonds carry 317 basis points against 236 at national banks, their home corporate bonds 270 against 121, and their private credit 406 against 351. Home LGFV bonds are thus the largest single source of smaller banks' extra yield, and the rest of this subsection examines them.

Asset allocation. Two higher-yielding investments are central to the analysis, marketable LGFV bonds and private credit. Before the program, local banks' WMPs relied more on LGFV bonds, whereas national banks' WMPs already made substantial use of private credit. We express each holding as a share of included debt holdings. This is the combined value of identified bonds and private credit with known borrower locations, and it excludes central-government, central-bank, and policy-bank obligations. This denominator isolates the

credit whose borrowers and their locations we observe.[6] LGFV bonds accounted for 43.1% of included debt holdings at local banks and 12.6% at national banks, while private credit accounted for 9.3% and 24.1% (Table D1). We then estimate the initial size gradient across national and local banks

$$\text{Asset share}_{iq} = \beta_S \,\text{Log bank size}_b + \gamma' \,\text{Controls}_{iq} + \alpha_q + u_{iq}. \tag{2}$$

The outcome is the total LGFV bond share or the total private credit share over 2022Q1–2023Q3, in percentage points of included debt holdings. Log bank size$_b$ is log FY2021 parent-bank interest-bearing assets in RMB trillion, which the products of a bank and its subsidiary share. The ten variables in Controls$_{iq}$ are three product characteristics (size, risk rating from 1 to 5, and redemption terms), three bank characteristics (nonperforming-loan ratio, cost–income ratio, and net interest margin, in percent), and four home-city characteristics (GDP, GDP growth, LGFV debt/GDP, and urban income).[7]

Smaller banks' WMPs allocate more to LGFV bonds and less to private credit (Table 4). A bank half as large has an estimated LGFV bond share about 4.9 percentage points higher and a private credit share 4.7 points lower. Private credit is also scarce at the smallest third of local banks, where it is only 1.1% of included debt holdings, against 11.4% at the largest third (Table D1, Panel B). Smaller banks thus entered the program more dependent on marketable LGFV bonds and with little use of the privately arranged alternative.

Home bias. Local banks also concentrate their investment in their home provinces. Home LGFV bonds account for more than one-fifth of included debt holdings in every local-bank size group (Table D1, Panel B). For the home LGFV bond share, equation (2) gives a size coefficient of −5.46 percentage points per log point of bank size (Table 4, column 3), so a

[6] Internet Appendix D.2 reconciles the samples and denominators, including bond-only shares and shares of all disclosed assets.

[7] Regressions weight by included debt value and cluster by parent bank.

bank half as large has a home LGFV bond share about 3.8 points higher. Most of smaller banks' additional LGFV allocation is at home.

Large home holdings could reflect a large local bond market rather than a preference for nearby borrowers, so we also measure geographic concentration within each asset class. For LGFV bonds, the home bias within the asset class is the product's home LGFV holdings divided by all its LGFV holdings, minus the fraction of nationwide disclosed LGFV holdings issued by borrowers in that province. We define it in the same way for other bonds and for private credit. Over 2022Q1–2025Q3, we estimate

$$\begin{aligned}\text{Home bias within class}_{iq}^{n} = \beta_n \, \mathbb{1}\{\text{Local bank}\}_b \\ + \gamma_n' \, \text{Controls}_{iq} + \alpha_q + \varepsilon_{iq}.\end{aligned} \tag{3}$$

Here n identifies the asset class, and β_n measures how much larger the home bias within class n is at local banks than at national banks.[8]

Local banks' WMPs invest disproportionately at home on this measure as well (Table 5). Their home bias within each asset class exceeds that of national-bank products by 32.0 percentage points for LGFV bonds, 16.1 points for other bonds, and 48.9 points for private credit. Their specialization is geographic as well as a choice of asset class.

In addition, local banks place more of their WMP assets in products that restrict withdrawals. Periodic-open and closed-end products hold 80.2% of local banks' WMP assets, compared with 47.3% at national banks (Table A4). Such products give banks more room to hold investments that are difficult to sell quickly. To connect withdrawal terms to investment, we divide a product's home holdings in an asset class by all its included debt holdings and subtract the corresponding nationwide share. This portfolio home overweight reflects both

[8] Internet Appendix D.4 lists the controls and gives the details of this specification and of equation (4).

the amount invested in the asset class and its geographic concentration. We estimate

$$\begin{aligned}\text{Portfolio home overweight}_{iq}^{n} = \beta_1 \, \mathbb{1}\{\text{Local bank}\}_b \\ + \beta_2 \, \mathbb{1}\{\text{Restricted redemption}\}_i \\ + \beta_3 \, \mathbb{1}\{\text{Local bank}\}_b \times \mathbb{1}\{\text{Restricted redemption}\}_i \\ + \gamma' \, \text{Controls}_{iq} + \alpha_q + \varepsilon_{iq}.\end{aligned} \tag{4}$$

$\mathbb{1}\{\text{Restricted redemption}\}_i$ identifies products without daily withdrawals. The coefficient β_1 measures the local–national allocation gap among daily-open products, and β_3 measures how much wider that gap is among restricted-redemption products.

The gap between local and national banks' home investment lies in products that restrict withdrawals (Table A4). Among daily-open products, the local-bank coefficient is not statistically significant, at −6.3 percentage points (standard error 6.0) for home investment pooled across asset classes and 0.9 points (standard error 3.5) for LGFV bonds. The interaction with restricted redemption is 20.0 percentage points for pooled home investment, of which 7.7 points are in LGFV bonds and 7.8 points in private credit.

Returns. Before the program, home LGFV bonds earned the largest return premium of the asset classes we examine. To separate home LGFV bonds from bonds elsewhere and from private credit, we relate monthly product returns to holdings at the preceding quarter end $h(m)$ in

$$\begin{aligned}\text{Monthly return}_{im} = \beta' \, \text{Asset shares}_{i,h(m)} \\ + \delta' \, \text{Asset shares}_{i,h(m)} \times \mathbb{1}\{\text{Post}\}_m \\ + \gamma' \, \text{Controls}_{i,h(m)} + \lambda \, \mathbb{1}\{\text{Local bank}\}_b + \alpha_m + \varepsilon_{im}.\end{aligned} \tag{5}$$

The outcome Monthly return$_{im}$ is the change in log distribution-adjusted NAV, in basis points. The five fractions in Asset shares$_{i,h(m)}$ are the shares of included debt holdings in

home LGFV bonds, home private credit, home other bonds, non-home LGFV bonds, and non-home private credit. Non-home other bonds are the omitted class. Controls are measured at the same date, and Post starts in October 2023.[9] Before implementation, β measures each asset's return premium relative to non-home other bonds. The interactions δ allow the premiums to change afterward, which Section 5.1 examines.

Reallocating 20 percentage points from non-home other bonds to home LGFV bonds is associated with 20.5 annual basis points more product return, compared with 7.7 basis points for LGFV bonds from other provinces (Table 6, column 2). The home premium is more than two and a half times the non-home premium, and their difference is statistically significant ($p = 0.003$). Private credit also earns more at home. The same 20-point allocation is associated with 17.6 annual basis points for home private credit, against 8.2 for private credit elsewhere. The premium on home other bonds is not statistically significant. This pattern ties the higher returns to LGFV bonds and private credit rather than to home investment generally.

Home LGFV bonds make up a large part of local banks' included debt holdings, with an initial mean allocation of 26% and a 90th percentile of 72% (Table A3). With the largest return premium and this weight in portfolios, home LGFV bonds were the largest single source of the extra yield with which smaller banks competed, and they tied their products to the borrowers of a single province. Section 4.2 asks whether savers could see that risk.

4.2 Hidden risk

Smaller banks earned their higher returns by holding riskier debt, and the largest source of their extra yield was the debt of LGFVs in their home provinces (Section 4.1). They also hid the risk of this debt from their savers. Their products held more risk than larger banks' products with the same redemption type and risk rating. Yet these products locked savers in

[9]The eleven controls add time-varying log bank assets to the ten controls in equation (2). The regression includes a local-bank indicator and month effects, and errors cluster by parent bank. It uses returns from April 2022 to December 2025, winsorized at the 1st and 99th percentiles of all monthly returns.

with restricted redemptions, reported returns that barely moved, and more often carried a safe rating from the bank itself. In the bond sell-off of November and December 2022, their reported returns only dipped to about zero while market yields and spreads implied a loss of about 1.6% on their holdings.

A saver sees a product's label and the returns it reports. The label is the product's redemption type and its risk rating. We therefore compare products with the same label and ask how what they hold and what they report vary with bank size. The sample comprises products that the filings classify as fixed-income and that have disclosed holdings and at least nine monthly returns between April 2022 and June 2023. For product i of bank b, we estimate

$$\text{Outcome}_i = \beta \,\text{Log bank size}_b + \alpha_{t(i),r(i)} + \varepsilon_i. \tag{6}$$

Redemption-type-by-risk-rating effects $\alpha_{t(i),r(i)}$ compare products with the same redemption type (closed-end, periodic-open, or daily-open) and risk rating, which a saver cannot tell apart by label. The sample includes national and local banks, and a negative β means that the outcome is larger at smaller banks. We also compare smaller with larger banks (Table D3). Smaller banks are those at or below the median size of this sample, and every national bank is among the larger banks. Each bank receives equal total weight, because the bank designs and rates its products.[10]

We first show that smaller banks' products hold more risk and then show how smaller banks hide it. We measure the risk a product holds in four ways, and each uses its disclosed holdings rather than what the bank reports. The first is the home LGFV credit share used in Section 3, which is LGFV bonds and private credit to LGFVs of the bank's home province as a share of disclosed holdings. The second is the issue spread of the bonds held. A bond's issue spread is its coupon minus the same-maturity Treasury yield at issue, and it is the compensation the market required for the bond's risk. The third is the all-in spread of

[10]Errors cluster by parent bank. Internet Appendix D.3 gives the sample construction, and Table D2 reports alternative specifications.

Section 4.1, which extends this compensation to the whole portfolio. The fourth is the volatility that the holdings imply. To compute it, we mark every disclosed asset each month to the market series that fits it. An LGFV bond moves with the Treasury yield of its maturity plus the LGFV spread of the issuer's province. Government and policy-bank bonds move with the Treasury yield of their maturity, and other bonds move with a credit-bond index of the same rating and maturity. Private credit to LGFVs moves like an LGFV bond of the borrower's province, and other private credit moves with a credit-bond index at its remaining maturity. Asset-management plans and funds move with a credit-bond index, and cash, deposits, and repurchase agreements do not move. Price changes omit the interest income that reported returns contain. We therefore add to each product's price changes the average difference between its reported returns and its price changes outside November and December 2022. We call the resulting monthly returns implied returns and take their standard deviation.

All four measures are higher at smaller banks (Table 7, Panel A, and Figure 3, Panel A). A bank half as large holds 3.7 percentage points more home LGFV credit, and the bonds it holds were issued at spreads 16 basis points higher. Across its whole portfolio it earns a spread 9 basis points higher, and its holdings imply 5% more return volatility. Compared with larger banks' products, smaller banks' products hold 15.5 percentage points more home LGFV credit (Table D3, Panel A). Their bonds were issued at spreads 50 basis points higher, their all-in spread is 33 basis points higher, and their holdings imply 17% more volatility. These gradients keep their sign and significance when we add a local-bank indicator and compare banks only with others of the same type (Table D2, column 2).

Issuers' official credit ratings do not show this risk. Of the rated LGFV bonds issued between 2021 and mid-2023, 62% carry issuer ratings of AA+ or AAA, and the rest carry AA. Yet AAA-rated LGFV bonds were issued at an average spread of 170 basis points, against 75 for other AAA-rated issuers (Figure 4). Smaller banks hold bonds priced as riskier than their ratings indicate, as Becker and Ivashina (2015) find for the bonds that insurers choose within a rating. Among bonds with the same issue quarter, issuer rating, administrative level,

maturity, and placement type, those held by a bank half as large were issued at spreads 12 basis points higher (Table 7, Panel A, column 3).

Smaller banks put this risk in products that lock savers in. Such products need not sell their holdings when markets fall. In the sample of Table 7, the average smaller bank restricts withdrawals on 98% of its products, against 89% at the average larger bank. The local–national gap in home LGFV bonds is 7.7 percentage points wider in restricted products than in daily-open ones (Section 4.1).

The returns that smaller banks' products report smooth out the swings in the value of their holdings. The products of a bank half as large report 10% lower return volatility (Table 7, Panel B, column 1), and smaller banks' products report 43% less volatility than larger banks' products with the same label (Table D3, Panel B, column 1, and Figure 3, Panel B). Relative to the volatility their own holdings imply, the volatility they report is 15% lower at a bank half as large and 51% lower at smaller banks than at larger banks (column 2 of Panel B of Table 7 and Table D3).

The bond sell-off of November and December 2022 shows how much stayed hidden. Smaller banks' products showed a third of the fall in the value of their holdings, and larger banks' products showed three-fifths. Smaller banks' holdings implied returns of −80 and −76 basis points in the two months, while their products reported −3 and −4 basis points (Figure 5). By construction, the two series have the same average in the other months, about 35 basis points. Their reported returns therefore fell 39 basis points a month below this usual level, against 114 for the returns implied by their holdings (Table 8, Panel A). For 2022Q4 as a whole, these products still reported a small gain. Larger banks' holdings implied −72 and −57 basis points, and their products reported −27 and −21. The difference between the two groups holds among products with the same redemption type and risk rating. Among such products, the part of the fall that reported returns did not show was 33 basis points a month larger at smaller banks than at larger banks, and 8.1 basis points larger at a bank half as large (column 4 of Panel B of Table D3 and Table 7). The result holds for the holdings

that products disclosed before the sell-off. When we mark only those of 2022Q2 and 2022Q3, the difference at a bank half as large is 8.9 basis points (Table D2, column 8). Panel C of Figure 3 shows the same gap bank by bank. Smaller banks' products showed less of the losses, and they also had smaller outflows. In the market-wide withdrawals of 2022Q4 and 2023Q1, the average smaller bank's products had net outflows of 9% and 8% of beginning assets, against 14% and 19% at the average larger bank (Figure 5, Panels C and D). This matches what we find within each bank, where the products that reported larger losses lost more in the following quarter (Section 3).

Products that lock savers in showed the least of the fall (Table 8). The part of the fall that reported returns did not show was 34 basis points a month larger in closed-end products and 12 larger in periodic-open products than in daily-open products of the same bank (Panel B). Smaller banks' closed-end products showed a quarter of the fall in their holdings' value and still reported positive returns in the sell-off months. The closed-end products of larger local banks and of national banks showed about half of the fall (Panel A). Within each redemption type, smaller banks' products showed less of the fall than the products of larger local banks, although the holdings of the two groups fell about as much. The difference therefore follows bank size rather than the line between local and national banks.

Smaller banks also rate more of their products as safe, although they hold riskier portfolios (Figure 3, Panels A and D). The issuing bank assigns the rating, and a safe rating is what lets a bank sell a product to its most cautious savers (Section 2). Among all fixed-income products that reported returns in this period and have the same redemption type, a bank half as large gives safe ratings to 3.0 percentage points more of its products (Table 7, Panel B, column 6). At the average smaller bank, 90% of fixed-income products carry a safe rating, against 78% at the average national bank (Figure 1, Panel D). The same rating also covers very different risk. A rating of 2 is the most common safe rating. Among products with this

rating, smaller banks' products imply 13% more volatility than larger banks' products with the same redemption type, yet they report 44% less.[11]

What savers do see is the higher return. Smaller banks' products reported 4.0 basis points a month more than larger banks' products with the same label, which is about 0.5 percentage points a year (Table D3, Panel B, column 5). Across banks, this return rises with the risk held (Figure 6, Panel A). Each 100 basis points of all-in spread in a bank's holdings comes with reported returns 6.4 basis points a month higher (standard error 1.0). The volatility implied by the holdings rises with the spread, while the volatility that products report does not (Panel B). In the sell-off, a bank with 100 basis points more spread had holdings whose implied returns fell 19 basis points a month more, yet its products reported returns 17 basis points a month higher.

The risk that smaller banks took thus does not appear in the returns and ratings their savers see, while the extra return does. This risk was large and concentrated. Home LGFV bonds made up 20% of all the holdings that smaller banks' products disclosed, against 1% at national banks (Table 3, Panel A). They made up 31% of smaller banks' bonds and private credit (Figure 1, Panel C). This exposure depended on the LGFVs of a single province, whose repayment rests on the same local fiscal resources and expected government support. Smaller banks' higher returns rested above all on this risk, which their savers could not see.

4.3 Reaching for yield, not information

Smaller banks hold more home LGFV credit than larger banks (Section 4.2). This concentration at home might seem to reflect better knowledge of local borrowers, which is the advantage that small banks are often thought to have (Stein 2002, Berger et al. 2005). However, a larger bank headquartered in the same city operates in the same local market and has more branches there, so a smaller bank should have no information advantage over it. We therefore

[11] With redemption-type effects, bank-equal weights, and parent-bank clustered errors, the coefficient on the smaller-bank indicator is 0.125 (standard error 0.054) for log implied volatility and −0.572 (0.144) for log reported volatility.

compare banks headquartered in the same city, first in home LGFV bonds and then in claims where neither bank has a local relationship. In both, smaller banks hold riskier claims with higher yields than larger banks from the same city.

Banks from the same city. For local bank b in quarter q, we group its LGFV bond holdings by issuer city c. A comparison cell g contains the banks headquartered in the same city that hold bonds of issuer city c in quarter q and have the same branch presence in c, which is either a branch there or none. Each cell has one location $\ell(g)$, which is the banks' own city, another city of their home province, or another province. Table 9 reports the comparisons in other cities of the home province and in other provinces, and Table 10 compares holdings in the banks' own city at the bank level. For bank b in cell g, we estimate

$$\text{Bond characteristic}_{bg} = \sum_{\ell} \beta_{\ell} \, \text{Log bank size}_b \times \mathbb{1}\{\ell(g) = \ell\} + \alpha_g + \varepsilon_{bg}. \tag{7}$$

The outcome is the value-weighted mean over the bonds bank b holds in cell g. Cell effects α_g absorb everything common to the headquarters city, issuer city, quarter, and branch presence, so β_{ℓ} compares the bonds that banks of different size hold in the same place with the same presence.[12] The yield outcome is the coupon in basis points relative to the mean coupon of held LGFV bonds with the same issue year and maturity group. The risk outcomes indicate county-level issuers, issuers rated below AAA, and issuers with a default, rating warning, or guarantor event between 2022Q4 and 2025Q4.

At home, smaller banks hold riskier LGFV bonds than larger banks from the same city (Table 9, Panel A). In other cities of its home province, a bank half as large holds bonds with coupons 24 basis points higher. Its bonds come from county-level issuers 9.2 percentage points more often and from issuers rated below AAA 2.8 points more often.

[12] Each bank receives equal total weight within each location, and errors cluster by parent bank. Holdings are 2022Q1–2023Q2 unless noted. Internet Appendix D.3 gives the construction.

These differences do not come from better local access. Among banks headquartered in the same city, a bank twice as large has about 16% more branches there (Table 10, column 1). Larger and smaller banks hold similar shares of their portfolios in own-city LGFV bonds (columns 2 and 3), but smaller banks hold more of the riskiest ones. A bank half as large holds about 1.3 percentage points more of its portfolio in own-city LGFV bonds of county-level issuers, which is a fifth more than the mean of 6.0% (column 4). With its own-city branches held fixed, the difference is 1.25 points (column 5).

Assets without local relationships. The same preference appears where no local relationship exists. In other provinces, a bank half as large holds LGFV bonds with coupons 53 basis points higher than a same-city bank with the same presence in the issuer city. Its bonds come from county-level issuers 4.8 percentage points more often and from issuers rated below AAA 2.6 points more often (Table 9, Panel A). This risk materialized. The out-of-province issuers held by a bank half as large were 0.85 percentage points more likely to have a credit event by 2025Q4, against a mean of 2.2% in these cells (column 4). The preference persisted after debt resolution and the 2024 guidance. In 2024Q3–2025Q2, a bank half as large still held LGFV bonds paying 27 basis points more in other cities of its home province and 20 more in other provinces (column 5). By then LGFV bonds offered about half their earlier extra yield over government bonds (Section 5.1).

Smaller banks also prefer weaker issuers in assets unrelated to local government finance. We first look at their holdings of other banks' tier-2 and perpetual capital bonds. Among same-city banks in the same quarter, a bank half as large invests 13 percentage points more of these holdings in the bonds of city and rural banks, and it earns coupons 17 basis points higher (Table 9, Panel B, columns 1 and 2). Interbank negotiable certificates of deposit give a second test, because a local bank has no relationship with issuers in other provinces. Among same-city banks in the same quarter and maturity group, a bank half as large holds 13.6

percentage points more of its out-of-province certificates in paper of city and rural banks rather than national banks, against a mean of 37% (Table 9, Panel B, column 3).

Although smaller banks prefer riskier claims everywhere, most of their risk and most of their extra yield are in their home provinces (Section 4.1). About three-fifths of local banks' LGFV bonds are issued at home (Table D1, Panel A), and home LGFV bonds carried the largest return premium (Section 4.1). Smaller banks lacked the reputations of larger banks, so they competed for savings by reaching for yield, and these home bonds were the largest source of that yield. Savers rewarded the yield (Section 3) without seeing its risk (Section 4.2). Both the yield and its risk depended on expected government support for LGFVs, so a change in that support would move them together.

5 Debt Resolution and the Loss of Yield

China's 2023 debt-resolution program removed the main source of the yield on which smaller banks competed, because the government support that lowered the credit risk of LGFV debt also lowered its yield. Sections 3 and 4 show that savers reward what they see and that smaller banks reached for yield. The largest source of their extra yield was home LGFV credit, and their savers could not see its risk. The strategy needed both the risky yield and the concealment of its risk. It would therefore fail if the products had to report at market prices, and it would also fail if the risk and its yield were removed. The debt-resolution program described in Section 2 did the second. The program reduced refinancing pressure and thereby strengthened expected government support for LGFVs. This lowered both the credit risk of local banks' home holdings and the yield those holdings could earn. Section 5.1 documents the loss of the yield, Section 5.2 shows which banks lost the most from it, and Section 5.3 shows where the borrowers and their risk went.

5.1 The loss of yield

The program removed most of the extra return that home LGFV bonds had supplied. We document the compression of LGFV spreads, the loss of the WMP-return premium on home LGFV bonds, and local banks' retreat from these bonds.

Spreads and yields. LGFV bonds lost about half of the extra yield they offered over government bonds. Among newly issued fixed-rate LGFV bonds, the average spread over the Treasury yield fell from 200 basis points in 2022Q1–2023Q3 to 97 basis points in 2023Q4–2025Q4.[13] The fall followed the program. Spreads fell more where special refinancing was larger relative to pre-program GDP. In a quarterly event study with province and quarter effects, this difference relative to 2023Q2 emerges with implementation and deepens through 2024 (Figure 7). For WMPs competing for savers, home LGFV bonds therefore offered much less extra yield.

Private credit retained more of its yield and offered a substitute. From 2023Q2 to 2024Q4, average yields on new private credit contracts fell from 6.6% to 5.9%, while yields on new LGFV bonds fell from 4.6% to 2.7%. The gap between the two thus widened from about 2.0 to 3.2 percentage points (Figure 8). The widening holds for the same borrower. We compare the private credit and the bonds of the same legal borrower in the same financing month, with borrower–month effects and a separate maturity slope in each quarter.[14] The gap widens from 1.90 percentage points in 2023Q2 to 2.86 points in 2024Q4 and remains 1.06 points above its 2023Q2 level in 2025Q3 (Figure D1).

Return premiums. In basis points, home LGFV bonds lost a larger WMP-return premium than any other asset class, while private credit kept more of its premium. Section 4.1 estimated the initial premiums as β in equation (5). The Post interactions δ in the same regression

[13] The averages weight issues equally and do not condition on WMP ownership.
[14] Internet Appendix D.4 gives this specification.

measure the change in each asset's premium, and $\beta + \delta$ gives its post-program premium (Table 6, column 2).

For the common 20-point allocation, the annual home LGFV bond premium falls from 20.5 to 5.0 basis points. This is a loss of 15.5 basis points, or 76%. The non-home LGFV premium falls from 7.7 to 1.5 basis points. The asset with the largest initial premium thus lost the most. The home private credit premium falls from 17.6 to 9.4 annual basis points, which is a loss of 8.2. The non-home private credit premium rises from 8.2 to 9.6. The decline in the home LGFV premium is statistically larger than the change in each of the other asset classes.[15] This difference matters for substitution, because private credit from local borrowers still offered a positive return premium after much of the bond premium had disappeared.

Lower income from the bonds that WMPs hold explains more than half of the loss. When we add each portfolio's average bond issue yield and its Post interaction to the return regression, the estimated annual loss of the home LGFV bond premium falls from about 16 to 7 basis points (Table D6, Panel A). Four further specifications give the same results. They use mature products only, add a control for return volatility, average returns within product and month, or drop the winsorization. In each, home LGFV bonds lose their premium and home private credit retains its premium (Table D6, Panel B).

Portfolio retreat. As home LGFV bonds lost their return advantage, local banks investing at home pulled back more than other buyers of bonds from the same city. Lower holdings alone could reflect fewer local bonds rather than a change in banks' preference for them. We therefore compare changes in the allocations of home buyers and other WMP buyers within the same issuer city and quarter, and we estimate

$$\begin{aligned} \text{LGFV bond share}_{acq} = \beta\, \mathbb{1}\{\text{Home buyer}\}_{ac} \times \mathbb{1}\{\text{Post}\}_q \\ + \mu_{ac} + \mu_{aq} + \mu_{cq} + \varepsilon_{acq}. \end{aligned} \tag{8}$$

[15] Parent-bank clustered tests using the regression's full covariance matrix give $p = 0.016$ for the difference in changes between home and non-home LGFV bonds, $p = 0.041$ between home LGFV bonds and home private credit, $p = 0.016$ against home other bonds, and $p < 0.001$ against non-home private credit.

Each observation is a buying institution a, an LGFV issuer city c, and a quarter q. The outcome is that city's LGFV bonds as a percentage of the institution's total WMP bond holdings. Home equals one for a local bank investing in a city of its headquarters province, and Post begins in 2023Q4. City–quarter effects μ_{cq} absorb changes in local bond supply, buyer–quarter effects μ_{aq} absorb changes across a buyer's portfolio, and buyer–city effects μ_{ac} absorb persistent investment ties.[16]

Home buyers cut their allocation by an additional 0.45 percentage points per issuer city relative to other buyers. This cut is about 12% of their pre-program mean allocation of 3.91% (Table 11). The decline is concentrated in bonds from their own city, and net buying and new purchases also fall. The fall in new purchases is larger for cities whose LGFV bonds carried higher spreads before the program (Table 11, column 5), and home buyers show no statistically significant differential trend before it (column 6). In an event study for the destination province as a whole, home buyers' allocation relative to other buyers' falls by 2.3 percentage points from 2023Q2 to 2024Q1 and by 6.1 to 2025Q3, while the pre-announcement confidence intervals include zero (Figure 9). The retreat also scaled with the program. Where a province received more special refinancing, the local institutions that had been more concentrated at home cut their home LGFV bond shares by more.[17] Within two years of the program, home LGFV bonds had lost most of the extra return on which smaller banks had competed for savings, and the banks that had relied on them were leaving them.

5.2 Replacing the lost yield

Private credit kept a higher yield, but arranging and monitoring it requires a fixed investment in sourcing, evaluation, and contracting (Section 2). This investment does not shrink with

[16] Errors cluster by bank city.

[17] We measure refinancing intensity as cumulative special-refinancing bond issuance divided by 2020 LGFV debt in the home province. When it is ten points higher, an institution whose initial home overweight (Section 4.1) is ten percentage points higher cuts its home LGFV bond share by 0.46 percentage points more (wild-bootstrap $p = 0.010$). Internet Appendix D.4 gives this specification, which compares local-bank products in the same home province and quarter.

the size of the position, so it favors banks with more savers. Larger local banks therefore replaced part of the lost home LGFV bond yield with private credit, whereas the smallest third of local banks did not, even in new products. Smaller banks fell behind in returns and WMP balances. We measure a bank's reliance on home LGFV bonds by its initial home LGFV bond share. Once this reliance and the bank's substitution into private credit are held fixed, size is no longer statistically significant for either the return losses or the lower balances of newly launched products.

Substitution into private credit. Local banks moved into private credit, which had been concentrated at national banks. Before implementation, national banks held 82.7% of the identified value of private credit in our sample, and most of it was outside their headquarters provinces (Table 12). Local banks' share then rose from 17.3% to 31.3%. Within their portfolios, home private credit increased from 6.1% to 13.9% of included debt holdings, which is more than the increase outside their provinces.

This shift was concentrated among larger local banks. Among banks observed in both periods, the largest third raised home private credit from 7.45% to 18.26% of included debt holdings. This increase more than offset their decline in home LGFV bonds, so the two together rose from 28.1% to 33.2%. At the smallest third, home private credit remained negligible and fell from 0.72% to 0.31%, and the combined share fell from 25.8% to 21.1%. The size groups use fixed FY2021 interest-bearing assets, so no bank is reclassified after the program. The new private credit went largely to LGFVs, so the change was in the form of financing rather than in the sector financed. We match borrowers to LGFV bond issuers and screen their names for terms typical of LGFVs. These two steps identify 85% of the largest local banks' post-program home private credit value as LGFV-linked (Internet Appendix D.1).

The size difference remains after accounting for product characteristics, bank conditions, and local economic conditions. In product-quarter regressions for local banks with bank

and quarter effects, the largest FY2021 size tercile raises its home private credit share after implementation by 8.21 percentage points more than the smallest tercile and its total private credit share by 10.46 points more. Both differences are significant at the 1% level (Table 13).[18] The size difference also holds conditional on initial home overweight, which measures banks' initial orientation toward home investments. Among banks facing the same provincial conditions, a bank twice as large increases its home private credit share by about 4.1 percentage points more (Table 14, Panel B). Panel A shows the same pattern when size interacts with refinancing intensity instead of the post-program indicator. Refinancing intensity is cumulative special-refinancing bond issuance divided by 2020 LGFV debt in the home province.

New products. New products let banks choose a fresh portfolio, yet even there the smallest banks made almost no use of the substitute. Because existing portfolios adjust slowly, we compare the first disclosed debt portfolios of WMPs launched in 2023Q4–2025Q3 across the same fixed size groups. The smallest banks launched 253 new products, and only two of them held home private credit. Of the 5,248 new products that the largest banks launched, 1,118 held it, which is about 21% (Table 15). Home private credit made up just 0.78% of the smallest banks' new-product debt holdings, against 27.5% at the largest banks. Home LGFV bonds still accounted for 29.4% at the smallest banks. Legacy bond positions therefore do not explain the limited substitution. It follows the pattern that a bank-level fixed cost of private credit implies for a bank with fewer savers.

Returns and balances. After the program, smaller local banks fell behind larger ones in product returns and in total outstanding WMP balances, which include older products. A size-by-quarter event study relates monthly product returns to initial bank size and compares each quarter with 2023Q3 (Figure 10). It includes issuing-institution and home-province-

[18] Internet Appendix D.4 gives the specifications of this subsection that are described in words.

by-month effects.[19] The size–return slope shows no trend before the program and does not move in the first quarters after implementation. It then widens from 2024Q3, and the gap is statistically significant through 2025Q2. This timing matches the mechanism of Section 2. A fall in spreads first raises the prices of bonds already held, and it lowers returns only as products reinvest at lower yields.

Because savers' money follows reported returns (Section 3), the return gap should also slow the growth of the whole WMP business. To test this, for local bank b at half-year end s we estimate

$$\begin{aligned}\text{Log WMP balances}_{bs} = \beta_S \,\text{Log bank size}_b \times \mathbb{1}\{\text{Post}\}_s \\ + \alpha_b + \alpha_s + \varepsilon_{bs}.\end{aligned} \tag{9}$$

The outcome is log total WMP balances, and Post begins with the December 2023 report. Bank effects α_b remove persistent differences in WMP balances, and half-year effects α_s remove common growth.[20]

Smaller banks' WMP businesses also grew more slowly. After implementation, the WMP balance of a bank twice as large grows about 11% more (Table 16), and about 10.5% more when we compare banks within the same province and half-year. Across the fixed size groups, the largest third grows about 40% more than the smallest. Among the smallest banks reporting in both June 2023 and December 2024, aggregate WMP balances fall by 22.0%. In an event study, the gap for a bank twice as large is small in December 2023 and grows to roughly 11% by December 2024 and 26% by December 2025 (Figure 11). The difference remains when we keep only banks observed at every date, exclude the final period, or give banks with and without wealth-management subsidiaries separate time paths.

Reliance and substitution. The size gaps in returns and fundraising largely follow what banks held and added. Among local banks, reliance on home LGFV bonds does not rise

[19] Internet Appendix D.4 gives the specification.

[20] The panel covers 2022H1–2025H2, and errors cluster by parent bank.

steadily as size falls. The middle size tercile held the largest initial home LGFV bond share (Table 12, Panel C), so reliance and size can be separated. We do so by including the initial home LGFV bond share, subsequent private-credit expansion, and initial bank size together in one regression. For parent bank b in month m, the outcome is its average WMP log return in basis points, and we estimate

$$\begin{aligned}\text{Monthly return}_{bm} = {} & \beta_L \,\text{Initial home LGFV bond share}_b \times \mathbb{1}\{\text{Post}\}_m \\ & + \beta_P \,\Delta\text{Home PC share}_b \times \mathbb{1}\{\text{Post}\}_m \\ & + \beta_S \,\text{Log bank size}_b \times \mathbb{1}\{\text{Post}\}_m \\ & + \beta_H \,\text{Initial home PC share}_b \times \mathbb{1}\{\text{Post}\}_m \\ & + \gamma' \,\text{Controls}_{b,h(m)} + \alpha_b + \alpha_m + u_{bm}.\end{aligned} \tag{10}$$

The initial home LGFV bond and private-credit shares are the bank's home holdings of each as a share of its included debt holdings, with holding values aggregated over 2022Q1–2023Q2. The expansion measure ΔHome PC share$_b$ is the bank's average quarterly home private-credit share in 2023Q4–2025Q3 minus its average in 2022Q1–2023Q2. Bank effects absorb the levels of these four bank-level measures, month effects absorb Post, and the eight prior-quarter bank and home-city controls enter through Controls$_{b,h(m)}$.[21]

On its own, size predicts the return losses. A bank twice as large loses about 15 annualized basis points less in returns after the program ($p = 0.017$, Table 17, column 1). Both investment measures are statistically significant once they enter. A ten-point higher initial home LGFV bond share predicts about 12 basis points more annualized return decline ($p = 0.017$), while a ten-point larger expansion in home private credit is associated with about 14 basis points less decline ($p = 0.023$, column 2). With these two measures held fixed, the size–Post coefficient halves and is no longer statistically distinguishable from zero, so the return losses follow what

[21]The sample is July 2022–September 2025, excluding July–September 2023. Each bank-month receives equal weight, and errors cluster by parent bank.

banks held and added more than their size. The initial home LGFV bond share also predicts losses without conditioning on banks' subsequent substitution (Internet Appendix D.4).

Banks with a higher initial home LGFV bond share also raised less money through newly launched WMPs. Our fundraising proxy is the log of the summed balances of the observed products that local bank b launched in quarter q, measured at the end of quarter $q+1$. We estimate the same two specifications, with launch-quarter effects in place of month effects (Table 17, Panel B). On its own, size again predicts the losses. A bank twice as large raises about 22% more through new products after the program ($p = 0.002$, column 3). With the investment measures added (column 4), a ten-percentage-point higher initial home LGFV bond share predicts 15.3% lower new-product balances ($p = 0.002$), and a ten-point larger increase in home private credit is associated with 14.6% higher new-product balances ($p = 0.034$). The size coefficient falls by more than half and is no longer statistically significant. Because the June 2024 guidance asked banks without a subsidiary to shrink their WMP business (Section 2), we also end the sample before it. The reliance estimate is nearly unchanged, at 14.9% lower balances ($p = 0.001$).

The banks that had competed most on home LGFV yield thus lost the most when debt resolution compressed it. Smaller banks had relied on that yield to compete, and because they had fewer savers to cover the fixed cost of private credit, they made little use of it when the yield fell.

5.3 Where the risk went

Debt resolution removed the main source of the yield on which smaller banks competed, and it also absorbed the credit risk that their products had hidden from savers. This subsection shows who took over the borrowers that the most reliant banks left and how far the market's price of the credit risk in their holdings fell. We focus on local banks without a wealth-management subsidiary. This group includes nearly all smaller local banks, and the 2024 guidance asked it to reduce its WMP business (Section 2).

The borrowers. As the most reliant banks withdrew, national banks' products took over their borrowers at lower coupons. We split local banks without a subsidiary at the median initial home LGFV bond share into high-reliance and low-reliance banks. For each LGFV borrower, we measure the fractions of its WMP holdings that each of the two groups held in 2022Q4–2023Q1. The outcome is the inverse hyperbolic sine of the borrower's bonds held by the WMPs of one group of banks. This transformation approximates the log and keeps zero holdings. We relate it to the two fractions, each interacted with indicators for the periods after implementation and after the June 2024 guidance. Borrower effects remove persistent differences in WMP funding, and borrower-province-by-quarter effects compare borrowers in the same province and quarter.[22]

Consider a borrower funded entirely by high-reliance banks before the program. After implementation, holdings of its bonds by local banks without a subsidiary fell by 1.89 log points, while holdings by national banks' WMPs rose by 1.83 and holdings by local banks with a subsidiary rose by 0.74 (Table 18, Panel A). Both the withdrawal and the national-bank replacement are significantly larger than for borrowers of low-reliance banks ($p = 0.005$ and $p = 0.010$), and private credit follows the same pattern (Panel B). Relative to borrowers that local banks without a subsidiary did not fund, the coupons of new bonds fell by 1.17 percentage points for a borrower funded entirely by high-reliance banks and by 0.62 for one funded by low-reliance banks ($p = 0.001$ for the difference, column 5). The same support that compressed the yield on these borrowers' bonds lowered their funding costs.

The credit risk. A year after the program, the market priced far less credit risk in these banks' home LGFV holdings. We weight each province's LGFV spread over Treasury yields by the home LGFV credit that local banks without a subsidiary held in 2022Q4–2023Q2. This weighted spread fell from 297 basis points in June 2023 to 87 in June 2024.

[22] Internet Appendix D.4 gives the specification and further results, and Internet Appendix D.3 the construction.

Debt resolution moved the credit risk that smaller banks' products had accumulated to provincial governments and to national banks' products, which now hold these borrowers at lower yields. Smaller banks' WMP business was fragile either way. Without resolution, credit losses would have fallen on products that savers had bought as safe. With it, these banks lost the return advantage on which they competed. Since June 2024, regulators in several provinces have asked banks without a subsidiary to shrink their WMP business, in some cases to zero. This retreat has so far coincided with lower funding costs for the borrowers these banks financed.

6 Model

The model explains how banks with little reputation can compete for savings by reaching for yield and hiding its risk, and why more reputable banks take less risk and report it in full. It also explains why smaller banks lose more of their return and their savers when debt resolution lowers the yield. Banks choose how much to invest in risky bonds and whether to hide part of their bonds' variance from their savers. Reputation determines which savers a bank attracts, and the savers most responsive to its offer determine how much risk it takes. Hiding is worth more to a bank with a large risky position, but it puts the bank's reputation at stake.[23] Internet Appendix B derives the results and solves the full payoff numerically, and Internet Appendix C gives proofs.

6.1 Savers and banks

A bank in the model is its WMP business, and the investments below belong to its products and not to its balance sheet. There are two banks, a larger bank L and a smaller bank S. Each saver in a unit mass invests one unit with one of the two banks or with an outside

[23] Savers choose institutions and institutions choose assets, and this two-layer demand follows Darmouni et al. (2024). Heterogeneous saver demand draws on Xiao (2020) and random-coefficient models (Berry et al. 1995), fee-based portfolio incentives follow La Spada (2018), and reputation as a restraint on risk taking follows Keeley (1990).

saving option. Savers differ in risk aversion γ_h, which is drawn from a distribution G on a bounded positive interval. Saver h obtains utility

$$U_{hb} = \gamma_h a_b + \beta\Big(R_b - \frac{\gamma_h}{2} V_b\Big) + \varepsilon_{hb} \tag{11}$$

from bank b. Here R_b and V_b are the mean and variance of the excess returns that the bank's products report in normal periods, and a_b is the bank's reputation and sales network. Savers take R_b and V_b at face value and do not price the risk that the reports leave out, as Section 3 documents. The parameter $\beta > 0$ measures how strongly savings move with the risk-adjusted reported return. We assume that cautious savers weight reputation more, because a bank's standing matters more to savers who fear losses. With logit taste shocks ε_{hb}, the share of savers with risk aversion γ choosing bank b is

$$s_b(\gamma) = \frac{\exp\big(\gamma a_b + \beta(R_b - \gamma V_b/2)\big)}{1 + \sum_j \exp\big(\gamma a_j + \beta(R_j - \gamma V_j/2)\big)}, \tag{12}$$

and the bank's investor share is $s_b = \int s_b(\gamma)\, dG(\gamma)$. The larger bank has the higher reputation, $a_L > a_S$. Both banks face the same investment opportunities, so differences in their choices come from the savers they attract and not from better information about borrowers.

Each bank places a share $x_b \in [0, 1]$ of its products' funds in LGFV bonds and the rest in a safe asset. LGFV bonds stand for the risky credit that was the largest single source of smaller banks' extra yield (Section 4.1). They pay an excess return $\mu_B > 0$, and at market prices their return has variance σ_B^2. A bank also chooses how it reports, $H_b \in \{0, 1\}$. A bank with $H_b = 0$ reports transparently, and a bank with $H_b = 1$ hides risk by reporting only a share $1 - c$ of the variance of its bonds' market returns, with $0 < c < 1$. The choice is binary for clarity. A continuous degree of hiding with an exposure cost proportional to it gives the same solution. For a given portfolio the payoff is linear in the degree of hiding, so a bank hides either as much as its products allow or not at all. In practice, products that restrict redemptions need not sell when prices fall, their reported returns smooth out the swings in

bond prices, and the bank rates them as safe (Section 4.2). The bank's products therefore report

$$R_b = \mu_B x_b, \qquad V_b = (1 - c\,H_b)\,\sigma_B^2 x_b^2. \tag{13}$$

Two risks restrain the bank, and both work through its reputation. First, LGFV bonds carry a credit risk that reported returns do not show. With probability θ, the credit losses in the bank's portfolio become salient to its savers. Savers then lower its reputation by $\kappa x_b^2/2$. Second, hiding can become public, for example through a supervisory finding or press coverage of the bank's valuation practices. This happens with probability ρ, and savers then lower the bank's reputation by δ. Savers continue to take reported performance at face value, so the exposure affects their choices only through the bank's reputation. WMPs do not guarantee principal, so the losses fall on savers, and the bank bears them only through its reputation. A bank earns a fee f per yuan of savings. Let $\mathcal{S}_b(a)$ be the bank's investor share at reputation a, holding its choices and its rivals' offers and reputations fixed. The bank chooses its bond share and its reporting to maximize

$$f\,\mathrm{E}\Big[\mathcal{S}_b\Big(a_b - \frac{\kappa}{2}x_b^2\,\mathbb{1}_C - \delta\,H_b\,\mathbb{1}_E\Big)\Big], \tag{14}$$

where $\mathbb{1}_C$ and $\mathbb{1}_E$ indicate the credit event and the exposure, which are independent with probabilities θ and ρ.

6.2 Reaching for yield and hiding risk

A bank's choices depend on the savers most likely to switch when its offer improves. Weighting each saver type by the response of its choice to the bank's offer gives the average risk aversion of these marginal savers,

$$\bar{\gamma}_b = \frac{\int \gamma\, s_b(\gamma)\big(1 - s_b(\gamma)\big)\,dG(\gamma)}{\int s_b(\gamma)\big(1 - s_b(\gamma)\big)\,dG(\gamma)}. \tag{15}$$

Because savers weight reputation by their risk aversion, a loss of reputation costs a bank more savers when its marginal savers are more cautious. To obtain closed-form comparative statics, we approximate the bank's expected investor share to first order in the reputation losses and evaluate both reporting choices with the same marginal savers. The bank then chooses as if it maximized

$$Q_b = R_b - \frac{\bar{\gamma}_b}{2}\left(V_b + \varphi\, x_b^2\right) - \bar{\gamma}_b\, \psi\, H_b, \qquad \varphi = \frac{\theta\kappa}{\beta}, \qquad \psi = \frac{\rho\delta}{\beta}, \tag{16}$$

which is the reported return, adjusted for risk as its marginal savers value it and net of the expected reputational losses. Here φ is the credit risk that the bank bears through its reputation, in units of reported variance, and ψ is the expected cost of exposure, in units of reported return. The bank adds bonds until their excess return equals their marginal cost to its marginal savers, $\mu_B = \bar{\gamma}_b\, y_{H_b}\, x_b$, so that

$$x_b = \frac{\mu_B}{\bar{\gamma}_b\, y_{H_b}}, \qquad y_H = (1 - c\, H)\, \sigma_B^2 + \varphi. \tag{17}$$

The effective risk y_H adds the reported variance and the credit risk borne through reputation. We consider banks whose bond shares are interior, $\mu_B < \bar{\gamma}_b\, y_1$, while the numerical solution does not need this restriction. Hiding lowers the reported variance of each bond, relaxes the restraint, and raises the bond share.

The two reputational costs do different work. Credit risk limits the bond position even when reported variance is low, and the cost of exposure decides whether hiding is worth it. Without the first, a bank that hides would expand its position until it reported more volatility than a transparent bank. With it, a bank that hides can report less volatility than a more reputable bank that holds far fewer bonds, which requires $\varphi > \sqrt{1-c}\,\sigma_B^2$ (Internet Appendix B).

Whether a bank hides depends on the caution of its marginal savers. When we compare the best portfolios with and without hiding, the gain from hiding is

$$\frac{c\,\sigma_B^2\,\mu_B^2}{2\,\bar{\gamma}_b\,y_0\,y_1} - \bar{\gamma}_b\,\psi, \tag{18}$$

which is nonnegative if and only if $\bar{\gamma}_b \leq \gamma^* = \mu_B\sqrt{c\,\sigma_B^2/(2\psi y_0 y_1)}$, with ties resolved in favor of hiding (Proposition B.1). At a given bond position, cautious savers value lower reported variance more. But their bank holds fewer bonds and so has less variance to hide. Once the portfolio adjusts, the gain from hiding falls with the caution of the bank's marginal savers, while the expected reputational cost rises with it, because cautious savers weight reputation most.

Reputation determines which banks hide. Suppose that each bank's share of savers is small and that the reputation gap exceeds $\beta\sigma_B^2/2$, which we call the sorting condition. Then more reputable banks attract more cautious marginal savers (Proposition B.2). A bank with little reputation therefore holds more bonds, and when the cutoff γ^* lies between the two banks' values of $\bar{\gamma}_b$, it is the one that hides. Reaching for yield and hiding reinforce each other, because the larger position makes hiding more valuable and hiding permits a larger position. Hiding also makes the strategy exploitative. A bank that hides reports less risk than its savers bear, and the lower reported risk draws more cautious savers than its portfolio would attract at market prices.

6.3 Debt resolution and fragility

Debt resolution strengthens expected support for LGFVs. We represent it as a proportional fall in the probability of a credit event and in the bond's excess return, $\theta_1 = (1-\tau)\theta_0$ and $\mu_{B,1} = (1-\tau)\mu_{B,0}$ with $0 < \tau < 1$, so the credit risk borne through reputation falls to $\varphi_1 = (1-\tau)\varphi_0$. The market variance σ_B^2 and the hiding and exposure parameters are

unchanged. Subscripts 0 and 1 denote values before and after resolution, and $y_{H,t}$ is y_H evaluated at φ_t.

Resolution has two opposing effects on the incentive to hide. The lower yield reduces the return from taking risk, and the lower credit risk relaxes the reputational restraint on it. Under the proportional decline specified here, the first effect dominates, so the hiding threshold falls from γ_0^* to $\gamma_1^* = (1-\tau)\sqrt{y_{0,0}\, y_{1,0}/(y_{0,1}\, y_{1,1})}\gamma_0^* < \gamma_0^*$. Banks whose $\bar{\gamma}_b$ lies between the two thresholds stop hiding, and they cut their bonds by more than banks that do not change how they report (Proposition B.3). Reporting at market prices also undoes the strategy. A bank that is required to stop hiding cuts its bonds and its reported return by the fraction $c\,\sigma_B^2/y_0$. The strategy is fragile in this sense, because it depends on both the yield and the hiding. Figure 12 shows the two banks' choices in an illustrative equilibrium.

Prediction 1. Smaller banks reach for yield and hide its risk. Under the sorting condition and with the cutoff γ^* between the two banks' values of $\bar{\gamma}_b$, a bank with less reputation holds more LGFV bonds and hides part of their variance, while a more reputable bank reports all of it. The bank with less reputation reports a higher return, and its reported volatility is lower when the two banks' marginal savers are close enough in caution, which requires $\varphi > \sqrt{1-c}\,\sigma_B^2$. We take bank size as the empirical counterpart of reputation. Table 3 shows that smaller banks hold more LGFV bonds and that home LGFV bonds are the largest source of their extra yield, and Table 7 shows that smaller banks' products hold more risk yet report less of it.

Prediction 2. Debt resolution cuts risky holdings, most at banks that stop hiding. A bank that keeps its reporting choice cuts its bonds in proportion to its initial position, and a bank that stops hiding cuts them by more. Home buyers retreated from LGFV bonds after the program (Table 11 and Figure 9), and local banks more concentrated at home cut their home LGFV bond shares by more where a province received more special refinancing (Section 5.1).

Prediction 3. Smaller banks lose more product return and savings business. Suppose the smaller bank hides before resolution and stops afterward, while the larger bank does not hide.

Then

$$\Delta R_L - \Delta R_S = \lambda_S R_{S,0} - \lambda_L R_{L,0} > 0, \tag{19}$$

where $\lambda_S = 1 - (1-\tau)^2 y_{1,0}/y_{0,1}$ and $\lambda_L = 1 - (1-\tau)^2 y_{0,0}/y_{0,1}$ are the proportional falls in the two banks' reported returns. The smaller bank loses a larger share of a larger return. Its larger bond position exposes it more to the fall in yield, and it gives up the hiding that let it hold that position. Its return advantage over the larger bank therefore narrows, and its share of savers falls relative to the larger bank's (Proposition B.4). Because smaller banks both cut their LGFV holdings and lose savers, more of the LGFV credit held by WMPs moves to larger banks' products, as in Section 5.3. In the data, banks with a larger initial home LGFV bond share lost more return and raised less money in new products (Table 17), and smaller local banks grew more slowly (Table 16).

7 Conclusion

Small banks must compete with large banks for household savings. In China's market for wealth-management products, smaller banks compete through a strategy that is both exploitative and fragile. Chinese households choose wealth-management products by the returns they report and by how much those returns fluctuate, not by the risk of the assets behind them. In the 2022 bond sell-off they ran from the losses that products reported, not from the losses their holdings took. Smaller banks lack the reputations of larger banks, so they can compete for savings only by exploiting this bias. They reached for yield in the dark. Their products held riskier debt than larger banks' products, and the largest source of their extra yield was concentrated LGFV credit from their home provinces. Yet these banks rated the products as safe, locked savers in with restricted redemptions, and reported returns that barely moved. Larger banks from the same city have more branches there, yet they held safer bonds, so smaller banks relied on taking risk, not on better knowledge of local borrowers.

Smaller banks needed this strategy to survive. It was fragile because it needed both the risky yield and the concealment of its risk. Savers would have run if these products had shown the losses that their holdings took in the 2022 sell-off. Instead, China's 2023 debt-resolution program absorbed much of the risk and removed much of the yield with it. Larger local banks replaced part of the lost yield with private credit, whose fixed costs favor scale. Smaller banks relied on that yield the most and could not replace it, so they lost returns and fundraising. National banks' products took over the borrowers that the most reliant banks left.

Our model traces this pattern to reputation. Hiding risk attracts savers who take reports at face value, but it puts the bank's reputation at stake. It pays only for a bank that holds enough risk to hide. Banks with the least reputation attract the least cautious marginal savers and hold the most risk, so they reach for yield and hide its risk, while more reputable banks take less risk and report it in full. China's regulators intend small banks to serve local small firms, yet in competing for savings these banks channeled household money into LGFV debt whose risk their savers could not see. Support for these borrowers absorbed this risk before savers bore it, and it removed the main source of the yield on which smaller banks competed. Their WMP business is winding down, and the borrowers they financed now borrow at lower cost. When banks have little reputation to lose, they compete through a fragile strategy that exploits the behavioral bias of households, who care about returns and observed volatility but not underlying risk.

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Table 1
Past returns, reported volatility, holdings risk, and investor flows

Dependent variable	Net flow (% initial assets)		
	(1)	(2)	(3)
$\mathbb{1}\{\text{Return quintile} = 2\}$	1.790^{***}	1.821^{***}	1.188^{**}
	(0.605)	(0.605)	(0.554)
$\mathbb{1}\{\text{Return quintile} = 3\}$	2.873^{***}	2.917^{***}	2.172^{***}
	(0.789)	(0.790)	(0.747)
$\mathbb{1}\{\text{Return quintile} = 4\}$	5.185^{***}	5.255^{***}	4.622^{***}
	(1.152)	(1.156)	(1.077)
$\mathbb{1}\{\text{Return quintile} = 5\}$	12.008^{***}	12.126^{***}	9.832^{***}
	(2.671)	(2.682)	(2.071)
Home LGFV credit share (per SD)	—	-0.540	-0.417
		(0.344)	(0.363)
Reported return volatility (per SD)	—	—	-2.276^{***}
			(0.360)
Product, bank, and city controls	Yes	Yes	Yes
Observations	129,577	129,577	106,058
Category × quarter fixed effects	Yes	Yes	Yes
Standard errors	Parent bank	Parent bank	Parent bank

Notes: Each column estimates equation (1) and adds the regressors shown, and each regressor enters per standard deviation. Return quintiles rank prior-quarter returns within risk rating, redemption terms, and quarter, and the bottom quintile is omitted. Home LGFV credit share is home-province LGFV bonds and private credit to home-province LGFVs as a percentage of disclosed holdings at the end of the prior quarter, and it is zero for products that hold none (SD 16.7 percentage points). Reported return volatility is the log standard deviation of the product's monthly reported returns over the prior 12 months (SD 1.00). Net flows are capped at the 1st and 99th percentiles. Columns 1 and 2 use all product-quarters of the flow sample, and column 3 those that also have a 12-month return history. Controls are those of equation (1) and a local-bank indicator. Parentheses contain parent-bank clustered errors. Stars indicate 10%, 5%, and 1% significance. Estimation sample: 2022Q3–2025Q3.

Table 2
Reported returns and investor flows

Dependent variable	Net flow in the same quarter		Net flow in the next quarter	
	(1)	(2)	(3)	(4)
Panel A. Daily-open products, by reported quarterly return				
Loss of more than 0.5%	−47.5***	−11.8*	−33.9***	−3.9
	(9.0)	(6.8)	(10.3)	(7.8)
Loss of 0.1% to 0.5%	−46.1***	−23.2***	−32.2***	−12.8**
	(8.6)	(8.2)	(7.3)	(6.1)
Loss of less than 0.1%	−51.1***	−31.5***	−28.1***	−12.2**
	(7.4)	(7.9)	(5.2)	(4.6)
Gain of less than 0.25%	−51.7***	−35.7***	−27.0***	−14.7**
	(9.7)	(10.6)	(5.7)	(5.6)
Gain of 0.25% to 0.5%	−43.4***	−30.4***	−18.5***	−15.7***
	(4.8)	(5.3)	(2.3)	(2.7)
Gain of 1% or more	17.6***	0.7	52.5***	31.2***
	(5.1)	(7.4)	(11.8)	(8.2)
Observations	22,106	21,449	22,106	21,449
Panel B. All products, by redemption type				
Return × daily-open	52.5***	20.7**	57.7***	26.6**
	(16.4)	(8.6)	(18.7)	(11.7)
Return × periodic-open	3.8***	−2.0**	8.2***	3.2***
	(0.8)	(0.9)	(1.4)	(1.1)
Return × closed-end	−0.3	1.1	−0.7	−0.5
	(0.9)	(0.8)	(0.7)	(0.6)
Observations	129,554	122,194	129,554	122,194
Product, bank, and city controls	Yes	Yes	Yes	Yes
Risk rating × redemption type × quarter fixed effects	Yes	Yes	Yes	Yes
Bank × quarter fixed effects	Yes	Yes	Yes	Yes
Product fixed effects	No	Yes	No	Yes
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank

Notes: The dependent variable is net flow, the asset growth net of investment gains as a percentage of beginning assets. It is capped at its 1st and 99th percentiles. The reported return is that of the same quarter in columns 1 and 2 and of the preceding quarter in columns 3 and 4. In Panel A, each row is an indicator for the product's reported quarterly return, and the omitted group reported 0.5% to 1%. In Panel B, the reported quarterly return is interacted with redemption-type indicators. It is in percentage points and capped at its 1st and 99th percentiles. All regressions include the lagged controls of Table 1, risk rating by redemption type by quarter effects, and bank-by-quarter effects, and columns 2 and 4 add product effects. Parentheses contain parent-bank clustered errors. Stars indicate 10%, 5%, and 1% significance. Estimation sample: 2022Q3–2025Q3.

Table 3
Where smaller banks' extra yield comes from

Panel A. Holdings and spreads by asset class

	Share of holdings (%)		Spread (bp)		Contribution (bp)
	Smaller local (1)	National (2)	Smaller local (3)	National (4)	Difference (5)
Home LGFV bonds	19.8	0.9	317	236	60.8
Other LGFV bonds	10.9	8.0	202	210	5.2
Private credit	5.4	10.2	406	351	−14.0
Local banks' capital bonds	8.6	1.7	191	153	13.8
National banks' capital bonds	3.6	8.0	125	113	−4.5
Other financial bonds	5.7	8.1	126	126	−3.0
Home corporate bonds	5.8	1.5	270	121	13.9
Other corporate bonds	3.5	8.1	136	142	−6.7
Government and policy-bank bonds	5.9	3.5	24	11	1.0
Certificates of deposit	3.7	2.7	112	88	1.8
Asset-management plans and funds	22.9	30.6	193	186	−12.4
Cash, deposits, and repos	4.1	16.7	0	0	0.0
All holdings	100.0	100.0	205	149	55.8

Panel B. Size gradient of each asset class's contribution to the all-in spread (bp)

Dependent variable	Home LGFV bonds (1)	Other LGFV bonds (2)	Private credit (3)	Local banks' capital bonds (4)	National banks' capital bonds (5)	Other financial bonds (6)
Log bank size	−17.0***	−2.7*	8.2***	−4.3***	1.1*	0.6
	(3.3)	(1.4)	(2.6)	(0.9)	(0.6)	(0.8)
Observations	4,165	4,165	4,165	4,165	4,165	4,165
Dependent variable	Home corporate bonds (7)	Other corporate bonds (8)	Government and policy-bank bonds (9)	Certificates of deposit (10)	Asset-management plans and funds (11)	All holdings (12)
Log bank size	−4.6***	1.4***	−0.4**	−0.4	5.2	−12.8***
	(1.4)	(0.5)	(0.2)	(0.4)	(3.4)	(2.9)
Observations	4,165	4,165	4,165	4,165	4,165	4,165
Redemption type × risk rating fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank	Parent bank	Parent bank

Notes: The sample is the products of Table 7, with holdings averaged over 2022Q2–2023Q2. Each holding is priced at its credit spread over Treasury yields. Bonds carry their issue spread, and private credit carries its contract rate over the Treasury yield of its remaining maturity. Asset-management plans and funds carry the mean issue spread of AA+ bonds with one to three years to maturity (AAA for bond funds and rate products). Certificates of deposit carry the spread of short-term bonds of the issuer's rating, and cash, deposits, and repurchase agreements carry zero. The all-in spread is the value-weighted mean, which equals the sum over classes of holding share times spread. Panel A averages products within a bank and then banks, and its column 5 is the contribution, holding share times spread, at smaller local banks minus that at national banks. In Panel B, each column is a separate regression of one class's contribution on log bank size, with the weights, fixed effects, and clustering of Table 7. The class coefficients sum to the all-in spread coefficient in column 12. Cash, deposits, and repurchase agreements contribute zero and have no column. Capital bonds are perpetual and tier-2 bonds, and home corporate bonds are bonds of non-LGFV, non-financial issuers from the home province. Parentheses contain parent-bank clustered errors. Stars indicate 10%, 5%, and 1% significance.

Table 4
Initial credit specialization by bank size

Dependent variable	Total LGFV bond share (pp) (1)	Total private credit share (pp) (2)	Home LGFV bond share (pp) (3)
Log bank size	−7.130***	6.831***	−5.458***
	(1.778)	(1.803)	(1.123)
Log product assets	−3.028***	−3.480***	−0.677
	(0.828)	(0.914)	(0.431)
𝟙{Restricted redemption}	2.405	16.943***	−0.264
	(2.138)	(4.113)	(1.323)
Product risk rating	−3.065**	4.162	0.392
	(1.316)	(2.956)	(0.867)
Bank nonperforming-loan ratio	−7.968	10.855**	−3.180
	(6.171)	(5.471)	(4.021)
Bank cost–income ratio	−0.245*	−0.582***	−0.085
	(0.131)	(0.184)	(0.092)
Bank net interest margin	−5.013	9.229	2.795
	(7.203)	(8.397)	(4.126)
Home-city GDP (RMB trillion)	0.064	−1.051	0.810
	(3.585)	(1.945)	(2.118)
Home-city GDP growth (%)	−0.381	−0.182	0.527
	(1.211)	(1.046)	(0.768)
Home-city LGFV debt/GDP	27.483**	3.752	22.502***
	(11.977)	(4.867)	(6.930)
Home-city income (RMB 10,000)	1.325	3.696**	−1.132
	(3.213)	(1.595)	(2.049)
Wild p: log bank size	0.002	0.009	< 0.001
Observations	68,331	68,331	68,331
Quarter fixed effects	Yes	Yes	Yes
Standard errors	Parent bank	Parent bank	Parent bank

Notes: Each column regresses the stated share of included debt holdings (pp) of national- and local-bank products on log FY2021 parent-bank interest-bearing assets in RMB trillion, without standardization. Products without holdings of a class enter with a zero share, and all ten controls are displayed. Fixed FY2021 size takes the place of the time-varying log bank assets that other tables include as a control. The regressions include quarter effects but no local-bank indicator or bank effects, and they weight by included debt value. Parentheses contain parent-bank clustered errors, which group each bank with its wealth-management subsidiary. Stars indicate 10%, 5%, and 1% significance. Wild tests use 4,999 Rademacher draws. Estimation sample: 2022Q1–2023Q3.

Table 5
Home bias within each asset class by bank type

Asset class	All included debt	LGFV	PC	Other bonds
Dependent variable	Home bias within class (fraction) (1)	Home bias within class (fraction) (2)	Home bias within class (fraction) (3)	Home bias within class (fraction) (4)
$\mathbb{1}\{\text{Local bank}\}$	0.2713***	0.3205***	0.4885***	0.1611***
	(0.0316)	(0.0449)	(0.0594)	(0.0292)
Log product assets	−0.0004	−0.0019	0.0016	−0.0083*
	(0.0051)	(0.0092)	(0.0123)	(0.0048)
$\mathbb{1}\{\text{Restricted redemption}\}$	−0.0182	−0.0189	−0.0285	−0.0107
	(0.0147)	(0.0218)	(0.0321)	(0.0093)
Product risk rating	0.0059	0.0035	0.0067	−0.0021
	(0.0089)	(0.0155)	(0.0182)	(0.0087)
Log(1 + disclosed positions)	−0.0965***	0.0077	0.0001	−0.0629
	(0.0337)	(0.0543)	(0.0298)	(0.0386)
Total disclosed weight (%)	0.0007*	0.0016**	0.0008	0.0003
	(0.0004)	(0.0007)	(0.0006)	(0.0004)
Log average product maturity	0.0203**	0.0337**	0.0178	0.0217***
	(0.0086)	(0.0145)	(0.0187)	(0.0080)
Average product risk rating	−0.0513	−0.1744	0.1068	−0.1189***
	(0.0598)	(0.1515)	(0.0931)	(0.0443)
Bank nonperforming-loan ratio (%)	−0.0371	0.0296	0.0490	−0.0142
	(0.0374)	(0.0651)	(0.0745)	(0.0367)
Bank cost–income ratio (%)	−0.0008	−0.0003	0.0019	−0.0001
	(0.0015)	(0.0023)	(0.0032)	(0.0016)
Bank net interest margin (%)	0.0939	0.2127***	0.2460***	0.0412
	(0.0577)	(0.0785)	(0.0926)	(0.0458)
Home-city GDP growth (%)	0.0188*	0.0310*	0.0158	0.0088
	(0.0105)	(0.0160)	(0.0188)	(0.0103)
Mean of dependent variable	0.1851	0.2562	0.1944	0.1305
Observations	186,731	108,313	43,803	158,812
Quarter fixed effects	Yes	Yes	Yes	Yes
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank

Notes: PC denotes private credit. Estimates use equation (3). The unit of observation is a product-quarter with positive holdings in the class. The dependent variable is the product's home-province share within the asset class minus the home province's share of nationwide disclosed holdings in the class, as a fraction. The first column pools all included debt, and the others restrict it to LGFV bonds, private credit, and other bonds. Local bank equals one for banks other than state-owned and joint-stock commercial banks. Product size and the number of disclosed positions enter as $\log(1+x)$. Parentheses contain parent-bank clustered errors. Stars indicate 10%, 5%, and 1% significance. Estimation sample: 2022Q1–2025Q3.

Table 6
Home credit exposure and product returns before and after the program

Dependent variable	Monthly NAV return (bp)	
	(1)	(2)
Home LGFV bond share	10.734***	8.543***
	(1.967)	(1.916)
Home LGFV bond share × 𝟙{Post}	−6.728***	−6.473***
	(1.720)	(1.667)
Home private credit share	8.262***	7.328***
	(1.590)	(1.450)
Home private credit share × 𝟙{Post}	−3.719**	−3.408**
	(1.715)	(1.542)
Home other-bond share	1.944	1.310
	(1.760)	(1.626)
Home other-bond share × 𝟙{Post}	−1.222	−0.978
	(1.863)	(1.723)
Non-home LGFV bond share	3.603***	3.188***
	(1.377)	(1.208)
Non-home LGFV bond share × 𝟙{Post}	−2.925**	−2.559*
	(1.416)	(1.362)
Non-home private credit share	3.243***	3.434***
	(1.109)	(0.969)
Non-home private credit share × 𝟙{Post}	0.616	0.580
	(0.936)	(0.873)
𝟙{Local bank}	1.700*	0.355
	(0.934)	(0.771)
Log product assets	−0.665***	−0.603***
	(0.165)	(0.152)
𝟙{Restricted redemption}	3.930***	3.721***
	(0.774)	(0.711)
Product risk rating	−2.686***	−2.391***
	(0.809)	(0.694)
Log bank assets	—	−0.160
		(0.456)
Bank nonperforming-loan ratio	—	1.329
		(1.277)
Bank cost–income ratio	—	0.061***
		(0.023)
Bank net interest margin	—	1.000
		(1.573)
Home-city GDP (RMB trillion)	—	−0.006
		(0.601)
Home-city GDP growth (%)	—	−0.065
		(0.296)
Home-city LGFV debt/GDP	—	3.750**
		(1.658)
Home-city income (RMB 10,000)	—	−0.338
		(0.612)
Observations	567,855	567,855
Month fixed effects	Yes	Yes
Standard errors	Parent bank	Parent bank

Notes: Monthly log adjusted-NAV returns (bp) are regressed on the five asset shares at the preceding quarter end, their Post interactions, and controls. Returns are winsorized at the 1st and 99th percentiles of all monthly returns. Month effects absorb Post, which starts in October 2023. Column 1 uses product controls, and column 2 adds bank and home-city controls. Both columns use the column 2 sample. Shares are fractions of included debt holdings, and non-home other bonds are the omitted class. Multiply coefficients by 0.2 for a 20-point allocation difference and by 12 to annualize. Column 2's post-program premiums are 2.070 (SE 1.809) for home LGFV bonds and 3.920 (1.070) for home private credit. The difference in their premium changes is −3.065 (1.489). Centrally administered issuers are never home. Parentheses contain parent-bank clustered errors. Stars indicate 10%, 5%, and 1% significance. Estimation sample: April 2022–December 2025.

Table 7
Bank size, risk held, and risk reported

Panel A. What the product holds						
Dependent variable	Home LGFV credit share (pp)	Issue spread of bonds held (bp)	Issue spread relative to same rating (bp)	All-in spread of holdings (bp)	Log volatility implied by holdings	
	(1)	(2)	(3)	(4)	(5)	
Log bank size	−5.394***	−22.374***	−17.079***	−12.792***	−0.074***	
	(0.874)	(3.511)	(4.084)	(2.866)	(0.016)	
Observations	4,165	3,991	3,263	4,165	4,165	
Redemption type × risk rating fixed effects	Yes	Yes	Yes	Yes	Yes	
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank	Parent bank	
Panel B. What the product reports						
Dependent variable	Log SD of monthly return	Log reported minus log implied SD	Return, Nov.–Dec. 2022 (bp)	Reported minus implied, Nov.–Dec. 2022 (bp)	Mean monthly return (bp)	Safe rating, 1 or 2 (pp)
	(1)	(2)	(3)	(4)	(5)	(6)
Log bank size	0.154***	0.228***	−6.228***	−11.656***	−1.413***	−4.266***
	(0.036)	(0.040)	(1.669)	(2.320)	(0.297)	(1.595)
Observations	4,165	4,165	4,165	4,165	4,165	11,580
Redemption type × risk rating fixed effects	Yes	Yes	Yes	Yes	Yes	No
Redemption type fixed effects	No	No	No	No	No	Yes
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank	Parent bank	Parent bank

Notes: Each column is a separate regression of the outcome on log bank size, which is log FY2021 parent-bank interest-bearing assets in RMB trillion. The sample is fixed-income products with disclosed holdings and at least nine monthly returns, except in column 6 of Panel B. Holdings are averaged over 2022Q2–2023Q2. Column 6 of Panel B uses every fixed-income product with a monthly return in the period, and its outcome equals 100 for a safe rating of 1 or 2. Home LGFV credit share is LGFV bonds and private credit to LGFVs of the bank's home province as a percentage of disclosed holdings. Issue spread is the coupon minus the same-maturity Treasury yield at issue, and column 3 of Panel A nets the mean of bonds with the same issue quarter, issuer rating, administrative level, maturity group, and placement type. The all-in spread is defined in Table 3. Implied returns mark every disclosed asset to its market series (Section 4.2), and column 2 of Panel B is log reported minus log implied volatility. Reported returns are monthly changes in log distribution-adjusted NAV, and each series is winsorized at the 1st and 99th percentiles of all its monthly values. Each bank receives equal total weight. Parentheses contain parent-bank clustered errors. Stars indicate 10%, 5%, and 1% significance. Estimation sample: April 2022–June 2023. Table D3 compares smaller with larger banks, and Table D2 reports alternative specifications.

Table 8
Sell-off losses reported, by redemption type

Panel A. Fall below the usual monthly return, November–December 2022 (basis points a month)

	Products	Reported	Implied by holdings	Share reported
Closed-end products				
Smaller local banks	601	−28.5	−117.9	0.24
Larger local banks	422	−54.7	−115.3	0.47
National banks	727	−51.8	−100.4	0.52
Periodic-open products				
Smaller local banks	533	−47.1	−106.8	0.44
Larger local banks	420	−63.6	−107.9	0.59
National banks	1,035	−60.8	−87.0	0.70
Daily-open products				
Smaller local banks	6	−22.7	−89.3	0.25
Larger local banks	24	−40.7	−71.2	0.57
National banks	397	−44.0	−61.8	0.71
All products				
Smaller local banks	1,140	−39.4	−113.6	0.35
Larger local banks	866	−56.3	−109.0	0.52
National banks	2,159	−57.5	−87.9	0.65

Panel B. Reported minus implied return in the sell-off, relative to other months, within bank

Dependent variable	Reported minus implied return (bp a month)	
	(1)	(2)
𝟙{Closed-end}	33.9***	29.0***
	(6.5)	(6.4)
𝟙{Periodic-open}	12.3**	10.4*
	(6.0)	(5.8)
Mean, daily-open products	28.0	27.1
Observations	4,163	4,153
Bank fixed effects	Yes	No
Bank × risk rating fixed effects	No	Yes
Standard errors	Parent bank	Parent bank

Notes: The sample is the products of Table 7. Smaller local banks are at or below the median FY2021 size of the banks in that sample, larger local banks are the local banks above it, and every national bank is above it. In Panel A, the fall below the usual return is the mean monthly return in November and December 2022 minus the mean in the other months from April 2022 to June 2023. It averages products within a bank and then banks. Share reported is the reported fall divided by the implied fall. Returns are constructed as in Figure 5. In Panel B, the dependent variable is the reported minus implied return in November and December 2022 minus the same difference in the other months, in basis points a month. Daily-open products are the omitted group. Each bank receives equal total weight. Parentheses contain parent-bank clustered errors. Stars indicate 10%, 5%, and 1% significance.

Table 9
Bank size and the yield and risk of bonds held

Dependent variable	Coupon (bp)	County-level issuer (pp)	Issuer below AAA (pp)	Issuer credit event (pp)	Coupon after June 2024 (bp)
	(1)	(2)	(3)	(4)	(5)
Panel A. LGFV bonds held					
Log bank size × 𝟙{Rest of home province}	−34.293***	−13.241*	−4.093**	−1.686	−39.405***
	(11.030)	(6.986)	(1.954)	(1.601)	(13.656)
Log bank size × 𝟙{Other provinces}	−76.785***	−6.993***	−3.704***	−1.219**	−29.276***
	(22.493)	(2.566)	(1.367)	(0.549)	(6.992)
Observations	1,674	1,606	1,562	1,652	1,144
Comparison cell fixed effects	Yes	Yes	Yes	Yes	Yes
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank	Parent bank

Dependent variable	Capital bonds issued by city or rural banks (pp)	Coupon of capital bonds (bp)	Certificates of city or rural banks, other provinces (pp)
	(1)	(2)	(3)
Panel B. Other banks' capital bonds and certificates of deposit held			
Log bank size	−19.218***	−24.668***	−19.693***
	(3.125)	(3.637)	(7.095)
Observations	196	196	199
Home city × quarter fixed effects	Yes	Yes	No
Home city × quarter × maturity group fixed effects	No	No	Yes
Standard errors	Parent bank	Parent bank	Parent bank

Notes: Each entry is a coefficient on log bank size among local banks headquartered in the same city. Log bank size is log FY2021 interest-bearing assets in RMB trillion. In Panel A, a comparison cell holds the banks that hold LGFV bonds of the same issuer city in the same quarter with the same branch presence there. The outcome is the value-weighted mean over the bank's bonds in the cell, and banks receive equal total weight within each location. Coupon is net of the mean coupon of held LGFV bonds of the same issue year and maturity group. The three risk outcomes equal 100 for a county-level issuer, an issuer rated below AAA, and an issuer with a default, rating warning, or guarantor event between 2022Q4 and 2025Q4. Columns 1 and 2 of Panel B use other banks' tier-2 and perpetual bonds within the same home city and quarter, with coupons net of issue-year and instrument-type means. Column 3 of Panel B uses other provinces' negotiable certificates of deposit within the same home city, quarter, and maturity group, and its outcome is the percentage issued by city or rural banks (mean 37%). Parentheses contain parent-bank clustered errors. Stars indicate 10%, 5%, and 1% significance. Estimation sample: 2022Q1–2023Q2, and 2024Q3–2025Q2 in column 5 of Panel A.

Table 10
Own-city LGFV bonds and local presence

Dependent variable	Own-city branches (log)	Own-city LGFV bonds (pp of portfolio)	Own-city LGFV bonds (pp of portfolio)	Own-city county-level LGFV bonds (pp of portfolio)	Own-city county-level LGFV bonds (pp of portfolio)
	(1)	(2)	(3)	(4)	(5)
Log bank size	0.217**	−0.006	0.063	−1.934***	−1.804***
	(0.094)	(1.544)	(1.533)	(0.627)	(0.540)
Own-city branches (log)			−0.313		−0.593
			(1.881)		(1.177)
Mean of dependent variable	5.462	9.829	9.829	6.050	6.050
Observations	204	202	202	202	202
Home city × quarter fixed effects	Yes	Yes	Yes	Yes	Yes
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank	Parent bank

Notes: Each column is a separate regression that compares local banks headquartered in the same city in the same quarter. The unit of observation is a bank-quarter. Column 1 is log one plus the number of the bank's branches in its headquarters city, from a bank-branch registry. Columns 2 and 3 are the bank's LGFV bonds issued in its headquarters city as a percentage of its disclosed bonds and private credit, and columns 4 and 5 are the same percentage for bonds of county-level issuers. Columns 3 and 5 add own-city branches, log one plus their number, as a regressor. The sample comprises local banks headquartered in cities with at least two sample banks, and each bank receives equal weight. Log bank size is log FY2021 interest-bearing assets in RMB trillion. Parentheses contain parent-bank clustered errors. Stars indicate 10%, 5%, and 1% significance. Estimation sample: 2022Q1–2023Q2.

Table 11
Home-province LGFV holdings after the program

Specification	Baseline	Own-city split	Net buying	New purchases	City-risk interaction	Pre-period trend
Dependent variable	LGFV holdings / bond book (pp)	LGFV holdings / bond book (pp)	Net holdings change / current book (pp)	Log(1 + new purchases)	Log(1 + new purchases)	LGFV holdings / bond book (pp)
	(1)	(2)	(3)	(4)	(5)	(6)
$\mathbb{1}\{\text{Home buyer}\} \times \mathbb{1}\{\text{Post}\}$	-0.453^{**}	—	-0.127^{**}	-0.158^{**}	0.203^{*}	—
	(0.193)		(0.053)	(0.079)	(0.117)	
$\mathbb{1}\{\text{Own city}\} \times \mathbb{1}\{\text{Post}\}$	—	-1.936^{***}	—	—	—	—
		(0.626)				
$\mathbb{1}\{\text{Rest of home province}\} \times \mathbb{1}\{\text{Post}\}$	—	-0.242	—	—	—	—
		(0.179)				
$\mathbb{1}\{\text{Home buyer}\} \times \mathbb{1}\{\text{Post}\} \times$ city risk	—	—	—	—	-0.149^{***}	—
					(0.051)	
$\mathbb{1}\{\text{Home buyer}\} \times$ trend	—	—	—	—	—	0.003
						(0.051)
Mean of dependent variable	0.784	0.784	-0.052	0.937	0.945	0.816
Observations	73,089	73,089	73,089	73,127	72,196	32,540
Bank and city controls	*Absorbed*	*Absorbed*	*Absorbed*	*Absorbed*	*Absorbed*	*Absorbed*
Buyer × issuer city fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Buyer × quarter fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Issuer city × quarter fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Standard errors	Bank city	Bank city	Bank city	Bank city	Bank city	Bank city

Notes: Estimates use equation (8). The unit of observation is a buying institution (a bank or its wealth-management subsidiary), an LGFV issuer city, and a quarter. An institution that holds bonds of a city in some quarter enters with zero holdings in its other quarters (73,635 cells in 248 cities). Home equals one when a local bank invests in a city of its own province, and Post equals one from 2023Q4. Holdings are the institution's LGFV holdings in the city as a percentage of its total WMP bond holdings in the quarter. Net holdings change includes entries and exits and is winsorized at the 1st and 99th percentiles, and new purchases enter as $\log(1 + \text{RMB million})$. Column 2 separates the bank's own city from the rest of its home province. Column 5 interacts Home × Post with the city's pre-program spread of one-to-five-year LGFV bonds in percentage points, so its Home × Post coefficient applies at a zero spread. Column 6 replaces Post with a linear quarterly trend over 2022Q1–2023Q3. The mean holdings share in home cells is 3.91 before 2023Q4 and 3.43 afterward. Parentheses contain bank-city clustered errors. Stars indicate 10%, 5%, and 1% significance.

Table 12
Home LGFV bonds and private credit by bank type and size

	National banks		Local banks	
Asset class	Pre	Post	Pre	Post
Panel A. Composition of included debt holdings, percent				
Home-province LGFV bonds	0.4	0.3	24.7	17.4
Home-province private credit	2.1	1.9	6.1	13.9
Private credit from other provinces	22.0	26.6	3.2	5.7
Product-quarters	34,595	59,691	36,194	57,516
Panel B. Share of the sector's holdings in local banks' products, percent				
All private credit			17.3	31.3
Panel C. Local banks present in both periods, by pre-program size tercile				
Home-province private credit, smallest tercile			0.72	0.31
Home-province private credit, middle tercile			3.45	4.74
Home-province private credit, largest tercile			7.45	18.26
Home-province LGFV bonds, smallest tercile			25.1	20.8
Home-province LGFV bonds, middle tercile			37.7	25.5
Home-province LGFV bonds, largest tercile			20.7	15.0

Notes: Private credit includes trust loans, leasing, and other identified private credit. The sample is product-quarters with positive included debt holdings over 2022Q1–2025Q3. Pre ends in 2023Q3, and holdings are weighted by value. Panel A reports class shares of each group's included debt holdings, and Panel B reports the fraction of each class's holdings across all sample WMPs that local banks' products hold. Guangfa Bank is a nationwide joint-stock bank, so it and its subsidiary count as national. Panel C keeps local banks observed in both periods, in fixed terciles of FY2021 interest-bearing assets.

Table 13
Bank size groups and private-credit allocation

Dependent variable	Home PC share (pp) (1)	Total PC share (pp) (2)
𝟙{Middle tercile} × 𝟙{Post}	1.536	1.355
	(1.404)	(1.752)
𝟙{Largest tercile} × 𝟙{Post}	8.212***	10.464***
	(2.906)	(3.376)
Log product assets	−1.952**	−2.986***
	(0.862)	(1.091)
𝟙{Restricted redemption}	5.778*	10.572***
	(3.019)	(3.749)
Product risk rating	3.842	4.604
	(2.914)	(3.846)
Bank nonperforming-loan ratio (%)	−1.448	−3.888
	(5.666)	(6.473)
Bank cost–income ratio (%)	−0.891	−1.222*
	(0.581)	(0.725)
Bank net interest margin (%)	9.202	10.547
	(5.931)	(6.540)
Home-city GDP (RMB trillion)	3.357	20.673
	(16.132)	(19.721)
Home-city GDP growth (%)	−1.041*	−1.234*
	(0.611)	(0.669)
Home-city LGFV debt/GDP	24.427	28.158
	(22.700)	(22.963)
Home-city income (RMB 10,000)	17.075*	22.989*
	(10.240)	(11.976)
Log bank assets	−10.152	−25.650
	(15.857)	(19.351)
Observations	85,181	85,181
Bank fixed effects	Yes	Yes
Quarter fixed effects	Yes	Yes
Standard errors	Parent bank	Parent bank

Notes: PC denotes private credit. Each column is a product-quarter regression for local banks. Private-credit shares are percentages of included debt holdings, not total WMP assets. The omitted group is the smallest tercile of FY2021 interest-bearing assets, and the groups follow Table 12. Post begins in 2023Q4. Bank effects absorb fixed group indicators, and quarter effects absorb Post. Regressions use included debt value as weights, so the estimates describe the allocation of investment value. The displayed product, bank, and city controls include contemporaneous log bank assets. Parentheses contain parent-bank clustered errors. Stars indicate 10%, 5%, and 1% significance. Estimation sample: 2022Q1–2025Q3.

Table 14
Bank size and private credit substitution

Dependent variable	Panel A. Refinancing intensity		Panel B. Post-program	
	Home PC share (pp)	Non-home PC share (pp)	Home PC share (pp)	Non-home PC share (pp)
	(1)	(2)	(3)	(4)
Log bank size × refinancing intensity	24.205***	14.531***	—	—
	(8.531)	(4.161)		
Initial home overweight × refinancing intensity	5.021	13.152*	—	—
	(17.269)	(6.792)		
Log bank size × 𝟙{Post}	—	—	5.853***	2.329***
			(1.338)	(0.752)
Initial home overweight × 𝟙{Post}	—	—	7.157	4.363**
			(5.827)	(1.984)
Log product assets	−2.172**	−1.122***	−2.151**	−1.115***
	(0.872)	(0.418)	(0.870)	(0.417)
𝟙{Restricted redemption}	5.538*	4.252**	5.456*	4.218**
	(2.860)	(1.647)	(2.877)	(1.638)
Product risk rating	3.442	0.907	3.563	0.961
	(2.636)	(1.372)	(2.634)	(1.375)
Log bank assets	−30.179	−23.281**	−25.758	−22.166**
	(21.762)	(9.606)	(20.394)	(9.513)
Bank nonperforming-loan ratio	−2.621	−4.356*	−0.336	−3.246
	(4.782)	(2.225)	(4.101)	(2.211)
Bank cost–income ratio	−0.417	−0.407	−0.370	−0.376
	(0.582)	(0.306)	(0.510)	(0.309)
Bank net interest margin	16.147***	1.141	13.616***	0.567
	(5.003)	(2.505)	(4.406)	(2.474)
Home-city GDP (RMB trillion)	51.899***	15.888**	37.848**	11.336
	(17.370)	(7.969)	(16.216)	(7.128)
Home-city GDP growth (%)	−3.070***	−0.469	−1.711*	0.181
	(1.150)	(0.388)	(0.960)	(0.436)
Home-city LGFV debt/GDP	3.490	13.626*	−12.207	4.744
	(23.197)	(7.933)	(20.837)	(6.382)
Home-city income (RMB 10,000)	−43.525*	−17.132**	−21.873	−11.227
	(26.121)	(7.902)	(22.859)	(7.048)
Observations	83,027	83,027	83,027	83,027
Institution fixed effects	Yes	Yes	Yes	Yes
Home province × quarter fixed effects	Yes	Yes	Yes	Yes
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank

Notes: PC denotes private credit. Each column is a product-quarter regression for local-bank WMPs. Size is log FY2021 interest-bearing assets, and initial home overweight is the issuing institution's average portfolio home overweight over 2022Q1–2023Q3. Panel A interacts both with cumulative special-refinancing issuance divided by the home province's 2020 LGFV debt, and Panel B interacts them with Post. Institution effects absorb their levels, and home-province-by-quarter effects absorb refinancing intensity and Post. All eleven controls are displayed, and regressions weight by included debt value. Parentheses contain parent-bank clustered errors. Stars indicate 10%, 5%, and 1% significance. Estimation sample: 2022Q1–2025Q3.

Table 15
Private credit in new products, by bank size

First eligible portfolio	Smallest tercile	Middle tercile	Largest tercile
New products	253	1,877	5,248
Products holding home private credit	2	108	1,118
Products holding any private credit	2	134	1,586
Home private credit share (%)	0.78	6.31	27.46
Non-home private credit share (%)	0.00	2.73	14.74
Total private credit share (%)	0.78	9.04	42.20
Home LGFV bond share (%)	29.36	27.94	11.11
Home private credit share, first disclosure within one quarter (%)	0.59	5.75	27.29
Products in that restriction	234	1,737	4,611

Notes: This table compares the first observed portfolios of products launched after the program by local banks in each size tercile. Launches in 2023Q4–2025Q3 are identified from income-start dates, when products begin accruing returns, and not from first disclosure. Each product contributes its first eligible portfolio, which is the first disclosed portfolio with positive included debt holdings. Shares are holding-value weighted percentages of included debt holdings, and counts are unweighted. Size terciles use fixed FY2021 parent-bank interest-bearing assets and the common local-bank cutoffs. Products of a bank's wealth-management subsidiary are assigned to the parent bank. The final two rows restrict to products whose first portfolio is disclosed in the launch quarter or the next quarter.

Table 16
Bank size and WMP business growth after the program

Dependent variable	Log WMP balance		
Specification	Baseline	Province–time	Size terciles
	(1)	(2)	(3)
Log bank size × 𝟙{Post}	0.152***	0.144***	—
	(0.039)	(0.052)	
𝟙{Middle size tercile} × 𝟙{Post}	—	—	0.184*
			(0.093)
𝟙{Largest size tercile} × 𝟙{Post}	—	—	0.335***
			(0.088)
Wild-bootstrap p	< 0.001	0.008	< 0.001
Observations	571	494	571
Bank fixed effects	Yes	Yes	Yes
Half-year fixed effects	Yes	No	Yes
Home province × half-year fixed effects	No	Yes	No
Standard errors	Parent bank	Parent bank	Parent bank

Notes: Each column is a regression of log WMP balances across local banks and half-years. Balances combine the bank and its subsidiary. Size is log FY2021 interest-bearing assets in RMB trillion, and Post begins at year-end 2023. Column 2 replaces half-year effects with home-province-by-half-year effects, and column 3 uses fixed size terciles, with the smallest omitted. Parentheses contain parent-bank clustered errors. Stars indicate 10%, 5%, and 1% significance. Estimation sample: 2022H1–2025H2.

Table 17
Initial home LGFV bond share, private-credit expansion, and WMP performance

Dependent variable	Panel A. Returns		Panel B. New-product balances	
	Monthly WMP return (bp)		Log new-product balances	
	Size only	Add investments	Size only	Add investments
	(1)	(2)	(3)	(4)
Initial home LGFV bond share × 𝟙{Post}		−10.182**		−1.662***
		(4.186)		(0.528)
Home private-credit expansion × 𝟙{Post}		11.432**		1.363**
		(4.936)		(0.630)
Log FY2021 bank size × 𝟙{Post}	1.808**	0.900	0.292***	0.138
	(0.744)	(0.735)	(0.091)	(0.109)
Initial home private credit share × 𝟙{Post}		3.980		1.547**
		(5.231)		(0.711)
Lagged log bank assets	−1.274	−0.217	−0.390	−0.213
	(6.173)	(5.972)	(0.717)	(0.735)
Lagged nonperforming-loan ratio	−3.147	−3.269	−0.340	−0.504*
	(3.112)	(2.985)	(0.305)	(0.285)
Lagged cost–income ratio	−0.350	−0.383	0.007	−0.010
	(0.323)	(0.325)	(0.030)	(0.032)
Lagged net interest margin	2.816	2.177	0.309	0.252
	(3.359)	(3.293)	(0.290)	(0.289)
Lagged city GDP (RMB trillion)	13.398**	14.211**	1.175	1.697
	(6.205)	(6.319)	(1.071)	(1.025)
Lagged city GDP growth	−0.757**	−0.695*	−0.070	−0.080
	(0.364)	(0.391)	(0.051)	(0.052)
Lagged city LGFV debt/GDP	14.810	15.145	2.979***	3.033***
	(12.724)	(12.202)	(1.121)	(1.035)
Lagged urban income (RMB 10,000)	12.129***	9.822***	1.112**	0.894*
	(4.437)	(3.681)	(0.536)	(0.488)
Observations	1,851	1,851	641	641
Bank fixed effects	Yes	Yes	Yes	Yes
Month fixed effects	Yes	Yes	No	No
Launch-quarter fixed effects	No	No	Yes	Yes
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank

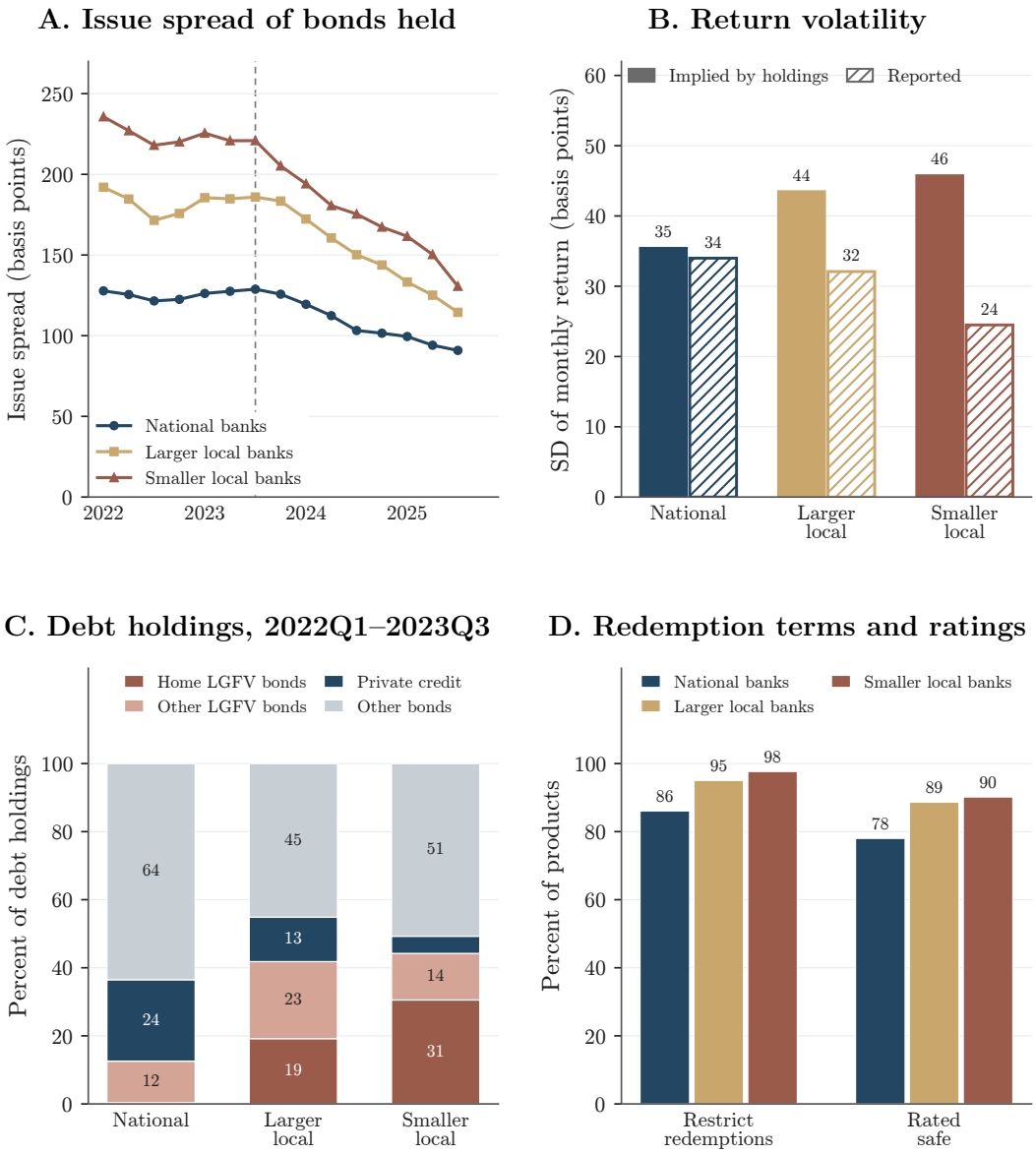
Notes: Each column is a separate regression. Panel A estimates equation (10) for bank-average monthly log returns over July 2022–September 2025, winsorized at the 1st and 99th percentiles. Panel B estimates equation (D.5) over 2022Q2–2025Q2. Its outcome is the log total balance of the products a local bank launched in a quarter, measured at the end of the following quarter, when they are 90 to 183 days old. A product's balance is its size, the median over its disclosed positions of value divided by portfolio weight, and bank-quarters without launches or without disclosed balances do not enter. Both omit 2023Q3, and Post begins in 2023Q4. Columns 1 and 3 include bank size and the lagged controls. Columns 2 and 4 add the initial home LGFV bond and private-credit shares from 2022Q1–2023Q2 holdings and the change in the home private-credit share to 2023Q4–2025Q3, all measured as fractions of included debt holdings. Size is log FY2021 interest-bearing assets. Bank effects absorb the levels, and time effects absorb Post. Parentheses contain parent-bank clustered errors. Stars indicate 10%, 5%, and 1% significance. Internet Appendix D.4 gives the construction.

Table 18
Who funded the borrowers that reliant local banks left

Dependent variable	Holdings by the WMPs of (inverse hyperbolic sine)				Coupon of new bonds (pp)
	Local banks without subsidiary	Local banks with subsidiary	National banks	All banks	
	(1)	(2)	(3)	(4)	(5)
Panel A. WMP holdings of the borrower's LGFV bonds, by holder					
High-reliance banks' share × 𝟙{Post}	−1.894***	0.735***	1.833***	−0.163	−1.166***
	(0.231)	(0.184)	(0.198)	(0.202)	(0.091)
Low-reliance banks' share × 𝟙{Post}	−0.904**	1.150***	1.190***	0.822***	−0.624***
	(0.352)	(0.258)	(0.276)	(0.318)	(0.160)
High-reliance banks' share × 𝟙{Late}	−0.899***	0.837***	1.065***	−0.475**	−0.141***
	(0.208)	(0.176)	(0.184)	(0.186)	(0.046)
Low-reliance banks' share × 𝟙{Late}	−0.764**	0.360	0.718***	−0.218	−0.019
	(0.304)	(0.228)	(0.261)	(0.287)	(0.059)
High = low reliance, Post (p)	0.005	0.054	0.010	0.002	0.001
Observations	17,271	17,271	17,271	17,271	12,541
Panel B. WMP holdings of the borrower's private credit, by holder					
High-reliance banks' share × 𝟙{Post}	−3.962***	0.809**	2.281***	−0.572	—
	(0.571)	(0.368)	(0.397)	(0.672)	
Low-reliance banks' share × 𝟙{Post}	−1.610***	0.286	1.140***	0.091	—
	(0.582)	(0.419)	(0.407)	(0.720)	
High-reliance banks' share × 𝟙{Late}	−1.739***	−0.229	0.360	−0.551	—
	(0.439)	(0.318)	(0.351)	(0.466)	
Low-reliance banks' share × 𝟙{Late}	−1.334*	0.580*	1.992***	1.457**	—
	(0.720)	(0.328)	(0.514)	(0.663)	
High = low reliance, Post (p)	0.004	0.336	0.038	0.495	—
Observations	8,244	8,244	8,244	8,244	—
Borrower fixed effects	Yes	Yes	Yes	Yes	Yes
Borrower province × quarter fixed effects	Yes	Yes	Yes	Yes	No
Borrower province × month fixed effects	No	No	No	No	Yes
Maturity group fixed effects	No	No	No	No	Yes
Standard errors	Borrower	Borrower	Borrower	Borrower	Borrower

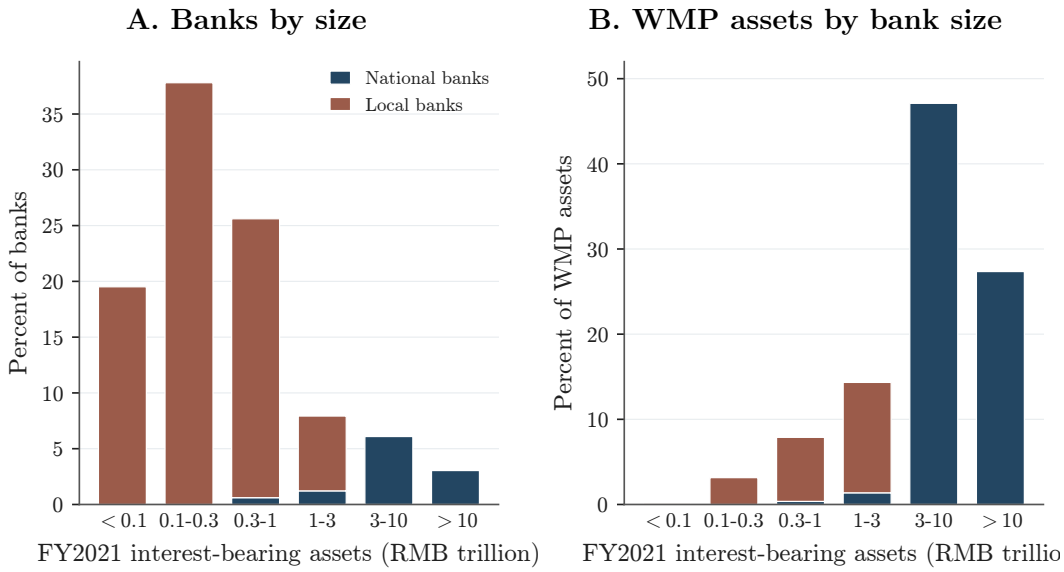
Notes: The unit of observation in columns 1–4 is a borrower-quarter. The outcome is the inverse hyperbolic sine of the borrower's bonds (Panel A) or private credit (Panel B) held by the stated group's WMPs, in RMB 10,000. High- and low-reliance banks' shares are the fractions of the borrower's WMP holdings held in 2022Q4–2023Q1 by local banks without a wealth-management subsidiary whose initial home LGFV bond share is above or below the median. Post begins in 2023Q4 and Late in 2024Q3, after the June 2024 guidance. Column 5 uses new LGFV bonds issued by the same borrowers and controls for log maturity and maturity-group fixed effects. Parentheses contain borrower-clustered errors. Stars indicate 10%, 5%, and 1% significance. Estimation sample: 2023Q2–2025Q2.

Figure 1
Holdings, return volatility, and product terms by bank group



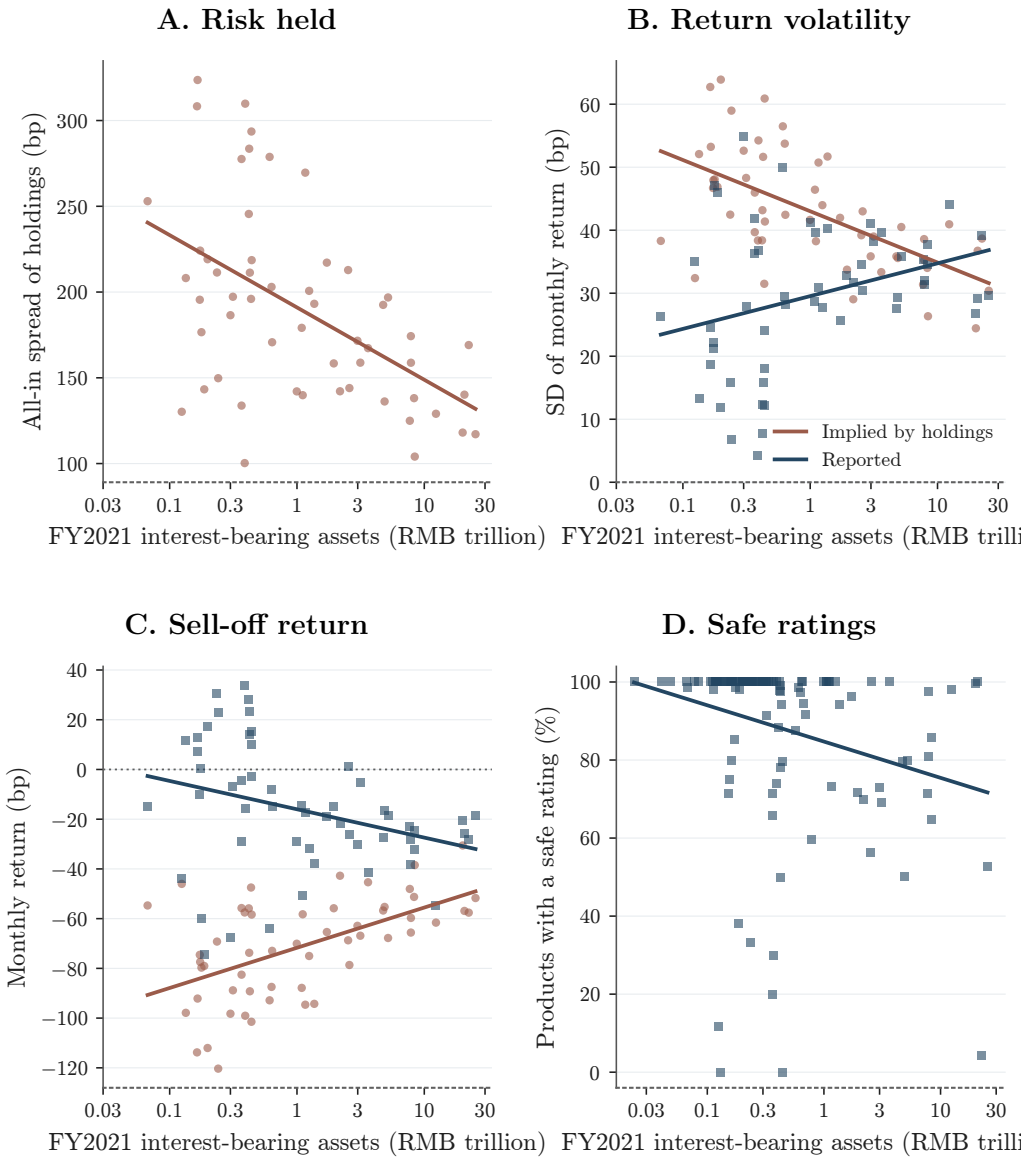
Notes: Smaller local banks are the banks at or below the median FY2021 parent-bank interest-bearing assets of the sample of Table 7, which is RMB 0.44 trillion. Larger local banks are the local banks above it. Panel A shows the value-weighted issue spread of the bonds held by each group's WMPs by quarter. The issue spread is the coupon minus the same-maturity Treasury yield at issue. Unlike the all-in spread of Table 3, it covers bonds only and leaves out central-government, central-bank, and policy-bank bonds. The dashed line marks the 2023Q3 announcement of the debt-resolution program. Panel B shows the implied and the reported standard deviation of monthly returns between April 2022 and June 2023. It uses the products of Table 7 and averages products within a bank and then banks. Implied returns mark every disclosed asset each month to a market series, and reported returns are monthly changes in log distribution-adjusted NAV. Each series is winsorized at the 1st and 99th percentiles of all its monthly values. Panel C shows the composition by value of the bonds and private credit with known borrower locations that each group's WMPs held over 2022Q1–2023Q3, as in Section 4.1. Panel D uses the fixed-income products that reported a monthly return between April 2022 and June 2023 and averages over banks. It shows the percentage of a bank's products that are closed-end or periodic-open, which restrict redemptions, and the percentage with a safe rating of 1 or 2 on the five-level product risk scale. Internet Appendix D.3 gives the construction of implied returns.

Figure 2
Bank size and WMP assets in the sample



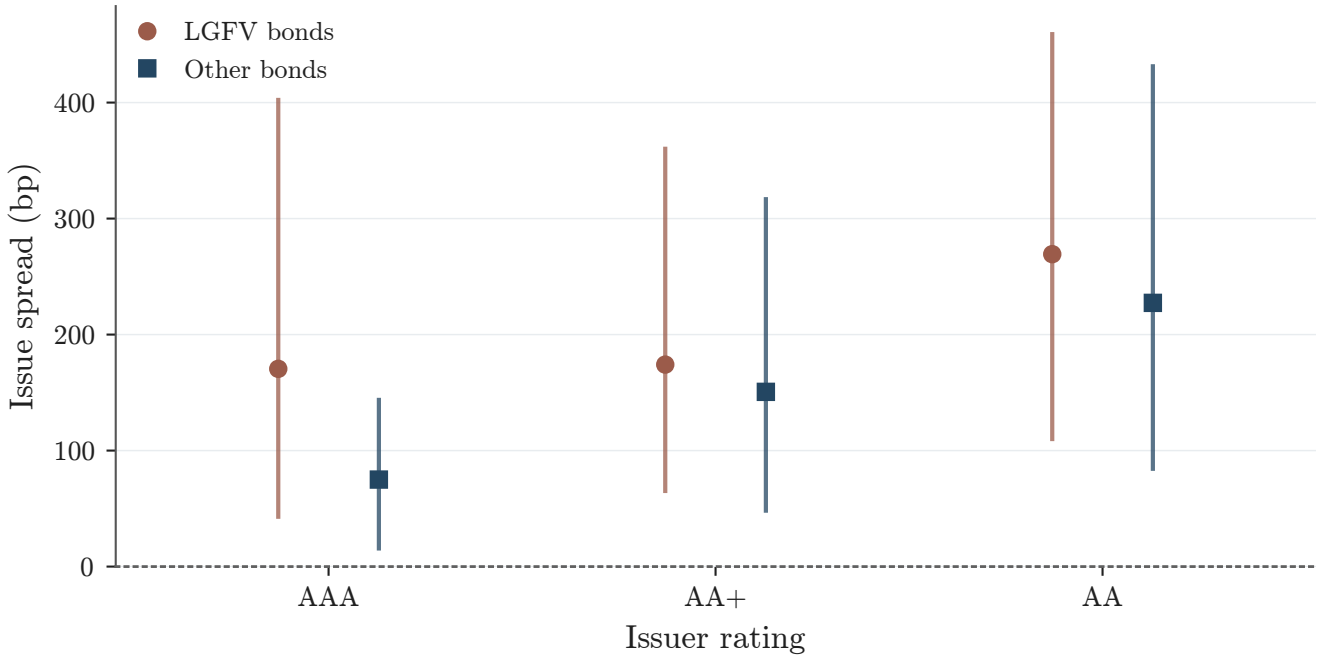
Notes: Both panels sort banks by FY2021 parent-bank interest-bearing assets and distinguish national from local banks. Panel A shows the percentage of banks in each asset range, and Panel B shows the percentage of sample WMP assets. A bank's WMP assets are the mean quarterly net assets of its products over 2022Q1–2023Q3, as implied by their disclosed holdings. Banks without FY2021 balance sheets are left out.

Figure 3
Risk held and risk reported, by bank size



Notes: Each panel plots one point per national or local bank against FY2021 parent-bank interest-bearing assets on a log scale, with linear fits. Panels A to C use the banks with at least five fixed-income products in the sample of Table 7. Panel D uses every bank with fixed-income products that reported a monthly return between April 2022 and June 2023, which is the sample of column 6 of Panel B of Table 7. Panel A shows the all-in spread of the bank's holdings, defined in Table 3. Panel B shows the standard deviation of monthly returns between April 2022 and June 2023 averaged over the bank's products, and Panel C shows the mean monthly return in November and December 2022. Panels B and C show both the value implied by holdings and the value reported. Panel D shows the percentage of the bank's fixed-income products with a safe rating of 1 or 2 on the five-level product risk scale. In bank-level fits to the plotted points, the all-in spread falls by 18.3 basis points per log point of these assets (standard error 3.7), and the safe-rating share falls by 4.0 percentage points (standard error 1.6). Implied returns mark every disclosed asset each month to a market series and add the product's mean difference between reported returns and price changes in the other months, which reflects interest income net of fees. Reported returns are monthly changes in log distribution-adjusted NAV, and each series is winsorized at the 1st and 99th percentiles of all its monthly values. Internet Appendix D.3 gives the construction.

Figure 4
Issuer credit ratings and issue spreads



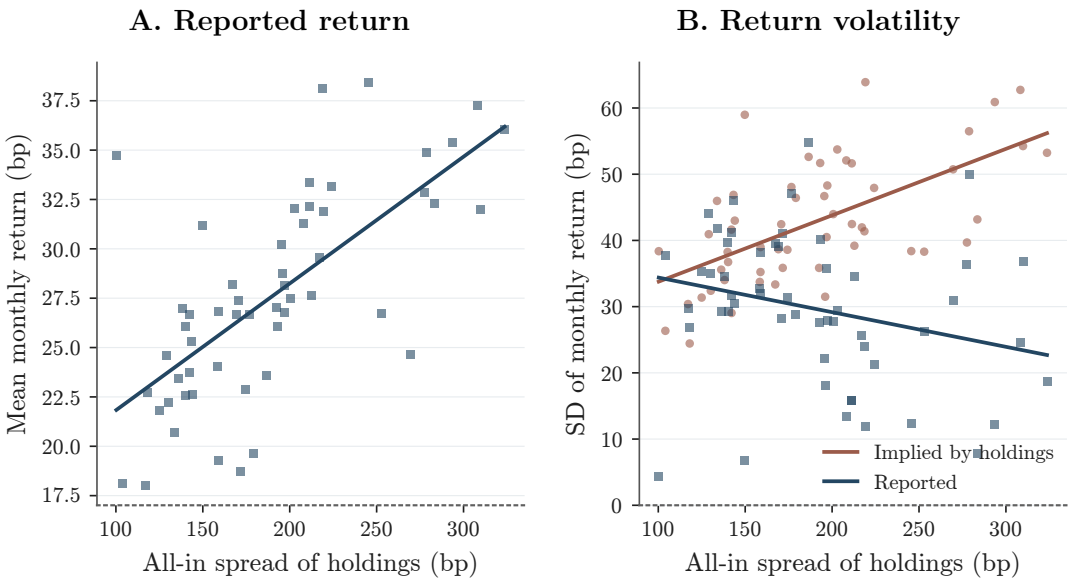
Notes: The figure shows the mean issue spread of LGFV bonds and of other bonds issued between 2021Q1 and 2023Q2 by issuer rating, together with the range from the 10th to the 90th percentile. The issue spread is the coupon minus the Treasury yield of the same maturity at issue. The sample comprises medium-term notes, short-term financing bills, and corporate bonds with an issuer rating of AAA, AA+, or AA. Internet Appendix D.3 gives the construction.

Figure 5
Returns implied by holdings, returns reported, and investor flows



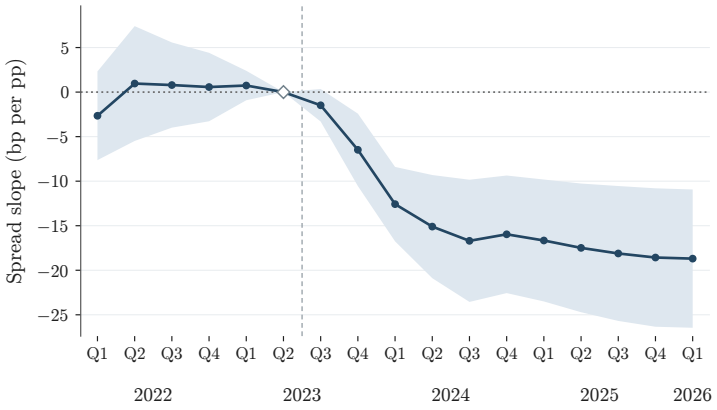
Notes: The panels plot the products in the sample of Table 7 between April 2022 and June 2023, separately for smaller and larger banks. Smaller banks are those at or below the median size. They are all local banks, and larger banks include all national banks. The panels average products within a bank and then banks. Panels A and B show monthly returns. Panels C and D show quarterly net flows for 2022Q2 to 2023Q2. Net flow is asset growth net of investment gains as a percentage of beginning assets, as in Table 1. Each quarterly flow is plotted at the middle month of its quarter and averages a bank's products weighted by beginning assets. Implied returns mark every disclosed asset each month to a market series. Each product's implied return adds to the month's price change the product's mean difference between reported returns and price changes in the other months. This difference reflects interest income net of fees, so the two series agree on average outside November and December 2022. Reported returns are monthly changes in log distribution-adjusted NAV. Reported and implied returns are each winsorized at the 1st and 99th percentiles of all their monthly values. Internet Appendix D.3 gives the construction.

Figure 6
What savers see and the risk held



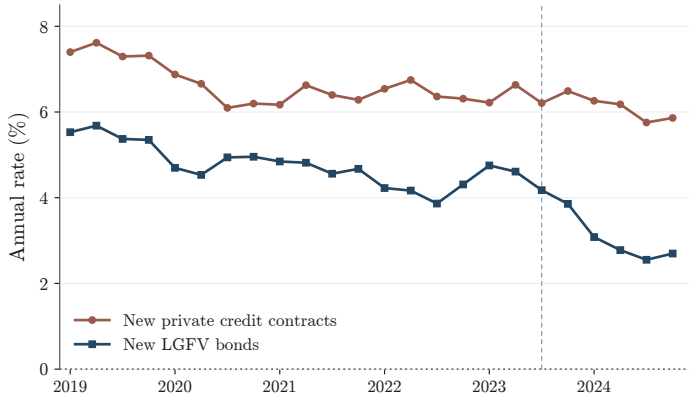
Notes: Each panel plots one point per national or local bank with at least five fixed-income products in the sample of Table 7 against the mean all-in spread of its products' holdings, with linear fits. The all-in spread is defined in Table 3. Panel A shows the mean reported monthly return between April 2022 and June 2023, and Panel B shows the standard deviation of monthly returns as implied by holdings and as reported. Per 100 basis points of all-in spread, the bank-level slope of the mean return is 6.4 basis points a month (standard error 1.0). In November and December 2022 the slope is −19 for the implied return and +17 for the reported return. Internet Appendix D.3 gives the construction.

Figure 7
Provincial LGFV spreads by special refinancing relative to GDP



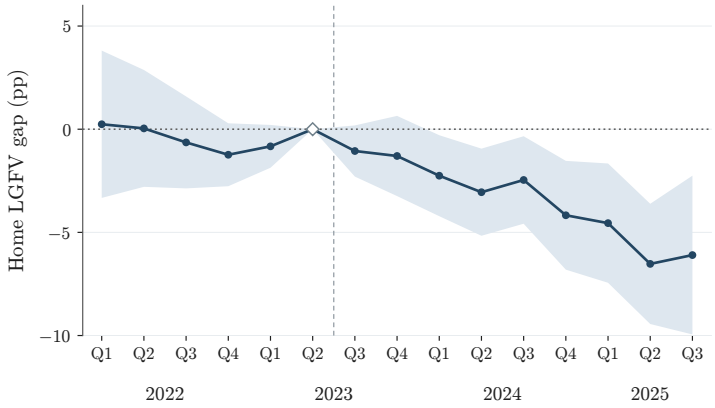
Notes: This event study plots quarter-specific coefficients β_k. Provincial LGFV secondary-market spreads in basis points are regressed on special refinancing relative to GDP interacted with quarter indicators. The regression includes province and quarter fixed effects, and 2023Q2 is the omitted quarter. Special refinancing relative to GDP is cumulative special-refinancing issuance at the end of the series as a percentage of 2020 provincial GDP. Coefficients are basis points per percentage point of special refinancing relative to GDP, without standardization. Spreads and special refinancing are winsorized at the 1st and 99th percentiles, and the sample covers 31 provinces over 2022Q1–2026Q1. Shading shows pointwise 95% confidence intervals clustered by province, the hollow diamond marks the omitted quarter, and the dashed line separates 2023Q2 and 2023Q3.

Figure 8
Contractual rates on new LGFV financing



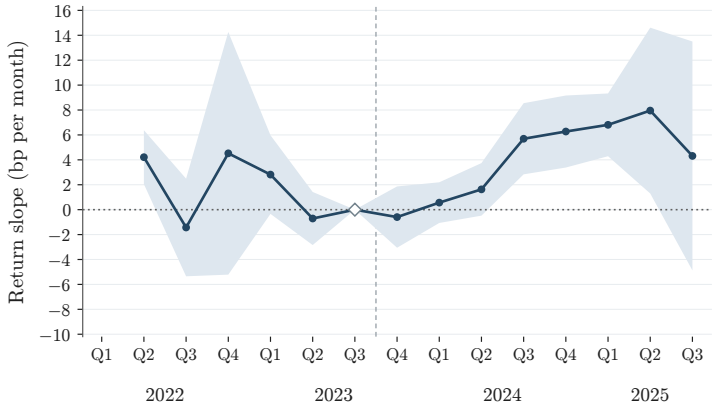
Notes: The series are annual contractual yields in percent on new private credit contracts and new LGFV bond issues, by quarter over 2019Q1–2024Q4. Each series uses the financing records of LGFV issuers, with duplicate records removed, and weights each observation with a reported rate by its original financing amount. Observations without a known amount are left out. Figure D1 compares the same borrowers. The dashed vertical line marks the 2023Q3 announcement. Incomplete 2025 observations are excluded.

Figure 9
Home-province gap in local banks' LGFV holdings by quarter



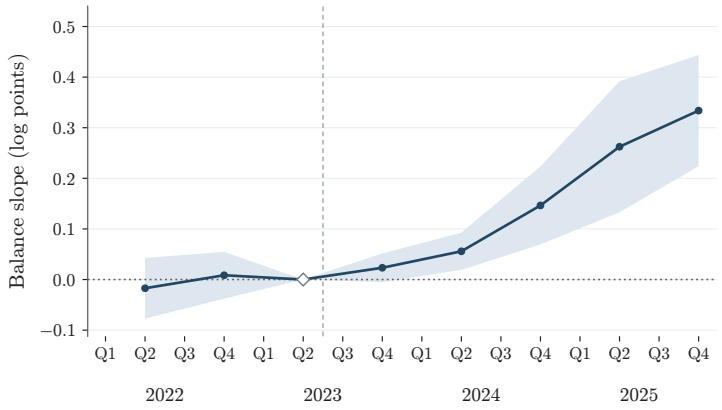
Notes: This event study plots quarter-specific home-bond allocation coefficients η_k. The outcome is a buying institution's LGFV holdings in a province as a percentage of its total bond holdings, winsorized at the 1st and 99th percentiles. It is regressed on Home interacted with quarter indicators. Buyer-by-province, buyer-by-quarter, and destination-province-by-quarter fixed effects enter, and 2023Q2 is omitted. The sample covers 2022Q1–2025Q3. The observation is a buyer, province, and quarter, so the coefficients measure changes in the home gap for an entire province rather than a single city as in Table 11. Home equals one for a local bank's own province, so the comparison includes national banks and local banks' holdings elsewhere. Shading shows pointwise 95% confidence intervals clustered by bank city, the hollow diamond marks the omitted quarter, and the dashed line separates 2023Q2 and 2023Q3.

Figure 10
Bank size and product returns by quarter



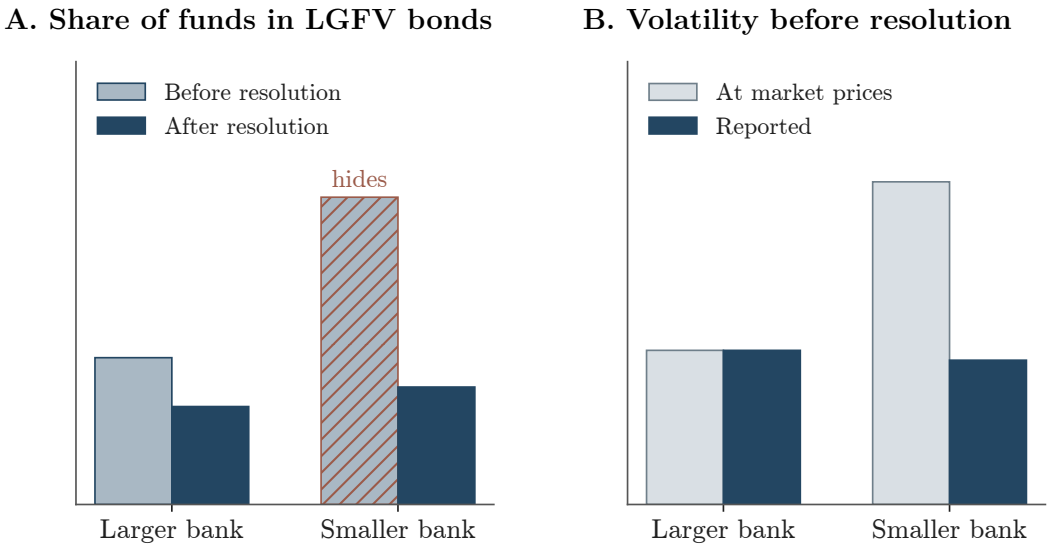
Notes: This event study replaces both refinancing interactions in equation (D.6) with log bank size interacted with quarter indicators. The outcome is monthly log adjusted-NAV return (bp) for local-bank products. Each coefficient measures the change in the size–return slope relative to 2023Q3, per log point of FY2021 parent-bank interest-bearing assets in RMB trillion, without standardization. The regression includes the eleven time-varying controls, institution effects, and bank-home-province-by-month effects, but no initial-home-overweight interactions. Returns are winsorized at the 1st and 99th percentiles of all monthly returns. The sample is 2022Q2–2025Q3 (212,016 NAV-series-months). Shading shows pointwise 95% confidence intervals clustered by parent bank, the hollow diamond marks the omitted quarter, and the dashed line separates 2023Q3 and 2023Q4.

Figure 11
Bank size and total WMP balances



Notes: This event study replaces the size–Post interaction in equation (9) with log bank size interacted with half-year reporting-date indicators. The quarterly axis locates actual midyear and year-end reports at Q2 and Q4. No Q1 or Q3 balances are imputed. The log-balance regression includes bank and half-year fixed effects and no additional controls, for local banks over 2022H1–2025H2 (571 bank-half-years). Each coefficient measures the change in the size–balance slope relative to the June 2023 report (2023Q2), per log point of FY2021 interest-bearing assets in RMB trillion, without standardization. Balances combine a bank's own products and those of its wealth-management subsidiary. Shading shows pointwise 95% confidence intervals clustered by parent bank, the hollow diamond marks the omitted report, and the dashed line separates 2023Q2 and 2023Q3. Source: institution-level WMP statistics from a private data vendor, corrected with bank disclosures.

Figure 12
The model's equilibrium before and after debt resolution



Notes: This figure shows the illustrative equilibrium of the model in Internet Appendix B.6 (Table B1). Panel A plots each bank's share of funds in LGFV bonds before and after debt resolution, and the hatched bar marks the bank that hides part of its bonds' variance. Panel B plots, before resolution, the standard deviation of each bank's portfolio return at market prices and the standard deviation that its products report. The smaller bank holds more bonds and reports less volatility than the larger bank, although its holdings are about twice as volatile. After resolution it stops hiding and cuts its bonds by more than the larger bank.

Internet Appendix

Reaching for Yield in the Dark
Bank Competition for Household Savings in China

The appendix contains supporting exhibits (Internet Appendix A), the model solution and proofs (Internet Appendices B and C), and specifications and supplementary results (Internet Appendix D).

Internet Appendix A
Supporting Evidence

A.1. Data and portfolio denominators

Table A1 defines the variables, Table A2 reports sample coverage, and Table A3 summarizes products, holdings, and issuers. Figure A1 compares WMPs with other Chinese asset-management products and shows their 2025 asset mix.

Figure A2 shows the estimated scale of the WMP–LGFV bond link, and Internet Appendix D.2 explains these industry estimates and reconciles them with the portfolio shares in Section 4.1.

A.2. Supporting estimates

Table A4 reports the home allocations by redemption terms discussed in Section 4.1, and Table A5 relates relative performance to subsequent investor flows (Section 3). Table A6 compares LGFV issuance yields with those of other issuers. Across 20,172 unique RMB fixed-rate bonds issued at par in 2022Q1–2023Q3, LGFV yields exceed other nonfinancial and financial issuers' yields by 122 and 149 basis points. Controlling for issue month, maturity, issue size, and call provisions leaves gaps of 91 and 108 basis points, and adding ratings gives 88 and 110.

Table A1
Variable definitions

Variable	Definition
$\mathbb{1}\{\text{Local bank}\}_b$	One for banks other than state-owned and nationwide joint-stock banks.
Home bias within class$^n_{iq}$	Home holdings / all holdings within class n, minus the home province's share of nationwide class-n holdings. Requires positive class holdings (equation (3)).
Portfolio home overweight$^n_{iq}$	Home class-n holdings / all included debt holdings, minus nationwide investment in class n in the home province / nationwide included debt holdings. Includes products holding none of class n. Without n, it sums across classes.
Initial home overweight$_a$	Product-assets-weighted pre-program mean of portfolio home overweight for issuing institution a, 2022Q1–2023Q3.
Log bank size$_b$	Natural log of FY2021 parent-bank interest-bearing assets (RMB trillion), without standardization.
Initial home LGFV bond share$_b$	Home LGFV bonds / included debt holdings, holding-value weighted over 2022Q1–2023Q2, without subtracting market weights.
Home LGFV credit share$_{iq}$	Home-province LGFV bonds and private credit to home-province LGFV borrowers / disclosed holdings (all reported positions, including equity and funds), in percent.
Disclosed bonds and private credit	All disclosed bond and private credit positions, including government and policy-bank bonds, which included debt holdings exclude.
$\mathbb{1}\{\text{Restricted redemption}\}_i$	One for products without daily redemption.
Monthly return	Log return computed from month-end adjusted NAVs in consecutive months, in basis points.
Reported return volatility	Log standard deviation of the product's monthly reported returns over the twelve months before the outcome quarter.
Net flow$_{iq}$	Quarterly asset growth net of investment gains, as a percentage of beginning assets.
$\mathbb{1}\{\text{Home buyer}\}_{ac}$	Equals one when issuing institution a is a local bank or its subsidiary and invests in issuer city c of the parent bank's headquarters province.
Refinancing$^{GDP}_{pq}$	Cumulative special-refinancing issuance / 2020 provincial GDP (percentage points). Final refinancing$^{GDP}_p$ is its value at the last observed date.
Refinancing$^{debt}_{pq}$	Cumulative special-refinancing issuance / 2020 provincial LGFV debt (fraction), the refinancing intensity of the allocation and return tests.
$\mathbb{1}\{\text{Post}\}$	Equals one from 2023Q4 in quarterly data and from October 2023 in monthly data.
Log WMP balances$_{bs}$	Log outstanding WMP balance of bank b at half-year end s, combining its own and its subsidiary's products.

Notes: Holdings cover 2022Q1–2025Q3. Included debt comprises bonds and identified private credit with known borrower locations, excluding central-government, central-bank, and policy-bank obligations. Issuer credit events include defaults, rating warnings, and guarantor events, not necessarily default on the held bond. Classes n are LGFV bonds, private credit (PC), and other bonds.

Table A2
Sample coverage

Panel A. Overall coverage

Measure	Count
Source holding records	2,288,894
Holding records with a usable product identifier	2,279,252
WMP products	60,396
Product-quarter observations	253,484
Bank headquarters provinces	30
Quarters	15

Panel B. Annual coverage

Year	Source holding records	WMP products
2022	421,072	19,514
2023	579,703	26,021
2024	708,907	30,512
2025 (Q1–Q3)	579,212	29,527

Panel C. Bank categories

Bank group	WMP products	Product-quarters
National banks	27,799	131,535
Large state-owned	9,571	49,593
Nationwide joint-stock	18,228	81,942
Local banks	32,597	121,949
City and provincial commercial	23,817	91,937
Rural banks	8,780	30,012
All banks	60,396	253,484

Notes: Coverage is 2022Q1–2025Q3. A holding record is a product-asset-quarter observation. Source-record counts include 9,642 records with blank or "NA" product registration codes. Usable-record, product, and product-quarter counts exclude these codes. A parent-bank crosswalk groups wealth-management subsidiaries and alternative bank names with their parent banks. Panel B counts products once per year, so annual counts need not sum to full-period counts. Panel C uses the source bank-type codes and assigns subsidiary products to the parent bank's category. The source does not separate city from provincial commercial banks. Rural banks include rural commercial and cooperative banks. Guangfa Bank's wealth-management subsidiary is classified with nationwide joint-stock banks. Estimation samples impose the restrictions stated in each regression table.

Table A3
Descriptive statistics

Variable	Observations	Mean	SD	Median	p90
Panel A. Holding-observation characteristics					
Position weight (%)	2,286,025	7.12	12.03	4.04	11.46
Holding value (RMB million)	2,278,335	59.22	345.74	10.39	113.89
Panel B. Bond characteristics					
Issue yield (%)	1,162,002	4.09	1.19	4.00	5.58
Issue maturity (years)	1,162,491	5.09	3.06	5.00	10.00
Panel C. Product-quarter characteristics in the NAV sample					
Log product assets	153,563	19.41	1.64	19.38	21.51
Product risk rating	153,563	2.15	0.44	2.00	3.00
Restricted redemption (0/1)	153,563	0.88	0.33	1.00	1.00
Panel D. Pre-implementation issuers' FY2021 assets (RMB billion)					
LGFV issuers	1,939	42.79	48.09	28.41	88.39
Other nonfinancial issuers	2,062	134.48	548.95	35.32	243.34
Financial issuers	570	525.79	2,765.50	52.90	560.29

Notes: Panel A describes holding records and Panel B bond holding records with the characteristic available. Panel C counts each product-quarter in the return sample once. Log product assets is $\log(1 + \text{inferred assets in RMB})$, product risk is the numerical rating, and p90 is the 90th percentile. The cross-sectional SD of products' average monthly returns is 10.5 bp. Among local-bank product-quarters before 2023Q4, mean home and non-home LGFV bond shares of included debt are 0.263 (0.722 at the 90th percentile for the home share) and 0.180. Mean issue rates are 5.62% for home LGFV holdings and 4.93% for other-province holdings, and the mean within-product gap is 0.33 pp. Panel D separately describes firms issuing RMB debt in a private data vendor's issuance file over 2022Q1–2023Q3, not only WMP-held issues. Each issuer receives equal weight and appears once, matched by normalized exact name to reported year-end 2021 total assets. No future balance-sheet values are substituted. The LGFV roster identifies the first group, and source issuer types identify financial institutions and other enterprises. Central and local governments, policy banks, and unclassified issuers are excluded. Assets are observed for 1,939 of 1,964 LGFV issuers (98.7%), 2,062 of 2,167 other nonfinancial issuers (95.2%), and 570 of 723 financial issuers (78.8%).

Table A4
Portfolio home overweight and redemption terms

Asset class	All included debt	LGFV	PC	Other bonds
Dependent variable	Portfolio home overweight (fraction) (1)	Portfolio home overweight (fraction) (2)	Portfolio home overweight (fraction) (3)	Portfolio home overweight (fraction) (4)
$\mathbb{1}\{\text{Local bank}\}$	−0.063	0.009	−0.021	−0.050
	(0.060)	(0.035)	(0.026)	(0.034)
Log bank assets	−0.016	−0.004	0.018	−0.030**
	(0.015)	(0.010)	(0.012)	(0.012)
$\mathbb{1}\{\text{Local bank}\} \times$ restricted redemption	0.200***	0.077***	0.078**	0.045**
	(0.056)	(0.030)	(0.036)	(0.023)
Log product assets	−0.001	0.001	0.000	−0.002
	(0.004)	(0.003)	(0.002)	(0.002)
$\mathbb{1}\{\text{Restricted redemption}\}$	−0.048***	−0.014**	0.010	−0.045***
	(0.009)	(0.007)	(0.008)	(0.008)
Product risk rating	0.015	0.000	0.012	0.003
	(0.014)	(0.008)	(0.012)	(0.008)
Bank nonperforming-loan ratio (%)	−0.074	−0.005	−0.044**	−0.022
	(0.055)	(0.041)	(0.020)	(0.027)
Bank cost–income ratio (%)	−0.001	−0.001	−0.000	−0.000
	(0.001)	(0.001)	(0.001)	(0.001)
Bank net interest margin (%)	0.040	0.026	0.007	0.004
	(0.060)	(0.045)	(0.027)	(0.031)
Home-city GDP (RMB trillion)	0.003	0.023	−0.023**	0.003
	(0.025)	(0.017)	(0.009)	(0.012)
Home-city GDP growth (%)	0.013	0.019***	−0.003	−0.003
	(0.010)	(0.007)	(0.005)	(0.006)
Home-city LGFV debt/GDP	0.189***	0.217***	0.019	−0.049
	(0.069)	(0.048)	(0.036)	(0.058)
Home-city income (RMB 10,000)	−0.048**	−0.044***	0.007	−0.010
	(0.023)	(0.016)	(0.009)	(0.012)
Mean of dependent variable	0.191	0.092	0.031	0.069
Observations	169,997	169,458	168,327	169,997
Quarter fixed effects	Yes	Yes	Yes	Yes
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank

Notes: PC denotes private credit. Estimates use equation (4). The unit of observation is a product-quarter. The dependent variable is the product's home-province holdings of the asset class divided by all its included debt holdings, minus nationwide holdings of the class in the home province divided by nationwide included debt holdings, as a fraction. Restricted redemption equals one for products without daily redemption, so the local-bank coefficient compares local and national banks among daily-open products and the interaction gives the additional difference among other products. Periodic-open and closed-end products account for 80.2% of local banks' product assets and 47.3% of national banks'. Parentheses contain parent-bank clustered errors. Stars indicate 10%, 5%, and 1% significance. Estimation sample: 2022Q1–2025Q3.

Table A5
Relative performance and subsequent investor flows

Dependent variable	Net flow (% initial assets) (1)	Net flow (% initial assets) (2)	Symmetric net flow (pp) (3)	Positive-flow probability (pp) (4)
Prior-quarter return quintile (bottom quintile omitted)				
$\mathbb{1}\{\text{Return quintile} = 2\}$	1.790***	1.793***	1.461***	1.860***
	(0.605)	(0.592)	(0.515)	(0.591)
$\mathbb{1}\{\text{Return quintile} = 3\}$	2.873***	2.860***	2.851***	4.600***
	(0.789)	(0.770)	(0.565)	(0.847)
$\mathbb{1}\{\text{Return quintile} = 4\}$	5.185***	5.396***	4.833***	6.102***
	(1.152)	(1.168)	(0.780)	(1.017)
$\mathbb{1}\{\text{Return quintile} = 5\}$	12.008***	12.604***	8.785***	8.036***
	(2.671)	(2.730)	(1.527)	(1.627)
Log beginning product assets	−7.375***	−7.615***	−4.594***	−3.783***
	(0.982)	(0.976)	(0.567)	(0.333)
Product risk rating	Absorbed	Absorbed	Absorbed	Absorbed
Restricted redemption	Absorbed	Absorbed	Absorbed	Absorbed
Log bank assets	3.970***	4.332***	3.308***	3.263**
	(1.104)	(1.304)	(0.826)	(1.406)
Bank nonperforming-loan ratio (%)	−1.279	−0.522	−0.827	1.228
	(1.958)	(3.322)	(1.543)	(2.371)
Bank cost–income ratio (%)	−0.136**	−0.117	−0.126**	−0.070
	(0.066)	(0.077)	(0.055)	(0.058)
Bank net interest margin (%)	0.898	−0.122	−0.787	2.815
	(1.903)	(1.725)	(1.248)	(2.508)
Home-city GDP (RMB trillion)	−0.751	0.527	−0.669	2.290
	(0.726)	(2.075)	(0.604)	(1.407)
Home-city GDP growth (%)	−0.217	−0.278	−0.069	0.106
	(0.420)	(0.605)	(0.330)	(0.660)
Home-city LGFV debt/GDP	−1.516	−2.940	1.622	4.631
	(2.830)	(3.758)	(1.952)	(5.471)
Home-city income (RMB 10,000)	0.003	−3.326*	−0.306	−2.861**
	(0.645)	(1.914)	(0.478)	(1.326)
$\mathbb{1}\{\text{Local bank}\}$	4.393*	3.779	5.296***	14.732***
	(2.600)	(3.203)	(1.993)	(3.860)
Mean of dependent variable	1.566	1.562	−4.763	42.946
Top-quintile wild-bootstrap p	< 0.001	—	< 0.001	< 0.001
Observations	129,577	129,570	129,577	129,577
Category × quarter fixed effects	Yes	Yes	Yes	Yes
Home province × quarter fixed effects	No	Yes	No	No
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank

Notes: Each column is one regression of next-quarter flows on prior-quarter return quintiles within risk and redemption categories, with the bottom quintile omitted. Controls are displayed and dated before flows. Category-by-outcome-quarter effects enter, and column 2 adds home-province-by-quarter effects. Parentheses contain parent-bank clustered errors, and the wild-bootstrap p is for the top-quintile coefficient. Net flows in columns 1 and 2 are capped at the 1st and 99th percentiles. Symmetric flows divide the same asset growth net of investment gains by the average of ending assets and beginning assets grown by the return, so they lie between −200 and 200 and are not capped. Positive flow is 100 times an indicator of net inflow. Stars indicate 10%, 5%, and 1% significance. Estimation sample: 2022Q3–2025Q3.

Table A6
LGFV bond issuance yields relative to other issuers

Panel A. Fixed-rate bonds issued at par, 2022Q1–2023Q3

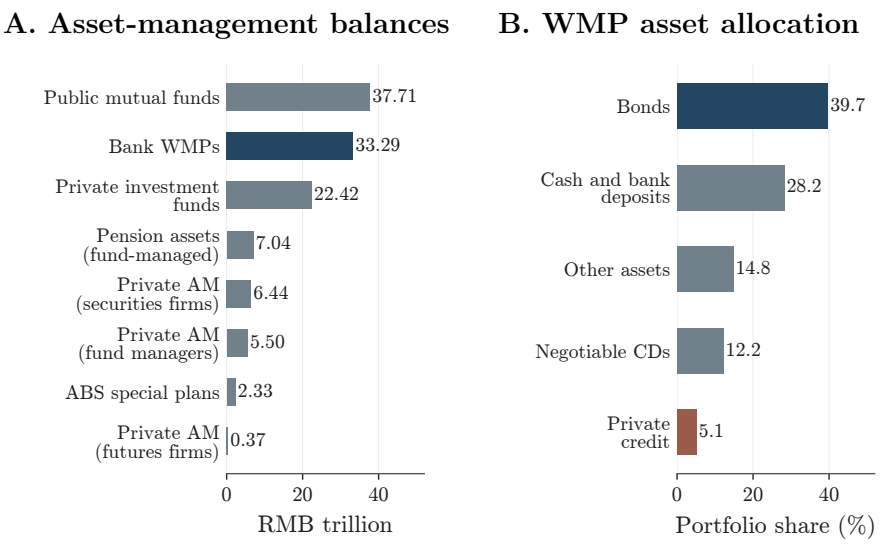
	LGFV	Other nonfinancial	Financial
Mean issuance yield (%)	4.26	3.03	2.77
Median issuance yield (%)	3.92	2.79	2.75
Mean maturity (years)	2.28	1.79	2.12
Bond issues	8,044	10,832	1,296
Issuers	1,581	1,523	205

Panel B. Yield differences and bond characteristics

Dependent variable	Issuance yield (pp) (1)	Issuance yield (pp) (2)	Issuance yield (pp) (3)
$\mathbb{1}\{\text{LGFV issuer}\}$	1.224***	0.910***	0.875***
	(0.069)	(0.089)	(0.093)
$\mathbb{1}\{\text{Financial issuer}\}$	−0.265***	−0.173***	−0.226***
	(0.054)	(0.057)	(0.063)
Maturity (years)	—	0.282***	0.277***
		(0.023)	(0.028)
Maturity squared	—	−0.009***	−0.009***
		(0.001)	(0.001)
Log issue amount	—	−0.280***	−0.264***
		(0.038)	(0.036)
$\mathbb{1}\{\text{Callable}\}$	—	0.378***	0.398***
		(0.099)	(0.099)
LGFV minus financial-issuer yield	1.489***	1.084***	1.101***
	(0.061)	(0.099)	(0.117)
Observations	20,172	20,172	20,172
Month fixed effects	No	Yes	Yes
Reported rating fixed effects	No	No	Yes
Standard errors	Issuer	Issuer	Issuer

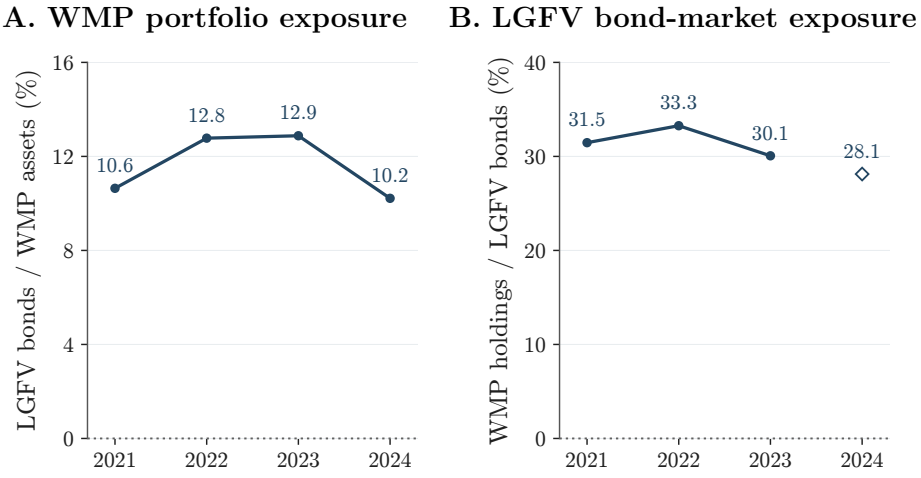
Notes: Panel A describes unique RMB fixed-rate bonds issued at par in 2022Q1–2023Q3. Panel B regresses issuance yields (pp) on issuer-type indicators, with other nonfinancial issuers omitted. All columns use the same sample, and later columns add the displayed issue characteristics, issue-month effects, and reported ratings (including a category for missing ratings). Unlike Panel D of Table A3, the sample does not require observed issuer assets. Certificates of deposit and convertible, exchangeable, structured, perpetual, subordinated, floating-rate, and reset-rate bonds are excluded. There is no winsorization. Parentheses contain issuer-clustered errors, and stars indicate 10%, 5%, and 1% significance. For these fixed-rate, at-par issues, the stated rate equals the contractual yield at issuance. Estimation sample: 2022Q1–2023Q3.

Figure A1
Chinese asset-management products and WMP asset allocation, 2025



Notes: Panel A reports assets under management of Chinese asset-management products at the end of 2025, in RMB trillion, and Panel B reports the asset allocation of bank WMPs, in percent. Balances overlap through cross-investment. Private AM denotes private asset-management plans. Sources: Asset Management Association of China and annual WMP market reports.

Figure A2
The scale of the WMP–LGFV bond link, 2021–2024



Notes: Both panels use estimated holdings $\widehat{H}_t = B_t q_t$ from Internet Appendix D.2, which multiply industry WMP bond assets B_t by the LGFV share q_t of bonds in disclosed year-end holdings. Panel A divides by official WMP net assets under management, and Panel B divides by the published LGFV bond balance. The open diamond marks the change in the source of the LGFV bond balance in 2024. Sources: holdings disclosures from a private data vendor, annual WMP market reports, Lianhe Ratings (2024), and Zhang (2025).

Internet Appendix B
Model Derivations and Numerical Solution

This appendix completes the model of Section 6 (Subsection B.1), derives its results (Subsections B.2–B.5), and solves the full payoff numerically (Subsection B.6). Internet Appendix C gives proofs. The propositions use a benchmark that is first order in the reputation losses, holds common marginal-saver weights under both reporting choices, and considers interior bond shares. Some sorting and investor-share results also treat each bank's share as small. Each proposition states which of these it uses. The numerical solution uses none of them.

B.1 The model in full

This subsection states the assumptions that Section 6 leaves implicit. Equation numbers without the prefix B refer to the main text.

Primitives. Savers' risk aversion γ has a nondegenerate distribution G on $[\underline{\gamma}, \overline{\gamma}]$, where $0 < \underline{\gamma} < \overline{\gamma} < \infty$. We assume $\beta > 0$, $f > 0$, $\mu_B > 0$, $\sigma_B^2 > 0$, $0 < c < 1$, $\theta, \rho \in (0, 1)$, and $\kappa, \delta > 0$, so $\varphi > 0$ and $\psi > 0$. The safe asset is a normalization. Each saver chooses the alternative with the highest utility, either a bank, with utility in equation (11), or the outside option, with utility ε_{h0}. The taste shocks are independent of risk aversion and independently and identically distributed type I extreme value across alternatives and savers, which yields equation (12) (Berry 1994, Xiao 2020). Savers base their choices on reputation and on reported performance. Product holdings are disclosed, but savers do not price the credit risk or the hidden variance that a full reading of the holdings would reveal (Section 3). Consistent with this, the exposure of hiding affects their choices only through the bank's reputation, and the reported variance in their utility is unchanged.

Timing and equilibrium. We compare two static franchise-value games, before and after an unanticipated reform. In each, banks simultaneously choose a bond share $x_b \in [0, 1]$ and a reporting choice $H_b \in \{0, 1\}$. Each bank values the demand generated by its offer and the expected damage to its own reputation, holding its competitors' offers and reputations fixed. Its two reputation events are independent, with probabilities θ and ρ, and the model period is normalized. A pure-strategy equilibrium is a profile at which no bank can raise its expected fee payoff in equation (14) by changing either choice. The investor shares reported in Table B1 are evaluated at undamaged reputations, while the objective also accounts for the bank's own event-contingent fee losses.

B.2 Savers and investor shares

Bounded support permits differentiation under the integral. Holding rivals' offers fixed,

$$\frac{\partial s_b(\gamma)}{\partial x_b} = \beta s_b(\gamma)\Big(1 - s_b(\gamma)\Big)\Big(\mu_B - \gamma(1 - cH_b)\sigma_B^2 x_b\Big), \qquad \frac{\partial s_b(\gamma)}{\partial a_b} = \gamma\, s_b(\gamma)\Big(1 - s_b(\gamma)\Big). \tag{B.1}$$

Hiding raises the utility that the bank offers to a saver of type γ by $\beta\gamma c\,\sigma_B^2 x_b^2/2$, which rises with the saver's risk aversion. The small-share approximation treats $s_b(\gamma)$ as small for every bank and saver type, so $s_b(\gamma)(1 - s_b(\gamma)) \approx s_b(\gamma)$, and a bank's marginal savers then have the same distribution of risk aversion as its savers. For any two banks and whatever their shares, equation (12) gives

$$\log \frac{s_b(\gamma)}{s_{b'}(\gamma)} = \beta\Big(R_b - R_{b'}\Big) + \gamma\Big[\Big(a_b - a_{b'}\Big) - \frac{\beta}{2}\Big(V_b - V_{b'}\Big)\Big]. \tag{B.2}$$

The bracket combines the reputation gap with the difference in reported risk. When it is positive, more risk-averse savers are relatively more likely to choose bank b. Hiding lowers V_b and therefore draws relatively more cautious savers to the bank that hides.

B.3 Portfolio choice and hiding

Let $k, e \in \{0, 1\}$ indicate the credit event and the exposure, π_{ke} their joint probability, and $W_{b,ke} = \int s_b(\gamma)(1 - s_b(\gamma))\, dG(\gamma)$ and $\bar{\gamma}_{b,ke}$ the demand slope and marginal risk aversion evaluated at the reputation $a_b - \kappa x_b^2 k/2 - \delta H_b e$. By equation (B.1), an interior optimum of the exact objective with reporting choice H satisfies

$$\sum_{k,e} \pi_{ke}\, W_{b,ke}\Big\{\beta\mu_B - \bar{\gamma}_{b,ke}\Big[\beta(1 - cH)\sigma_B^2 + \kappa k\Big]x_b\Big\} = 0. \tag{B.3}$$

The benchmark replaces the state-dependent weights by common local weights W_b and $\bar{\gamma}_b$, defined in equation (15), under both reporting choices. The expected fee loss in equation (14) is then $f\beta W_b\bar{\gamma}_b\Big(\varphi x_b^2/2 + \psi H_b\Big)$, with φ and ψ in equation (16). This expansion is first order in the magnitudes of the reputation losses. Products of the losses enter only at second order, so the event probabilities themselves need not be small. After division by $f\beta W_b$, the bank maximizes

$$Q_b(x, H) = \mu_B x - \frac{\bar{\gamma}_b}{2}\, y_H\, x^2 - \bar{\gamma}_b\, \psi\, H, \tag{B.4}$$

which is equation (16), with y_H in equation (17). For each H, $Q_b(\cdot, H)$ is strictly concave on $[0, 1]$ with maximizer $x_H^* = \min\{1, \mu_B/(\bar{\gamma}_b y_H)\}$. The dispersion of risk aversion among marginal savers is $\mathrm{Var}_b(\gamma) = W_b^{-1} \int (\gamma - \bar{\gamma}_b)^2 s_b(\gamma)(1 - s_b(\gamma))\, dG(\gamma)$.

Proposition B.1 (Reaching for yield and hiding)**.** *Fix common marginal-saver weights under both reporting choices and suppose both candidate bond shares are interior, $\mu_B < \bar{\gamma}_b y_1$. Then*

(i) the bond share with reporting choice H is $x_H = \mu_B/(\bar{\gamma}_b y_H)$, and $x_1 > x_0$.

(ii) the gain from hiding, $\Gamma(\bar{\gamma}_b) = \mu_B^2(1/y_1 - 1/y_0)/(2\bar{\gamma}_b) - \bar{\gamma}_b\psi$, is equation (18). It is strictly decreasing in $\bar{\gamma}_b$, and the bank hides if and only if $\Gamma \geq 0$, equivalently $\bar{\gamma}_b \leq \gamma^* = \mu_B\sqrt{c\,\sigma_B^2/(2\psi y_0 y_1)}$.

(iii) across banks in this interior region, the bond share and reported return decrease with $\bar{\gamma}_b$, and if γ^* lies in the region, they fall discretely to the fraction y_1/y_0 of their value where $\bar{\gamma}_b$ crosses γ^*. The reported variance is a share $1-c$ of the market variance if $\bar{\gamma}_b \leq \gamma^*$ and all of it otherwise.

Both terms of the gain from hiding are scaled by the caution of marginal savers, as both the reported variance that hiding removes and the reputation that exposure would damage are weighted by risk aversion. At a given bond position, cautious savers value lower reported variance more. What differs across banks is the risk there is to hide. Cautious marginal savers lead a bank to hold few bonds, so hiding gains it little and its expected reputational cost dominates. A bank with less cautious marginal savers holds many bonds, hiding gains it more than the expected cost, and hiding lets it hold still more. When a candidate bond share reaches one, the comparison uses Q_b at x_H^*, and the cutoff formula no longer applies.

A bank that hides reports a lower variance per bond but holds more bonds. For banks with $\bar{\gamma} \leq \gamma^* < \bar{\gamma}'$ and interior bond shares, the bank that hides reports the lower variance if and only if

$$(1-c)\left(\frac{\bar{\gamma}'\,y_0}{\bar{\gamma}\,y_1}\right)^2 < 1. \tag{B.5}$$

Because $\bar{\gamma}' > \bar{\gamma}$, this requires $\sqrt{1-c}\,y_0 < y_1$, which holds if and only if $\varphi > \sqrt{1-c}\,\sigma_B^2$. The credit risk borne through reputation is what keeps a bank that hides from expanding its position until it reports more variance than a transparent bank.

B.4 Sorting by reputation

Proposition B.2 (Sorting)**.** *In the small-share case, holding portfolios and reporting choices fixed, $\partial\bar{\gamma}_b/\partial a_b = \mathrm{Var}_b(\gamma)$ and $\partial \log s_b/\partial a_b = \bar{\gamma}_b$. At an interior optimum with a given reporting choice H, holding rivals' offers fixed and allowing the bank's bond share to respond,*

$$\frac{d\bar{\gamma}_b}{da_b} = \frac{\mathrm{Var}_b(\gamma)}{1-\beta(1-cH)\,\sigma_B^2\,x_b^2\,\mathrm{Var}_b(\gamma)/\bar{\gamma}_b}, \qquad \frac{dx_b}{da_b} = -\frac{x_b}{\bar{\gamma}_b}\frac{d\bar{\gamma}_b}{da_b}, \tag{B.6}$$

with $d\bar{\gamma}_b/da_b > 0$ and $dx_b/da_b < 0$ when the denominator is positive. In the small-share case, if $a_b - a_{b'} > \beta\sigma_B^2/2$, then $\bar{\gamma}_b > \bar{\gamma}_{b'}$ for every feasible pair of offers. If both banks choose their

reporting optimally and their candidate bond shares are interior, then also $H_b \leq H_{b'}$ *and* $x_b < x_{b'}$.

We call the condition $a_b - a_{b'} > \beta\sigma_B^2/2$ the sorting condition. It guarantees that the bracket in equation (B.2) is positive for every feasible offer. In equation (B.6), the portfolio response reinforces the sorting. A bank that holds fewer bonds reports less return risk, which attracts still more cautious savers. Savers do not observe the credit risk borne through reputation, so it does not enter this feedback, but it lowers the bond share at every reputation. Outside the small-share case, the ordering holds for all of a bank's savers, and the numerical equilibrium of Subsection B.6 confirms it for marginal savers without the small-share approximation. In the data, FY2021 interest-bearing assets proxy for the reputation and sales network that determine a bank's share.

B.5 Debt resolution and fragility

Debt resolution lowers the probability of a credit event from θ_0 to $\theta_1 = (1-\tau)\theta_0$ and the bond's excess return from $\mu_{B,0}$ to $\mu_{B,1} = (1-\tau)\mu_{B,0}$, with $0 < \tau < 1$, so the credit risk borne through reputation falls from φ_0 to $\varphi_1 = (1-\tau)\varphi_0$. The variance σ_B^2 and the hiding and exposure parameters are unchanged. We write $y_{H,t} = (1-cH)\sigma_B^2 + \varphi_t$ and γ_t^* for the threshold in Proposition B.1. In a more general shock, γ^* falls if and only if $\mu_{B,1}/\mu_{B,0} < \sqrt{y_{0,1}y_{1,1}/(y_{0,0}y_{1,0})}$, and a fall in credit risk alone raises it.

Proposition B.3 (Fragility and bond retreat)**.** *Fix common marginal-saver weights and suppose the candidate bond shares are interior in both states.*

(i) If a bank that hides is required to report transparently, holding the yield and credit risk fixed, its bond share and reported return fall by the fraction $c\,\sigma_B^2/y_0$.

(ii) Debt resolution lowers the hiding threshold to $\gamma_1^* = (1-\tau)\sqrt{y_{0,0}y_{1,0}/(y_{0,1}y_{1,1})}\gamma_0^* < \gamma_0^*$.

(iii) The bank changes its bond share by $\Delta x_b = -\nu_b\, x_{b,0}$*, where*

$$\nu_H = \frac{\tau(1-cH)\sigma_B^2}{(1-cH)\sigma_B^2 + (1-\tau)\varphi_0} \quad \text{and} \quad \nu^{stop} = 1 - (1-\tau)\frac{y_{1,0}}{y_{0,1}} \tag{B.7}$$

apply to a bank whose reporting choice H *is unchanged and to a bank that hides before resolution and not after* ($\gamma_1^* < \bar{\gamma}_b \leq \gamma_0^*$)*. These satisfy* $\nu^{stop} > \nu_0 \geq \nu_1$ *and* $\nu_0 \leq \tau$.

The strategy is fragile in this sense, because its value depends on both the hiding and the yield. Because the change in the bond share is proportional to the initial position, a larger initial position produces a stronger response, which Section 5.1 examines. The retreat is largest at banks that stop hiding, because they lose both the yield and the hiding that let them hold so many bonds. A bank that keeps hiding retreats proportionally less, because

hiding has already removed most of the reported risk that restrained its bonds. The exposure measure in Section 5.1 is Initial home overweight$_a$. It also includes other home credit and subtracts a nationwide benchmark, so it proxies initial exposure to the affected investment.

Proposition B.4 (Returns and investor shares)**.** *Fix common marginal-saver weights and suppose the bond shares are interior. Suppose $\bar{\gamma}_L > \bar{\gamma}_S$, the larger bank does not hide, and the smaller bank hides before resolution and stops afterward ($\gamma_1^* < \bar{\gamma}_S \leq \gamma_0^*$). Then*

$$\Delta R_L - \Delta R_S = \lambda_S \, R_{S,0} - \lambda_L \, R_{L,0} > 0, \tag{B.8}$$

with $\lambda_S = 1 - (1-\tau)^2 y_{1,0}/y_{0,1}$ and $\lambda_L = 1 - (1-\tau)^2 y_{0,0}/y_{0,1}$, which satisfy $\lambda_S > \lambda_L > 0$. This is equation (19). In the small-share case and to first order in utility changes, the larger bank's pre-event investor share grows relative to the smaller bank's,

$$\Delta \log s_L - \Delta \log s_S \simeq \beta \Big[\big(J_{S,0} - J_{S,1} \big) - \big(J_{L,0} - J_{L,1} \big) \Big] > 0, \tag{B.9}$$

where $J_b = R_b - \bar{\gamma}_b V_b/2$ is the risk-adjusted reported return at the bank's marginal savers.

The smaller bank loses a larger share of a larger return. It held more bonds before resolution, both because its marginal savers are less cautious and because it hid their risk, and it loses the hiding together with the yield. Its return advantage over the larger bank therefore narrows.

B.6 Numerical solution of the full payoff

We now compute a two-bank equilibrium of the full demand system for an illustrative parameterization, with the exact expected fee payoff in equation (14). Savers' risk aversion is lognormal with log mean 1.0 and log standard deviation 0.6, truncated to $[0.4, 15]$. We set $\beta = 60$, $f = 1$, $\mu_B = 0.009$, and $\sigma_B^2 = 0.006$. Credit losses become salient with probability $\theta = 0.9$ and lower reputation by $\kappa x_b^2/2$ with $\kappa = 0.4$, so $\varphi = 0.006$. A bank that hides reports a share $1 - c = 0.2$ of its bonds' market variance, which is 45% of their volatility. Its hiding is exposed with probability $\rho = 0.9$, which lowers its reputation by $\delta = 0.05$, so $\psi = 0.00075$. Each event moves reputation little, and the numerical solution evaluates the exact payoff directly. Reputations are $a_L = -1$ and $a_S = -1.5$, and resolution lowers the bond return and the probability of a credit event by $\tau = 0.5$. Each bank's choices are a numerically computed best response to its rivals' choices, with investor shares obtained by integrating equation (12) over the distribution of risk aversion.

Table B1
Full equilibrium of the model

	Before resolution		After resolution	
	Larger bank	Smaller bank	Larger bank	Smaller bank
Bond share x_b	0.404	0.847	0.270	0.324
Hides risk H_b	0	1	0	0
Pre-event investor share s_b	0.091	0.050	0.085	0.037
Marginal savers' risk aversion $\bar{\gamma}_b$	1.88	1.59	1.87	1.56
Reported excess return R_b (bp)	36.4	76.2	12.1	14.6
Market return volatility (%)	3.13	6.56	2.09	2.51
Reported return volatility (%)	3.13	2.93	2.09	2.51

Notes: The table reports the numerically computed equilibrium of the model in Section 6 with the illustrative parameters in the text. Investor shares are evaluated at undamaged reputations. The bank's objective separately averages fee income over its own reputation events, holding competitors' offers fixed, so these shares are not the expected fee shares in the objective. Market return volatility is the standard deviation of the portfolio's excess return at market prices, and reported return volatility is the standard deviation that its products report.

The three predictions hold in this example (Table B1). The larger bank's marginal savers are more cautious, and the first-order hiding threshold, 1.73, lies between the smaller bank's 1.59 and the larger bank's 1.88. The exact solution makes the same choices. Only the smaller bank hides. It holds 85% of its products' funds in bonds, against 40% at the larger bank, and it earns the higher reported return. Its holdings imply about twice the market volatility of the larger bank's, yet it reports less volatility, as $\varphi = 0.006$ exceeds $\sqrt{1-c}\,\sigma_B^2 = 0.0027$. In the equilibrium in which neither bank can hide, the smaller bank holds 49% in bonds, reports 44 basis points, and has a pre-event investor share 18% lower.

Debt resolution lowers the hiding threshold to 1.31, below the smaller bank's marginal savers, so the smaller bank stops hiding. Resolution lowers the larger bank's bond share by 33% and the smaller bank's by 62%. Proposition B.3 gives one third for a bank that stays transparent, 14% for a bank that keeps hiding, and 60% for a bank that stops hiding. The smaller bank's reported return advantage narrows from 39.8 to 2.4 basis points, and the log ratio of the larger bank's pre-event investor share to the smaller bank's rises by 0.23. The smaller bank's share of the LGFV bonds held by the two banks' products falls from 54% to 35%. Figure 12 shows these choices.

We check each bank before and after resolution under both reporting choices. Holding rivals' equilibrium choices fixed, a search over bond shares in steps of 0.0005 finds no higher value. Optimizations from four starting values agree within 10^{-15}. The solutions have first-order residuals below 10^{-9} and negative second derivatives. Increasing integration accuracy from 400 to 800 Gauss–Legendre nodes changes no reported entry. These checks support the computed best responses for the stated parameterization.

Internet Appendix C
Proofs

C.1. Proof of Proposition B.1

For a given H, $Q_b(\cdot, H)$ is a strictly concave quadratic. Under $\mu_B < \bar{\gamma}_b y_1$, both candidate maximizers $x_H = \mu_B/(\bar{\gamma}_b y_H)$ lie in $(0,1)$, with maxima $\mu_B^2/(2\bar{\gamma}_b y_H) - \bar{\gamma}_b \psi H$. Since $y_1 = y_0 - c\sigma_B^2 < y_0$, $x_1 > x_0$. The gain from hiding is

$$\Gamma(\bar{\gamma}_b) = \frac{\mu_B^2}{2\bar{\gamma}_b}\Big(\frac{1}{y_1} - \frac{1}{y_0}\Big) - \bar{\gamma}_b\psi = \frac{c\,\sigma_B^2\mu_B^2}{2\bar{\gamma}_b y_0 y_1} - \bar{\gamma}_b\psi,$$

which is equation (18). Its derivative, $-c\sigma_B^2\mu_B^2/(2\bar{\gamma}_b^2 y_0 y_1) - \psi$, is negative. Solving $\Gamma = 0$ yields the stated cutoff, and since Γ is strictly decreasing, $\Gamma \geq 0$ if and only if $\bar{\gamma}_b \leq \gamma^*$ on the interior region. The bank hides when $\Gamma \geq 0$, with ties resolved in favor of hiding. Part (iii) follows because $x_b = \mu_B/(\bar{\gamma}_b y_{H_b})$ and $R_b = \mu_B x_b$, with H_b switching from one to zero at γ^*, and because the reported variance is $(1 - cH_b)\sigma_B^2 x_b^2$.

For condition (B.5), the two reported variances are $(1-c)\sigma_B^2\mu_B^2/(\bar{\gamma}y_1)^2$ and $\sigma_B^2\mu_B^2/(\bar{\gamma}' y_0)^2$. Since $\bar{\gamma}'/\bar{\gamma} > 1$, the condition requires $\sqrt{1-c}\,y_0 < y_1$, that is, $\sqrt{1-c}\,(\sigma_B^2+\varphi) < (1-c)\sigma_B^2 + \varphi$, which rearranges to $\varphi\big(1-\sqrt{1-c}\big) > \sqrt{1-c}\,\sigma_B^2\big(1-\sqrt{1-c}\big)$, or $\varphi > \sqrt{1-c}\,\sigma_B^2$. □

C.2. Proof of Proposition B.2

In the small-share case, $s_b(\gamma) = \exp(u_b(\gamma))/D(\gamma)$, with $u_b(\gamma) = \gamma a_b + \beta(R_b - \gamma V_b/2)$ and $D(\gamma)$ the denominator in equation (12), whose dependence on bank b's own offer is negligible. The weights defining $\bar{\gamma}_b$ are then proportional to $\exp(\gamma t_b)\, m_b(\gamma)$, where $t_b = a_b - \beta V_b/2$ and $m_b(\gamma) = e^{\beta R_b}\, dG(\gamma)/D(\gamma)$ does not depend on a_b. This is an exponential tilting of a fixed distribution, so the derivative of its mean with respect to the tilt t_b is its variance, which gives $\partial\bar{\gamma}_b/\partial a_b = \mathrm{Var}_b(\gamma)$ holding the portfolio fixed. Similarly, $\partial \log s_b/\partial a_b = \int \gamma\, s_b(\gamma)\, dG / \int s_b(\gamma)\, dG = \bar{\gamma}_b$.

With reporting choice H, $V_b = (1-cH)\sigma_B^2 x_b^2$ and $x_b = \mu_B/(\bar{\gamma}_b y_H)$. Then $d\bar{\gamma}_b/da_b = \mathrm{Var}_b(\gamma)\, dt_b/da_b$, with $dt_b/da_b = 1 - \beta(1-cH)\sigma_B^2 x_b\, dx_b/da_b$ and $dx_b/da_b = -(x_b/\bar{\gamma}_b)\, d\bar{\gamma}_b/da_b$. Substituting the last expression into the first and solving gives equation (B.6).

For the final statements, equation (B.2) shows that $s_b(\gamma)/s_{b'}(\gamma)$ increases with γ when $(a_b - a_{b'}) - \frac{\beta}{2}(V_b - V_{b'}) > 0$. Every feasible offer has $0 \leq V \leq \sigma_B^2$, so $a_b - a_{b'} > \beta\sigma_B^2/2$ guarantees the inequality for every pair of offers. The distribution of bank b's savers then dominates that of bank b''s savers in the likelihood-ratio order, and in the small-share case $\bar{\gamma}_b > \bar{\gamma}_{b'}$. If both banks choose their reporting optimally with interior candidate bond shares, Proposition B.1 gives $H_b \leq H_{b'}$, so $y_{H_b} \geq y_{H_{b'}}$ and $x_b = \mu_B/(\bar{\gamma}_b y_{H_b}) < \mu_B/(\bar{\gamma}_{b'} y_{H_{b'}}) = x_{b'}$. □

C.3. Proof of Proposition B.3

Part (i) follows from $x_0/x_1 = y_1/y_0 = 1 - c\sigma_B^2/y_0$ and $R_b = \mu_B x_b$, where c is the strength of the hiding technology the bank used. For part (ii), $(\gamma_t^*)^2 = c\sigma_B^2\mu_{B,t}^2/(2\psi y_{0,t}y_{1,t})$, so $(\gamma_1^*/\gamma_0^*)^2 = (1-\tau)^2 y_{0,0}y_{1,0}/(y_{0,1}y_{1,1})$. Since $y_{H,1} = (1-cH)\sigma_B^2 + (1-\tau)\varphi_0 > (1-\tau)y_{H,0}$, this ratio is below one. The same expression with a general yield ratio gives the condition for a general shock in the text. For part (iii), a bank with an unchanged reporting choice has $x_{b,1}/x_{b,0} = (1-\tau)y_{H,0}/y_{H,1}$, so

$$\nu_H = 1 - (1-\tau)\frac{y_{H,0}}{y_{H,1}} = \frac{\tau(1-cH)\sigma_B^2}{(1-cH)\sigma_B^2 + (1-\tau)\varphi_0}.$$

This expression rises with $(1-cH)\sigma_B^2$, so $\nu_0 \geq \nu_1$, and $\nu_0 \leq \tau$ with equality if and only if $\varphi_0 = 0$. A bank that stops hiding has $x_{b,1}/x_{b,0} = (1-\tau)y_{1,0}/y_{0,1}$, which gives ν^{stop}. Since $y_{1,0} < y_{0,0}$, $\nu^{\text{stop}} > 1 - (1-\tau)y_{0,0}/y_{0,1} = \nu_0$. □

C.4. Proof of Proposition B.4

The smaller bank holds $x_{S,0} = \mu_{B,0}/(\bar{\gamma}_S y_{1,0})$ before resolution and $x_{S,1} = \mu_{B,1}/(\bar{\gamma}_S y_{0,1})$ after it, so $R_{S,1}/R_{S,0} = (1-\tau)^2 y_{1,0}/y_{0,1} = 1 - \lambda_S$. The same argument without hiding gives $R_{L,1}/R_{L,0} = 1 - \lambda_L$. Since $y_{1,0} < y_{0,0}$, $\lambda_S > \lambda_L$, and $\lambda_L > 0$ because $y_{0,1} - (1-\tau)^2 y_{0,0} = (1-(1-\tau)^2)\sigma_B^2 + \tau(1-\tau)\varphi_0 > 0$. Hence $\Delta R_L - \Delta R_S = -\lambda_L R_{L,0} + \lambda_S R_{S,0}$, which is positive because $\lambda_S > \lambda_L > 0$ and $R_{S,0} > R_{L,0}$, which holds since $\bar{\gamma}_S < \bar{\gamma}_L$ and $y_{1,0} < y_{0,0}$.

In the small-share case, $\log s_b(\gamma) = u_b(\gamma) - \log D(\gamma)$. $D(\gamma)$ does not change to first order, because both banks' shares are small. A change in the bank's choices then changes $\log s_b$ by β times the change in J_b. With reported variance rate $\sigma_H^2 = (1-cH)\sigma_B^2$, $J_b = \mu_B x_b - \bar{\gamma}_b \sigma_H^2 x_b^2/2 = \mu_B^2 \mathcal{J}(\sigma_H^2, \varphi)/(2\bar{\gamma}_b)$, with $\mathcal{J}(s,\varphi) = (s+2\varphi)/(s+\varphi)^2$ and $\partial\mathcal{J}/\partial s = -(s+3\varphi)/(s+\varphi)^3 < 0$. Then

$$J_{S,0} - J_{S,1} = \frac{\mu_{B,0}^2\mathcal{J}(\sigma_1^2,\varphi_0) - \mu_{B,1}^2\mathcal{J}(\sigma_0^2,\varphi_1)}{2\bar{\gamma}_S},$$
$$J_{L,0} - J_{L,1} = \frac{\mu_{B,0}^2\mathcal{J}(\sigma_0^2,\varphi_0) - \mu_{B,1}^2\mathcal{J}(\sigma_0^2,\varphi_1)}{2\bar{\gamma}_L}.$$

Writing $q = 1-\tau$, $\mu_{B,1}^2\mathcal{J}(\sigma_0^2,\varphi_1) = \mu_{B,0}^2 q^2 \mathcal{J}(\sigma_0^2, q\varphi_0)$, and

$$\frac{d\log\left(q^2\mathcal{J}(\sigma_0^2,q\varphi_0)\right)}{dq} = \frac{2}{q} - \frac{2q\varphi_0^2}{(\sigma_0^2+2q\varphi_0)(\sigma_0^2+q\varphi_0)} > 0,$$

so the numerator of $J_{L,0} - J_{L,1}$ is positive. The numerator of $J_{S,0} - J_{S,1}$ is larger because $\mathcal{J}(\sigma_1^2,\varphi_0) > \mathcal{J}(\sigma_0^2,\varphi_0)$, and $1/\bar{\gamma}_S > 1/\bar{\gamma}_L$, so $J_{S,0} - J_{S,1} > J_{L,0} - J_{L,1} > 0$, which gives equation (B.9). □

Internet Appendix D
Specifications and Supplementary Evidence

Subsection D.1 identifies the borrowers that receive private credit, and Subsection D.2 reconciles market-wide exposure with shares of included debt holdings. Subsection D.3 gives the designs behind the tests of holdings risk, reported risk, reaching for yield, and borrower funding, and Subsection D.4 states the specifications that the main text describes in words. Table D1 reports initial credit portfolios by bank type and size, and Figure D1 shows the gap between private credit and bond rates of the same borrower. Table D2 reports alternative specifications for Table 7, Table D3 compares smaller with larger banks, and Tables D4 and D5 report investor flows in and after the 2022 sell-off. Table D6 reports the part of the loss of the home LGFV return premium that bond income explains, together with sample and NAV checks.

D.1. Private-credit borrowers and the LGFV sector

To identify who receives private credit financing, we extract borrower names from asset descriptions, locate them using company registration records, and match them to LGFV bond issuers over 2022Q1–2025Q3. These matches account for about 60% of local banks' home private credit value before the program and 59% afterward. Adding a screen for borrower names with terms typical of LGFVs raises the shares to 79% and 86%. For the largest FY2021 size tercile, the corresponding shares with the keyword screen are 77% and 85%. Most of the shift into private credit therefore remains within the LGFV sector.

D.2. The scale of the WMP–LGFV bond link

A bond-only denominator gives a simpler benchmark for the LGFV shares in Section 4.1. We sum reported LGFV bond values and divide by the value of all bonds held by each bank group over 2022Q1–2023Q3, retaining government and policy-bank bonds and excluding private credit. This gives 38.5% for local banks and 13.2% for national banks. The product-quarters of Table D1 all have positive included debt holdings. In them, local banks' LGFV bonds account for 40.4% of all bonds, 47.5% of the bonds in included debt holdings, and 43.1% of included debt holdings. As shares of all disclosed assets over 2022Q1–2025Q3, LGFV bonds account for 16.4% of local banks' reported portfolio weights and 5.4% of national banks'.

For market context, we reconstruct the year-end 2021–2024 top-ten holdings cross-sections from the data vendor's original exports. We retain one representative record per product registration code and deduplicate positions within that record, preserving identical investments made by distinct products. We restrict amounts to yuan-denominated positions, identify bonds by asset type, and match issuers to the January 4, 2026 LGFV roster. Unidentified

bond issuers remain in the denominator but not the LGFV numerator. In 2024, the retained disclosures contain RMB 0.3859 trillion of LGFV bonds out of RMB 2.3459 trillion of bonds, including negotiable certificates of deposit. This ratio of sums is 16.45%, and it is not an equally weighted average of product ratios.

For each year t, let q_t be this disclosed LGFV-to-bond ratio, B_t the industry's bond assets including negotiable certificates of deposit, and A_t its net assets under management, from the annual WMP market reports. Estimated LGFV bond holdings are $\widehat{H}_t = B_t q_t$, and their share of WMP assets is $\widehat{H}_t/A_t$. The annual WMP market report published in 2025 gives year-end 2024 values of $B_t = 18.60$ and $A_t = 29.95$ trillion yuan. Using unrounded holdings gives $\widehat{H}_t = 3.06$ trillion yuan, or 10.2% of net assets under management. Equivalently, the rounded inputs give $16.45\% \times (18.60/29.95) = 10.22\%$. The extrapolation applies the disclosed bond mix to the industry's bond assets. It excludes private credit and indirect exposure through funds and other investment vehicles.

WMPs' share of outstanding LGFV bonds is $\widehat{H}_t/O_t$, where O_t is the published year-end LGFV bond balance. Lianhe Ratings (2024) reports balances of RMB 9.81, 10.62, and 11.48 trillion for 2021–2023. Zhang (2025) reports RMB 10.88 trillion for 2024, implying that WMPs held 28.1% of outstanding LGFV bonds that year. Figure A2 marks the 2024 point with an open diamond to show the change in source. These ratios describe two different economic exposures. Under the extrapolation, LGFV bonds account for about one-tenth of WMP assets in 2024, but WMPs hold more than a quarter of the LGFV bond market. A tenth of the assets of a large savings industry is thus more than a quarter of these borrowers' bond financing.

D.3. Samples and constructions behind the main tests

Risk held and risk reported. Table 7 uses products that the filings classify as fixed-income, with at least nine monthly returns between April 2022 and June 2023 and with disclosed holdings in 2022Q2–2023Q2. Monthly returns are changes in log distribution-adjusted NAV in basis points, as in equation (5). They cover every month in which the product has NAVs at the ends of consecutive months. The sell-off return is the product's mean monthly return in November and December 2022. Reported returns are winsorized at the 1st and 99th percentiles of all monthly NAV returns and implied returns at the 1st and 99th percentiles of their own distribution, so that the sell-off months remain in both. Redemption types are closed-end, periodic-open, and daily-open and come from the product filings. The risk rating is the product's modal rating. Holdings measures average the product's disclosed holdings over the quarters in which it reports. The home LGFV credit share combines LGFV bonds and private credit to LGFVs of the bank's headquarters province as a percentage of disclosed holdings. The all-in spread is defined in Section 4.1. It is computed position by position, as are the implied returns below. The issue spread of a bond is its coupon minus the Treasury

yield of the nearest half-year maturity on its issue date, and the product-level spread weights the bonds held by value. The spread relative to the same official rating uses medium-term notes, short-term financing bills, and corporate bonds issued since 2021. It subtracts from each bond's issue spread the mean spread of bonds with the same issue quarter, issuer rating, issuer administrative level, maturity group, and placement type. Log bank size is the FY2021 measure of Section 4.1, and products are weighted by the inverse of the number of products of their bank in the sample.

Implied returns. The implied return of a product in a month sums over its disclosed positions. Each term is the position's share of disclosed value times the month's price change of the market series assigned to the position. The shares are averaged over the quarters in which the product reports. These are price changes, which leave out interest income. LGFV bonds move by minus their modified duration times the sum of the month's changes in the Treasury yield of their maturity and in the LGFV spread of the issuer's province. Government and policy-bank bonds move by the Treasury term alone. Modified duration is that of a par bond with the asset's coupon and remaining maturity. Other bonds move with the China Securities Index (CSI) credit-bond index of their rating (AAA, AA+, or AA and below) and remaining-maturity bucket (up to 1, 1–3, 3–5, 5–7, and 7–10 years), scaled by the ratio of the bond's duration to that of the bucket. These other bonds include bank capital bonds and securities without the standard-bond flag that we find in the bond descriptions. The price part of an index is its monthly log return minus its median monthly log return over 2021–2023, which removes coupon accrual. Certificates of deposit move with the index for up to one year, at AAA for national banks and at AA+ otherwise. Bank preferred shares move with the 3–5 year AAA index, and asset-backed securities and public bond funds move with the 1–3 year AAA index. Private credit moves like an LGFV bond of the borrower's province when the data vendor classifies the borrower as an LGFV, when the borrower issues LGFV bonds, or when its name passes the screen of Subsection D.1. Other private credit moves with the AA+ index of its maturity bucket. In both cases the maturity is the balance-weighted remaining maturity of the borrower's contracts that are active at the quarter end. Other asset-management plans and funds, including trust plans and insurance asset-management products, move with the 1–3 year AA+ index, because most of them are fixed-income mandates. Cash, deposits, repurchase agreements, interbank lending, money-market funds, and futures margin do not move, and neither do equity-labeled positions, which are 0.4% of disclosed value. Reported returns also contain interest income net of fees, so we add to each product's implied return its mean difference between the reported return and the price change over the months other than November and December 2022 in which it reports a return. The two series then agree on average outside the sell-off. The adjustment is constant within a product, so it leaves implied volatility unchanged. Implied volatility is the standard deviation of the implied return over the months in which the product reports a return, and the implied sell-off return is its mean in November and December 2022.

The reported minus implied sell-off return thus measures how much more the reported return exceeded the price change of the holdings in the sell-off than in other months. Table D2 reports alternative specifications. They add a local-bank indicator, weight products equally, control for the shares of each asset class, keep products whose disclosed holdings are at least 60% bonds, or mark asset-management plans and funds with the AAA index instead. Two further variants mark only directly held bonds, so that every other asset contributes nothing, or mark only the holdings disclosed before the sell-off. When only bonds are marked, the implied-volatility gradient is more than three times as large, because larger banks hold more of the assets that this measure leaves unpriced. In Figure 3, Panels A to C use banks with at least five products in the sample, and Figure 5 averages products within a bank and then banks within each group.

Product design and ratings. Table 8 uses the products of Table 7. Its Panel B compares closed-end and periodic-open products with daily-open products of the same bank, with bank effects in column 1 and bank-by-risk-rating effects in column 2. Column 6 of Panel B of Table 7 uses all fixed-income products with a monthly return between April 2022 and June 2023 and a redemption type in the filings. It includes redemption-type effects and assigns each product to its parent bank by the issuer named in its filing. The bond-rating statistics use the medium-term notes, short-term financing bills, and corporate bonds issued between 2021Q1 and 2023Q2 with an issuer rating.

Comparison cells. Panel A of Table 9 uses local banks' LGFV bond holdings. Branch presence comes from a bank-branch registry matched to banks by name and headquarters city, and a bank has presence in an issuer city if it has at least one branch there in that quarter. The coupon outcome subtracts from each bond's coupon the mean coupon of all held LGFV bonds with the same issue year and maturity group, so it compares bonds issued under similar market conditions. Issuer administrative level and rating are from the issuance data. Credit events are defaults, rating warnings, and guarantor events recorded for the issuer between 2022Q4 and 2025Q4. A cell enters only if it contains at least two banks. Columns 1 and 2 of Panel B use tier-2 and perpetual bonds of other banks, classify issuers as city or rural banks by name, and compare bank-quarters with the same headquarters city and quarter. Column 3 of Panel B identifies negotiable certificates of deposit by name and code, classifies issuers as national banks or as city and rural banks by name, excludes the bank's own certificates, and compares bank-quarters with the same headquarters city, quarter, and maturity group.

Own-city holdings and local presence. Table 10 keeps local banks headquartered in cities with at least two sample banks and compares banks from the same city in the same quarter. Own-city branches come from the same registry as branch presence, and a bank with no recorded branch in its headquarters city counts as zero. Own-city LGFV bonds are those issued by LGFVs of the bank's headquarters city, and the county-level share uses the issuer's administrative level from the issuance data.

Borrower funding. In Table 18, X_f^H (X_f^L) is the fraction of borrower f's WMP-held bonds or private credit in 2022Q4–2023Q1 that was held by local banks without a subsidiary whose initial home LGFV bond share was above (below) the median. The borrower panel is balanced over 2023Q2–2025Q2, with zero holdings for groups that hold none of the borrower's debt. Column 5 uses new bonds issued by the same borrowers between 2023Q2 and 2025Q2, with coupons between 0.5% and 12% and known maturity. The new-issue premium estimates in Subsection D.4 use the same specification for the coupon minus the CSI valuation yield at listing.

Provincial spreads on home holdings. The spread comparison of Section 5.3 weights each province's month-end LGFV spread over Treasury yields by the disclosed value of the home LGFV bonds and private credit to home LGFVs that local banks without a subsidiary held over 2022Q4–2023Q2.

D.4. Specifications and details not displayed in the main text

This subsection states the regressions that the main text describes in words and gives the details of the displayed ones.

Flows (Section 3). The flow regression with holdings risk follows equation (1). It forms return quintiles within risk rating, redemption terms, and quarter. It includes the eleven lagged controls, a local-bank indicator, and category-by-quarter effects, and errors cluster by parent bank.

Allocation and home tilt (Section 4.1). In equation (2), the sample is 2022Q1–2023Q3. Products without holdings of an asset class keep a zero share, and shares are not benchmark-adjusted. Fixed size replaces contemporaneous bank assets, and quarter effects enter without bank effects or a local-bank indicator. In equation (3), the product controls are size, risk, withdrawal terms, disclosure count, and disclosure coverage. The bank controls are the bank's average product maturity and risk rating, nonperforming-loan ratio, cost–income ratio, and net interest margin, and the city control is home-city GDP growth. Errors cluster by parent bank. The two home measures use different denominators. For example, consider a WMP with RMB 100 in included debt holdings. It holds RMB 20 in LGFV bonds, and RMB 10 of these are from home. Its within-LGFV home share is 50%, but its home LGFV allocation is 10% of included debt holdings. Each measure subtracts the matching nationwide benchmark. Neither denominator is total WMP assets. In equation (4), the controls are those in equation (2) plus time-varying log bank assets. Redemption is not among them, because it enters separately. The sample is 2022Q1–2025Q3. Quarter effects enter, and errors cluster by parent bank. In equation (5), the pre-program home-minus-non-home LGFV contrast is 5.354 monthly basis points per unit share (parent-bank clustered standard error 1.773, $p = 0.003$). We calculate it from the same regression and its covariance matrix. At a 20-point allocation difference, the contrast is about 13 annual basis points.

Risk held and risk reported (Section 4.2). In Table 7, the coefficients on log bank size for the home LGFV credit share, the issue spread, the all-in spread, and log implied volatility are -5.39, -22.4, -12.8, and -0.074. A bank half as large differs by $\ln 2$ times each coefficient with the opposite sign. Among the LGFV issuers of Figure 4 rated AA+, average issue spreads range from 72 basis points in Guangdong to 449 in Tianjin. Holding the issue quarter, maturity, placement type, and issuer rating fixed, LGFV bonds were issued at spreads 36 basis points higher (standard error 13) than other bonds. Column 6 of Panel B uses every fixed-income product with a monthly return in the period, whether or not it has nine returns and disclosed holdings. Each product's rating is its modal rating in the returns data.

Reaching for yield (Section 4.3). In Table 9, bonds are weighted by value within a bank. Risk outcomes are indicators multiplied by 100. For other cities of the home province, the coefficients per log point of bank size are -34.3 basis points for the coupon, -13.2 percentage points for county-level issuers, and -4.1 points for issuers rated below AAA. In column 5, they are -39.4 basis points for other cities of the home province and -29.3 for other provinces. Multiplying by $\ln 2$ gives the comparisons for a bank half as large. For the capital bonds of other banks, the coefficients are -19.2 percentage points and -24.7 basis points per log point, with coupons measured relative to issue-year and instrument-type means. Three-quarters of the city and rural banks' capital bonds held by local banks without a wealth-management subsidiary were issued in other provinces. Certificates of deposit issued by city and rural banks pay higher yields than those of national banks.

Same-borrower financing rates (Section 5.1). We estimate the gap between private credit and bond yields separately for each quarter. For contract or bond issue j of borrower f in month m, we estimate

$$\begin{aligned}\text{Yield}_{jfm} = \alpha_{fm} + \sum_q \beta_q \, \mathbb{1}\{\text{Private credit}\}_j \mathbb{1}\{q(m) = q\} \\ + \sum_q \gamma_q \, \text{Maturity}_j \mathbb{1}\{q(m) = q\} + e_{jfm}.\end{aligned} \tag{D.1}$$

The outcome Yield_{jfm} is the annualized contractual borrowing rate, expressed in percent. The indicator $\mathbb{1}\{\text{Private credit}\}_j$ identifies private credit, Maturity_j is maturity in years, and $q(m)$ is the quarter containing month m. Borrower–month effects compare the two financing channels of the same legal borrower and exclude affiliates. A separate maturity slope in each quarter allows the relationship between maturity and yield to change over time. Thus β_q measures the quarter's private-credit-minus-bond yield gap in percentage points. The sums include every sample quarter, so they already span the private-credit and maturity main effects. Quarter indicators are absorbed by borrower–month effects. There are 17,303 observations for 589 borrowers over 2021Q1–2025Q3. Errors cluster by borrower province. Mixed-channel borrower-months identify the direct comparison, while other months inform

the maturity controls. To present the event study relative to 2023Q2, Figure D1 plots

$$\delta_q = \beta_q - \beta_{q_0}, \qquad q_0 = 2023\text{Q2}, \qquad \delta_{q_0} = 0. \tag{D.2}$$

Zero on the plot means that the gap is unchanged from the 1.90 percentage points of 2023Q2 and not that yields are equal. The gap reaches 2.86 points in 2024Q4, which is 0.96 points above the reference. Confidence intervals use the covariance of both gap estimates. The plot therefore shows how the relative financing terms of the same borrowers moved.

Retreat from home LGFV bonds and initial home overweight (Section 5.1). Comparing local-bank products in the same home province and quarter, we estimate

$$\begin{aligned}\text{Home LGFV bond share}_{iq} = \beta\, &\text{Initial home overweight}_a \times \text{Refinancing}^{debt}_{p_b,q} \\ &+ \gamma' \text{Controls}_{iq} + \alpha_a + \alpha_{p_b,q} + \varepsilon_{iq}.\end{aligned} \tag{D.3}$$

The outcome is the product's home LGFV bond share in percentage points. Initial home overweight is the issuing institution's product-assets-weighted 2022Q1–2023Q3 average portfolio home overweight, summed across bonds and private credit (Section 4.1). Refinancing intensity is $\text{Refinancing}^{debt}_{p_b,q}$, the cumulative special-refinancing bond issuance divided by 2020 LGFV debt in the parent bank's home province. Institution and home-province–quarter effects hold persistent portfolio differences and common regional conditions fixed. A negative β means that initially more home-oriented institutions reduce home bonds more as refinancing rises. Both interaction variables are fractions. The eleven time-varying controls enter, and errors and wild-bootstrap draws cluster by home province. The refinancing denominator is LGFV debt, not GDP as in the spread tests. Measuring initial home overweight before the announcement or allowing it separate quarterly paths gives similar estimates. The event study in Figure 9 repeats the home-versus-other-buyer comparison of equation (8) for an entire destination province rather than a single city. Adjustment can occur through sales or by letting bonds mature without replacing them.

Bank size and the shift into private credit (Section 5.2). For product i of local bank b in quarter q, we estimate

$$\begin{aligned}\text{Private credit share}_{iq} = \beta_M\, &\mathbb{1}\{\text{Middle tercile}\}_b \times \mathbb{1}\{\text{Post}\}_q \\ &+ \beta_L\, \mathbb{1}\{\text{Largest tercile}\}_b \times \mathbb{1}\{\text{Post}\}_q \\ &+ \gamma' \text{Controls}_{iq} + \alpha_b + \alpha_q + u_{iq}.\end{aligned} \tag{D.4}$$

The outcome is home or total private credit as a percentage of the product's included debt holdings. Banks remain in their FY2021 asset terciles, and Post begins in 2023Q4. The smallest tercile is the comparison group. Bank effects remove persistent allocation differences, and quarter effects remove common changes. Thus β_L measures how much more the largest group's private-credit share changes after implementation than the smallest group's

share. We weight each product-quarter by the value of its included debt holdings, giving larger portfolios more weight. The sample covers 2022Q1–2025Q3. The eleven controls are those in equation (5). Errors cluster by parent bank. Fixed group indicators and Post are absorbed by the fixed effects. The comparison that conditions on initial home overweight (Table 14) allows initial home overweight and log FY2021 interest-bearing assets to predict post-program adjustment separately. It retains institution effects, the eleven controls, and home-province–quarter effects.

Return losses and private-credit substitution (Section 5.2). For equation (10), we winsorize monthly log returns at the 1st and 99th percentiles of all monthly returns. We then average them across NAV series within product-month and across products within bank-month. Post begins in October 2023. The initial shares are raw fractions and not deviations from market weights, and size is log FY2021 interest-bearing assets. A negative β_L means that a higher initial home LGFV bond share predicts a larger return loss, given subsequent substitution and size. A positive β_P means that banks that expanded private credit preserved more return, given their initial portfolios and size. Including initial private credit distinguishes expansion from simply having held more of the instrument before the program. In column 2 of Table 17, the size–Post coefficient is 0.900 monthly basis points per log point of FY2021 interest-bearing assets (parent-bank clustered standard error 0.735). It is 1.808 (standard error 0.744) in column 1, where size enters with the controls alone. A further specification drops the private-credit interactions and includes bank effects, subsidiary-status-by-month effects, and the eight lagged controls. In it, a ten-point higher initial home LGFV bond share predicts about 10 basis points more annualized return decline (wild $p = 0.050$). The relationship survives province-by-month effects (wild $p = 0.010$) and separate quarterly size slopes.

Balances in newly launched WMPs (Section 5.2). New-product balances sum the balances of observed products launched in quarter q. They are measured at the end of $q + 1$, when products are roughly three to six months old. The measure reflects both the number of products launched and their balances. We estimate

$$\begin{aligned}\text{Log new-product balances}_{bq} = {} & \beta_L \,\text{Initial home LGFV bond share}_b \times \mathbb{1}\{\text{Post}\}_q \\ & + \beta_P \,\Delta\text{Home PC share}_b \times \mathbb{1}\{\text{Post}\}_q \\ & + \beta_S \,\text{Log bank size}_b \times \mathbb{1}\{\text{Post}\}_q \\ & + \beta_H \,\text{Initial home PC share}_b \times \mathbb{1}\{\text{Post}\}_q \\ & + \gamma' \,\text{Controls}_{b,q-1} + \alpha_b + \alpha_q + u_{bq}.\end{aligned} \tag{D.5}$$

Initial shares, private-credit expansion, and bank size have the same definitions as in equation (10), and the same eight bank and home-city controls lag one quarter. Bank effects α_b absorb the fixed bank-characteristic levels, and launch-quarter effects α_q absorb common changes, including Post. Post starts with launches in 2023Q4. Each bank–launch quarter

receives equal weight, and errors cluster by parent bank. Column 4 of Table 17 reports this specification, and column 3 keeps only bank size and the controls. The sample covers launches in 2022Q2–2025Q2 except 2023Q3, and missing bank–launch quarters are not assigned zero balances. The sample uses the bank–launch quarters with enough pre- and post-program holdings to measure expansion. In column 4, the size coefficient is 0.138 (standard error 0.109) and is not statistically significant, against 0.292 (standard error 0.091) in column 3. Ending the sample in June 2024 gives a coefficient on the initial home LGFV bond share of -1.613 (standard error 0.505, $p = 0.001$). Tables report unstandardized coefficients, and percentage effects in the text use $100[\exp(\beta\Delta x) - 1]$.

Bank size and product returns (Section 5.2). For product returns, each observation is a NAV-series–month, as in equation (5). The event study in Figure 10 builds on the regression

$$\begin{aligned}\text{Monthly return}_{im} = {} & \beta_S \,\text{Log bank size}_b \times \text{Refinancing}^{debt}_{p_b,m} \\ & + \beta_B \,\text{Initial home overweight}_a \times \text{Refinancing}^{debt}_{p_b,m} \\ & + \gamma' \,\text{Controls}_{i,h(m)} + \alpha_a + \alpha_{p_b,m} + \varepsilon_{im}.\end{aligned} \tag{D.6}$$

The size measure Log bank size_b, $\text{Initial home overweight}_a$, and home-province intensity retain their earlier definitions. The outcome $\text{Monthly return}_{im}$ is in basis points and winsorized at the 1st and 99th percentiles of all monthly returns. The eleven controls are measured at the preceding quarter end $h(m)$. Issuing-institution effects remove persistent performance differences, and home-province-by-month effects compare banks facing the same regional conditions.

Figure 10 replaces both refinancing interactions in equation (D.6) with log bank size interacted with quarter indicators and omits 2023Q3. The monthly return outcome, eleven controls, and fixed effects are unchanged, and initial home overweight is not included. The sample covers 2022Q2–2025Q3. Observations are unweighted, and errors cluster by parent bank. Institution effects absorb fixed size, and province–month effects absorb the quarter indicators. The pre-program coefficients show no trend. The plotted coefficients remain in monthly units. They are 5.69 and 7.96 basis points per log point of assets in 2024Q3 and 2025Q2. Multiplying by $\ln 2 \times 12$ gives the annualized doubling comparisons.

Total WMP balances (Section 5.2). In equation (9), size is log FY2021 interest-bearing assets in RMB trillion. A positive β_S means that balances grow proportionally more after implementation at initially larger banks. Column 2 of Table 16 replaces common half-year effects with province-by-half-year effects, and column 3 replaces continuous size with fixed size-tercile indicators interacted with Post. Fixed size is absorbed by bank effects and Post by time effects. The bank-level panel enters without additional controls. For an asset ratio a, the implied relative growth difference is $100[\exp(\beta_S \log a) - 1]\%$. Missing reports are not zeros. Balances measure the scale of the WMP business. They combine investment returns, product entry and maturity, and investor transactions. The raw size–Post coefficient is

0.152 log balance points per log point of assets, and 0.144 within the same province and half-year. The event study in Figure 11 replaces the size–Post interaction with size interacted with each half-year reporting date and omits June 2023. It retains the baseline's bank and half-year effects, bank-clustered errors, and absence of additional controls. The quarterly axis shows actual reports at Q2 and Q4, without interpolating Q1 or Q3 balances. The raw log-balance coefficients are 0.146 and 0.334 per log point of assets in December 2024 and December 2025, and the doubling comparisons are $100[\exp(0.146 \ln 2) - 1] \approx 11\%$ and $100[\exp(0.334 \ln 2) - 1] \approx 26\%$. The late-period difference exceeds the average 11% Post comparison because that average also includes the earlier reports when the gap was much smaller.

Borrower funding (Section 5.3). High-reliance and low-reliance banks are the local banks without a subsidiary whose initial home LGFV bond share in equation (10) was above and below the median. Let X_f^H and X_f^L be the fractions of borrower f's WMP holdings that these two groups held in 2022Q4–2023Q1. For borrower f in quarter q, we estimate

$$\begin{aligned}\text{Holdings}_{fq}^G = {} & \beta_H \, X_f^H \times \mathbb{1}\{\text{Post}\}_q + \beta_L \, X_f^L \times \mathbb{1}\{\text{Post}\}_q \\ & + \eta_H \, X_f^H \times \mathbb{1}\{\text{Late}\}_q + \eta_L \, X_f^L \times \mathbb{1}\{\text{Late}\}_q + \alpha_f + \alpha_{p(f),q} + \varepsilon_{fq}.\end{aligned} \tag{D.7}$$

The outcome is the inverse hyperbolic sine of the borrower's bonds held by the WMPs of holder group G, in RMB 10,000. Post begins in 2023Q4 and Late in 2024Q3, after the June 2024 guidance. Borrower effects α_f remove persistent differences in WMP funding, and borrower-province-by-quarter effects $\alpha_{p(f),q}$ compare borrowers in the same province and quarter, which absorbs provincial refinancing. The coefficient β_H compares a borrower funded entirely by high-reliance banks with one that neither group funded, and $\beta_H - \beta_L$ tests whether a higher initial home LGFV bond share mattered beyond being a local bank without a subsidiary. The sample covers borrower-quarters in 2023Q2–2025Q2, and errors cluster by borrower. Panel B of Table 18 repeats the regression for private credit. After June 2024, total WMP holdings of the borrowers of high-reliance banks fell by a further 0.48 log points. For a borrower funded entirely by high-reliance banks, private credit held by local banks without a subsidiary falls by 3.96 log points and national banks' holdings rise by 2.28, compared with 1.61 and 1.14 for borrowers of low-reliance banks.

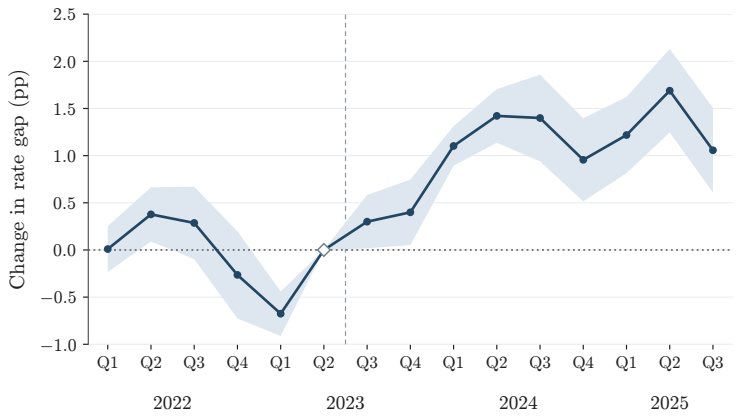
These borrowers also became a larger part of national banks' WMP portfolios. We say that local banks without a subsidiary mainly funded a borrower when they held at least half of its WMP holdings before the program. The share of national banks' LGFV bond holdings invested in such borrowers rose from 10.1% in 2022Q4 to 19.3% in 2025Q2. National banks earned more on these positions than on the rest of their LGFV bond holdings, with value-weighted coupons of 3.74% against 3.37% in 2025Q2. Column 5 of Table 18 relates the coupons of the borrowers' new bonds to the same funding shares. It compares bonds of the same borrower and province-month and controls for log maturity and maturity-group fixed effects. In the same specification, the coupon's premium over the CSI valuation yield at listing also fell for X_f^H, by 15.0 basis points (standard error 3.7) after implementation and by a further 9.0 (2.9) after June 2024.

Table D1
Initial credit portfolios by bank type and bank size

Share (%)	All provinces		Home province	
	LGFV bonds	PC	LGFV bonds	PC
Panel A. Bank type, all available banks				
National banks	12.6	24.1	0.4	2.1
Local banks	43.1	9.3	24.7	6.1
Panel B. FY2021 size terciles, local banks present in both periods				
Smallest tercile	38.6	1.1	25.1	0.7
Middle tercile	53.3	5.3	37.7	3.5
Largest tercile	40.6	11.4	20.7	7.4

Notes: PC denotes private credit. Entries are value-weighted percentages of included debt holdings over 2022Q1–2023Q3. Included debt excludes deposits, funds, other investment vehicles, and central-government, central-bank, and policy-bank obligations. Home means the parent bank's headquarters province. Panel A contains 34,595 national-bank and 36,194 local-bank product-quarters. Panel B uses the local banks and fixed FY2021 size terciles of Panel C of Table 12. Table 4 reports the controlled comparisons. Before implementation, local banks hold 65.0% of the sample's LGFV bonds and national banks hold 82.7% of its PC. Afterward these shares are 62.6% and 68.7%.

Figure D1
The private credit–bond rate gap within the same borrower



Notes: This within-borrower event study plots $\delta_q = \beta_q - \beta_{2023Q2}$ from equation (D.2). It is the change in the annual contractual gap between private credit and bond rates of the same borrower, in percentage points. Gap levels come from equation (D.1), which includes borrower-by-calendar-month effects and maturity interacted with every quarter. The sample is 17,303 equally weighted financing observations of 589 borrowers in 2021Q1–2025Q3. Shading shows pointwise 95% confidence intervals clustered by borrower province, the hollow diamond marks the omitted quarter, and the dashed line separates 2023Q2 and 2023Q3. Coverage falls to 58, 35, and 21 borrowers in 2025Q1–Q3.

Table D2
Alternative specifications for bank size, risk held, and risk reported

Specification	Baseline	Local-bank indicator added	Products weighted equally	Asset-mix controls	Bonds at least 60% of holdings	Plans and funds marked at AAA	Only bonds marked	Holdings before the sell-off
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A. What the product holds								
Home LGFV credit share (pp)	−5.394*** (0.874)	−4.280*** (1.345)	−5.730*** (1.092)	−4.048*** (0.755)	−6.012*** (1.097)	—	—	—
Issue spread of bonds held (bp)	−22.374*** (3.511)	−19.807*** (6.705)	−22.578*** (4.478)	−21.968*** (3.630)	−19.168*** (4.167)	—	—	—
Issue spread relative to same rating (bp)	−17.079*** (4.084)	−22.247*** (6.889)	−11.900*** (3.445)	−18.553*** (4.402)	−16.085*** (5.262)	—	—	—
All-in spread of holdings (bp)	−12.792*** (2.866)	−9.914** (4.753)	−10.570*** (3.893)	−13.556*** (2.437)	−17.082*** (3.576)	−14.642*** (2.941)	—	−14.165*** (2.995)
Log volatility implied by holdings	−0.074*** (0.016)	−0.054** (0.025)	−0.087*** (0.018)	−0.024* (0.013)	−0.038* (0.020)	−0.077*** (0.015)	−0.253*** (0.060)	−0.089*** (0.017)
Panel B. What the product reports								
Log SD of monthly return	0.154*** (0.036)	0.137** (0.055)	0.070** (0.035)	0.158*** (0.035)	0.191*** (0.051)	—	—	—
Log reported minus log implied SD	0.228*** (0.040)	0.191*** (0.062)	0.158*** (0.033)	0.183*** (0.039)	0.229*** (0.055)	0.231*** (0.040)	0.406*** (0.069)	0.243*** (0.040)
Return, Nov.–Dec. 2022 (bp)	−6.228*** (1.669)	−6.144** (2.594)	−1.697 (1.949)	−7.941*** (1.635)	−8.963*** (2.174)	—	—	—
Reported minus implied, Nov.–Dec. 2022 (bp)	−11.656*** (2.320)	−9.881** (3.898)	−8.394*** (2.070)	−10.124*** (2.168)	−13.200*** (3.050)	−12.205*** (2.226)	−16.357*** (2.155)	−12.870*** (2.372)
Mean monthly return (bp)	−1.413*** (0.297)	−1.357** (0.535)	−1.309*** (0.373)	−1.405*** (0.310)	−1.302*** (0.322)	—	—	—
Redemption type × risk rating fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank	Parent bank	Parent bank	Parent bank	Parent bank

Notes: Each entry is the coefficient on log bank size from a separate regression of the row's outcome. Column 1 repeats Table 7. Column 2 adds a local-bank indicator, so it compares banks of different size within each bank type. Column 3 weights products equally, and column 4 controls for the share of each asset class in disclosed holdings. Column 5 keeps the 1,406 products whose disclosed holdings are at least 60% bonds (1,234 for the spread relative to the same rating). Columns 6 to 8 change how holdings are valued, so they apply only to the all-in spread and the outcomes built from implied returns. Column 6 values asset-management plans and funds at AAA instead of AA+, column 7 marks only directly held bonds, and column 8 uses only the holdings disclosed in 2022Q2 and 2022Q3, before the sell-off (4,164 products). The other columns use the products of Table 7. Except in column 3, each bank receives equal total weight. Parentheses contain parent-bank clustered errors. Stars indicate 10%, 5%, and 1% significance. Estimation sample: April 2022–June 2023.

Table D3
Smaller banks, risk held, and risk reported

Panel A. What the product holds						
Dependent variable	Home LGFV credit share (pp)	Issue spread of bonds held (bp)	Issue spread relative to same rating (bp)	All-in spread of holdings (bp)	Log volatility implied by holdings	
	(1)	(2)	(3)	(4)	(5)	
$\mathbb{1}\{\text{Smaller bank}\}$	15.509***	50.431***	48.555***	32.983***	0.157***	
	(3.512)	(14.552)	(14.121)	(11.214)	(0.049)	
Observations	4,165	3,991	3,263	4,165	4,165	
Redemption type × risk rating fixed effects	Yes	Yes	Yes	Yes	Yes	
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank	Parent bank	
Panel B. What the product reports						
Dependent variable	Log SD of monthly return	Log reported minus log implied SD	Return, Nov.–Dec. 2022 (bp)	Reported minus implied, Nov.–Dec. 2022 (bp)	Mean monthly return (bp)	Safe rating, 1 or 2 (pp)
	(1)	(2)	(3)	(4)	(5)	(6)
$\mathbb{1}\{\text{Smaller bank}\}$	−0.565***	−0.722***	20.999***	33.023***	3.952***	6.982*
	(0.134)	(0.144)	(5.995)	(7.713)	(1.028)	(4.135)
Observations	4,165	4,165	4,165	4,165	4,165	11,580
Redemption type × risk rating fixed effects	Yes	Yes	Yes	Yes	Yes	No
Redemption type fixed effects	No	No	No	No	No	Yes
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank	Parent bank	Parent bank

Notes: Each column is a separate regression of the outcome on an indicator for smaller banks, those at or below the median FY2021 size of the banks in the main sample of Table 7, which is RMB 0.44 trillion. All smaller banks are local banks, and every national bank is among the larger banks. The products, outcomes, fixed effects, and weights are those of Table 7. Parentheses contain parent-bank clustered errors. Stars indicate 10%, 5%, and 1% significance. Estimation sample: April 2022–June 2023.

Table D4
Net flows by reported return in the 2022 sell-off quarter

Reported return, 2022Q4	Products	Median reported return (bp)	Share of beginning assets (%)	Net flow (% of beginning assets)
Loss of more than 0.5%	303	−86.6	13.5	−25.3
Loss of 0.1% to 0.5%	160	−33.8	12.5	−27.5
Loss of less than 0.1%	46	−4.8	3.3	−37.1
Gain of less than 0.5%	196	23.2	34.3	−17.6
Gain of 0.5% to 1%	114	56.7	36.3	8.9
Gain of more than 1%	7	144.5	0.0	39.4
All daily-open products	826	−25.5	100.0	−10.9

Notes: The table covers daily-open WMPs in 2022Q4, which is the quarter of the bond sell-off. It groups them by the return each product reported for the quarter. Net flow is end-of-quarter assets minus beginning assets grown by the reported return, which is the asset growth net of investment gains used in Table 1. It is summed within each group and divided by the group's beginning assets. Daily-open products allow withdrawals on any business day, so their outflows reflect savers' redemption decisions. The median reported return is over the quarter. In the preceding quarter of 2022Q3, daily-open products took in net flows of 6.2% of beginning assets.

Table D5
Flows after the 2022 sell-off, by reported and implied losses

Dependent variable	Net flow in 2023Q1 (% of beginning assets)		
Sample	All products	Daily- and periodic-open	Daily-open
	(1)	(2)	(3)
Reported return, Nov.–Dec. 2022 (per 10 bp a month)	1.16***	1.35**	8.22*
	(0.38)	(0.57)	(4.36)
Return implied by holdings, Nov.–Dec. 2022 (per 10 bp a month)	−0.45	−0.01	−0.79
	(0.28)	(0.37)	(1.20)
Mean reported return (bp a month)	−26.0	−29.4	−19.1
Mean implied return (bp a month)	−63.6	−58.1	−39.0
Observations	5,296	3,732	481
Product, bank, and city controls	Yes	Yes	Yes
Risk rating × redemption type fixed effects	Yes	Yes	Yes
Bank fixed effects	Yes	Yes	Yes
Standard errors	Parent bank	Parent bank	Parent bank

Notes: The unit of observation is a product of any type with disclosed holdings and at least nine monthly returns between April 2022 and June 2023 that reports assets in 2022Q4 and 2023Q1. The dependent variable is its net flow in 2023Q1. Net flow is asset growth net of investment gains as a percentage of beginning assets, as in Table 1. The two returns enter jointly. The reported return is the product's mean monthly reported return in November and December 2022, and the implied return is its mean monthly return in the same months when its disclosed holdings are marked to market prices (Internet Appendix D.3). The regressions include the eleven lagged controls of Table 1, fixed effects for risk rating by redemption type, and bank fixed effects, so products are compared with other products of the same bank. Parentheses contain parent-bank clustered errors. Stars indicate 10%, 5%, and 1% significance.

Table D6
Home credit exposure, bond income, and product returns

Panel A. Bond-income controls

Dependent variable	Monthly return (bp)			
Specification	Observed rates No yield control (1)	Observed rates Add yield control (2)	50% coverage No yield control (3)	50% coverage Add yield control (4)
Home LGFV bond share	8.831***	3.759**	9.021***	3.692**
	(1.964)	(1.663)	(2.083)	(1.815)
Home LGFV bond share × 𝟙{Post}	−6.731***	−2.886*	−7.245***	−3.222*
	(1.662)	(1.568)	(1.701)	(1.654)
Home private credit share	8.759***	6.895***	8.087**	6.370**
	(1.450)	(1.402)	(3.297)	(3.044)
Home private credit share × 𝟙{Post}	−3.639*	−2.227	1.280	1.926
	(1.892)	(1.859)	(3.346)	(3.142)
Mean bond issue yield (pp)	—	2.659***	—	2.773***
		(0.510)		(0.521)
Mean bond issue yield × 𝟙{Post}	—	−1.334***	—	−1.414***
		(0.452)		(0.487)
Implied effects from the same regression				
Post-program home-PC premium	5.120***	4.668***	9.367***	8.295***
	(1.490)	(1.353)	(2.978)	(2.637)
Product, bank, and city controls	Yes	Yes	Yes	Yes
Observations	524,670	524,670	449,162	449,162
Month fixed effects	Yes	Yes	Yes	Yes
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank

Panel B. NAV and sample checks

Dependent variable	Monthly return (bp)			
Specification	Mature products (1)	Add volatility (2)	Product-month average (3)	No winsorizing (4)
Home LGFV bond share	9.048***	8.442***	8.990***	8.512***
	(1.777)	(2.269)	(2.040)	(1.938)
Home LGFV bond share × 𝟙{Post}	−6.553***	−5.285***	−7.254***	−6.609***
	(1.811)	(1.978)	(1.662)	(1.718)
Home private credit share	6.951***	7.012***	7.859***	7.598***
	(1.514)	(1.412)	(1.559)	(1.467)
Home private credit share × 𝟙{Post}	−2.816	−2.619	−4.418**	−3.637**
	(1.865)	(1.644)	(1.797)	(1.559)
Implied effects from the same regression				
Post-program home-PC premium	4.136***	4.393***	3.441**	3.961***
	(1.502)	(1.217)	(1.365)	(1.089)
Product, bank, and city controls	Yes	Yes	Yes	Yes
Observations	380,984	383,198	448,645	567,855
Month fixed effects	Yes	Yes	Yes	Yes
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank

Notes: All regressors of Table 6 enter. PC denotes private credit, and the post-program home-PC premium is the home private credit share coefficient plus its Post interaction. Month effects absorb Post. In Panel A, columns 1–2 keep products with observed bond issue rates, and columns 3–4 require issue rates for at least 50% of included debt holdings. Columns 2 and 4 add mean bond issue yield and its Post interaction. In Panel B, column 1 uses mature products with age and remaining-maturity controls, column 2 adds volatility and its Post interaction, column 3 averages NAV series equally within product-month, and column 4 uses returns without winsorization. Parentheses contain parent-bank clustered errors. Stars indicate 10%, 5%, and 1% significance.